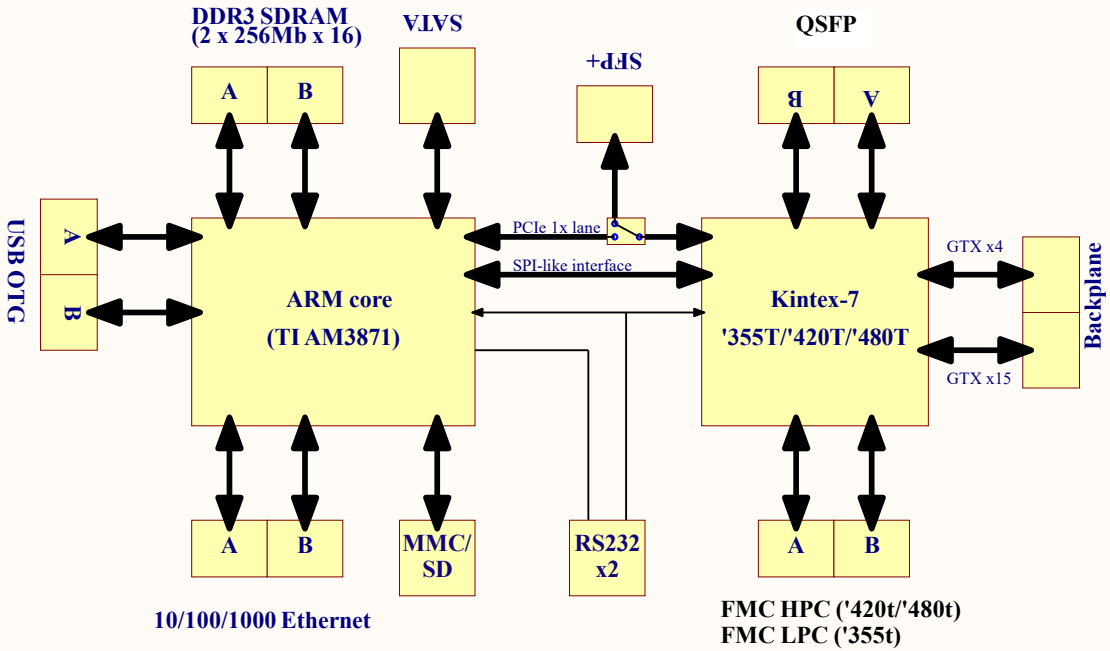
McGill Cosmology Instrumentation Library
ICE Board Schematics

These schemaatics represent the Rev4 E build



Filename	Page	Description
preamble_clocking.SchDoc	6	Cartoon: Clocking
preamble_addresses.SchDoc	7	Cartoon: I2C Bus Addressing
master.SchDoc	8	Top-Level Sheet
AM3871_master.SchDoc	9	ARM: Master Sheet
AM3871_configuration.SchDoc	10	ARM: Configuration
AM3871_ddr3_regulator.SchDoc	16	ARM: DDR3 termination regulator
AM3871_eth_cpu.SchDoc	17	ARM: Ethernet CPU
AM3871_eth_phy.SchDoc	18	ARM: Ethernet PHY
AM3871_eth_phy_single.SchDoc	19	ARM: Ethernet PHY (Single)
AM3871_fast_serial.SchDoc	20	ARM: Fast Serial
AM3871_unused.SchDoc	26	ARM: Unused Pins
FPGA_master.SchDoc	27	FPGA: Master Sheet
FPGA_configuration.SchDoc	28	FPGA: Configuration
FPGA_power.SchDoc	29	FPGA: Power
FPGA_ground.SchDoc	30	FPGA: Ground
FPGA_gtx111_gtx112_gtx113_gtx114_multiplexing.SchDoc	36	FPGA: GTX banks 111-114 (Multiplexing)
FPGA_gtx_mux4.SchDoc	37	FPGA: GTX 4-way multiplexer
FPGA_gtx115_gtx116_gtx117.SchDoc	38	FPGA: GTX banks 115, 116, 117
Backplane_master.SchDoc	39	Backplane Master Sheet
QSFP.SchDoc	40	QSFP Port
power_master.SchDoc	46	Power: Master Sheet
Power_BuckDirect.SchDoc	47	Power: Buck Regulators (via PosSupply)
Power_BuckIndirect.SchDoc	48	Power: Buck Regulators (via intermediate 5v5)
Power_BuckSelect.SchDoc	49	Power: Buck Regulators (selectable)
BuckRegulator.SchDoc	50	Power: Buck Regulator (single)
I2C_matrix.SchDoc	56	I2C: Interconnect Matrix
I2C_gpio.SchDoc	57	I2C: GPIOs
I2C_temp.SchDoc	58	I2C: Temperature Sensors
mechanical.SchDoc	59	Mechanical design elements (e.g. heat sinks)
ProtectionDiodes.SchDoc	60	Protection from Backplane faults

http://mcgillcosmology.ca

http://kingspeak.physics.mcgill.ca

Unmarked capacitor packages are 0402s
Unmarked resistor packages are 0603s

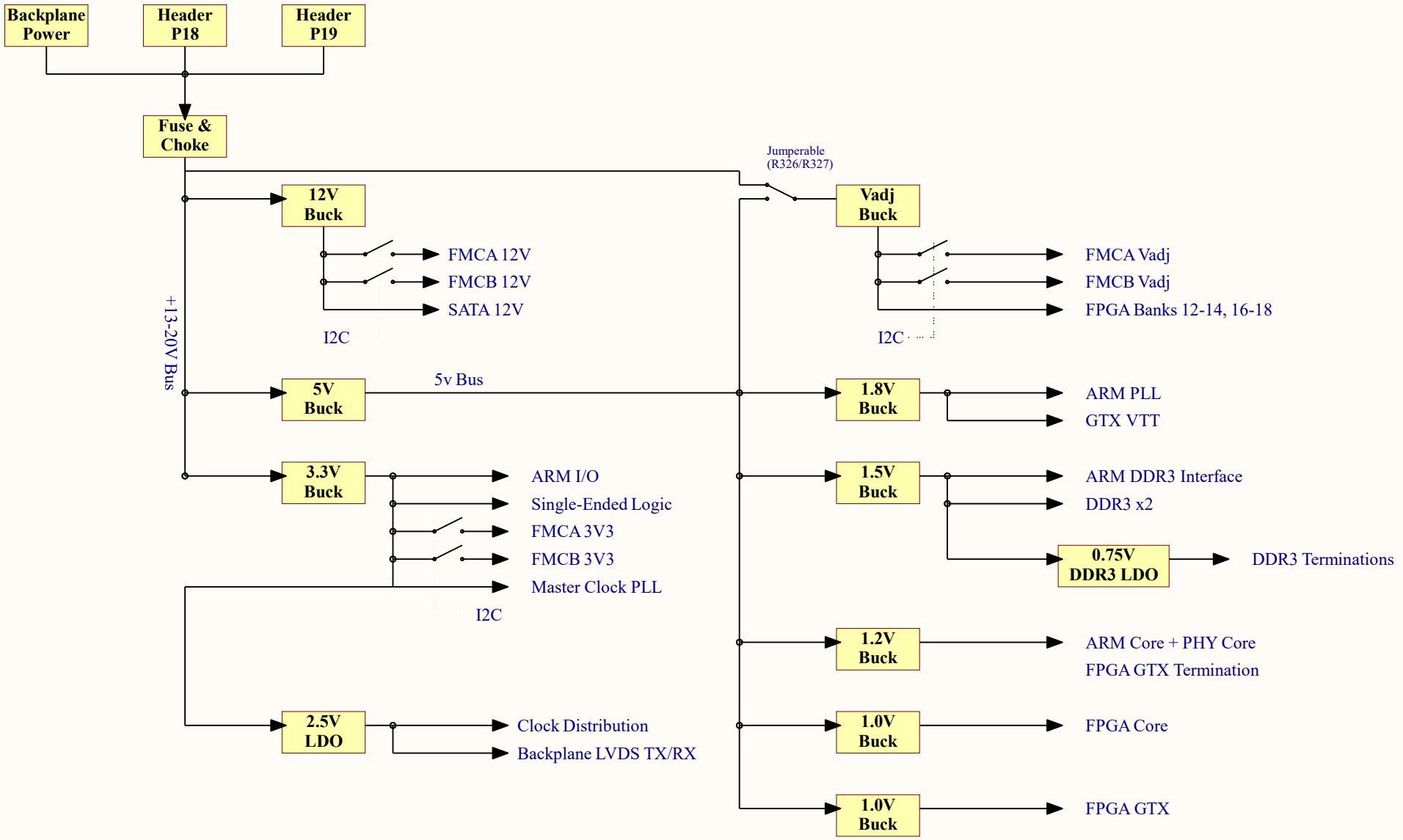
Title: Title Sheet		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet lof 60
File: addresses.SchDoc\title.SchDoc		Drawn By: Graeme Smecher

CONTAINER master.SchDoc;preamble_history.SchDoc;preamble_power.SchDoc;preamble_banking.SchDoc;preamble_gtx.SchDoc;preamble_clocking.SchDoc;preamble_addresses.SchDoc

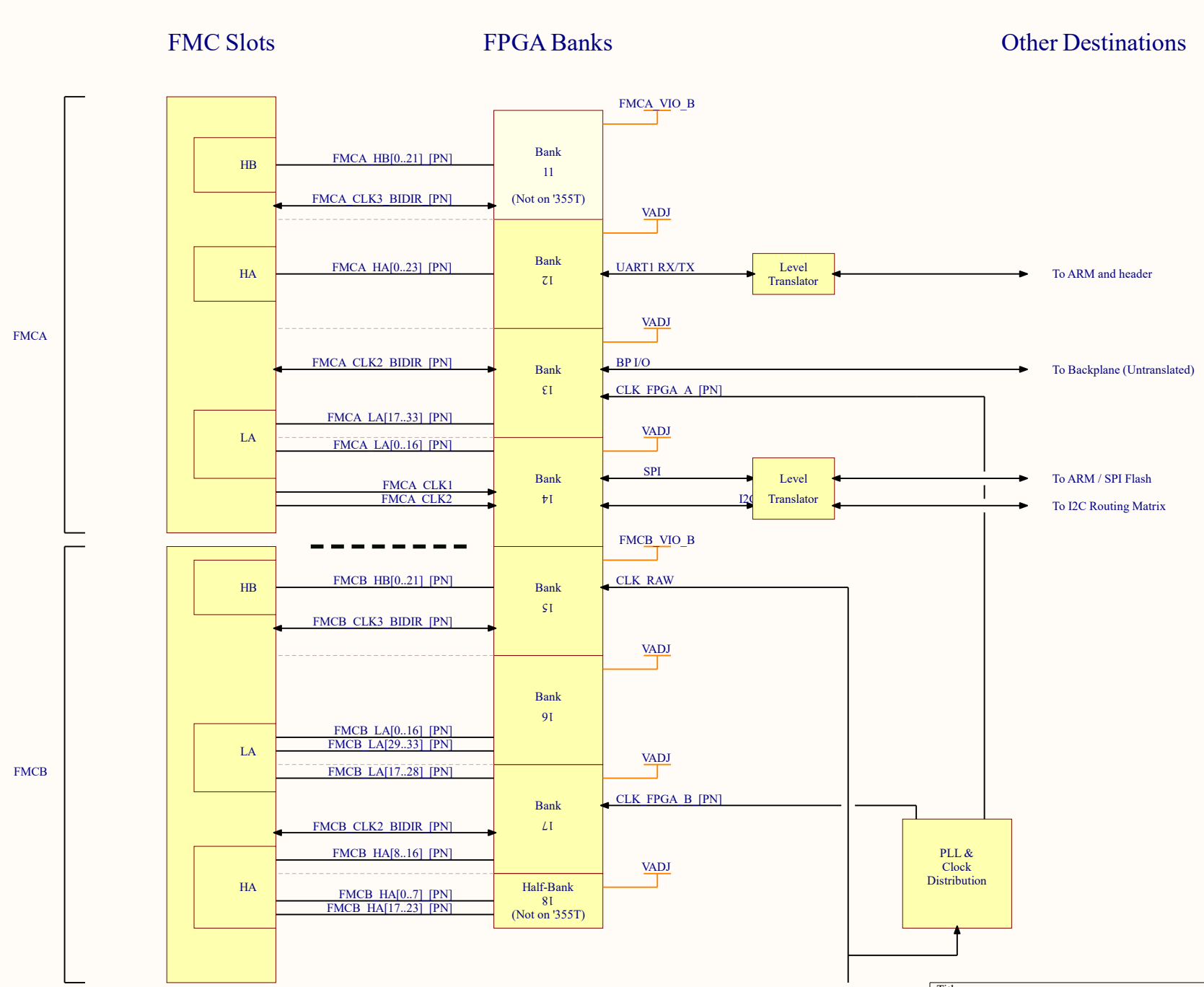
Date	Revision	Tag	Description
Feb. 22, 2013	0	rev0	Schematic capture complete. Preliminary placement. Fab run has significant limitations. FMCs rotated; IMPACT connector pads incorrectly manufactured. Several small, manual modifications required for successful power-up.
Nov. 15, 2013	2b	rev2b_1	Minor component modifications to Rev2 Found errors in Power stages - will recycle old PCBs and cut traces This commit has been used to generate the purchasing BOM
	3	rev3	Clocking stages were updated & LEDs changed
Jul. 21, 2014	4	rev4	Protection Diodes Added - as well as smoothing caps on backplane reset lines

Date	Revision	Tag	Description

Title: Revision History		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 2of 60
File: C:\Users\..\preamble_history.SchDoc		Drawn By: Graeme Smecher

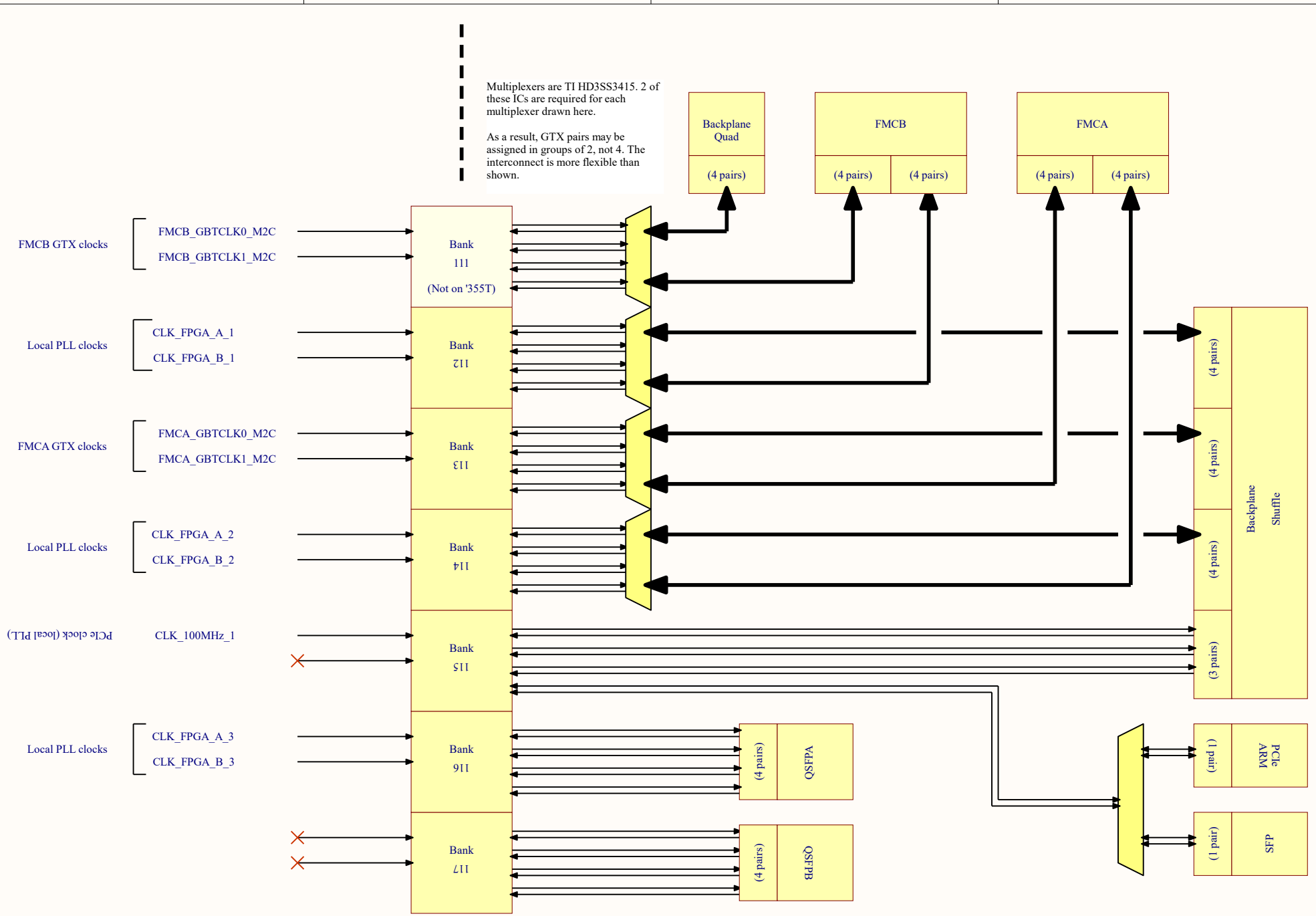


Title: Power Distribution		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 3of 60
File: C:\Users\..\preamble_power.SchDoc		Drawn By: Graeme Smecher



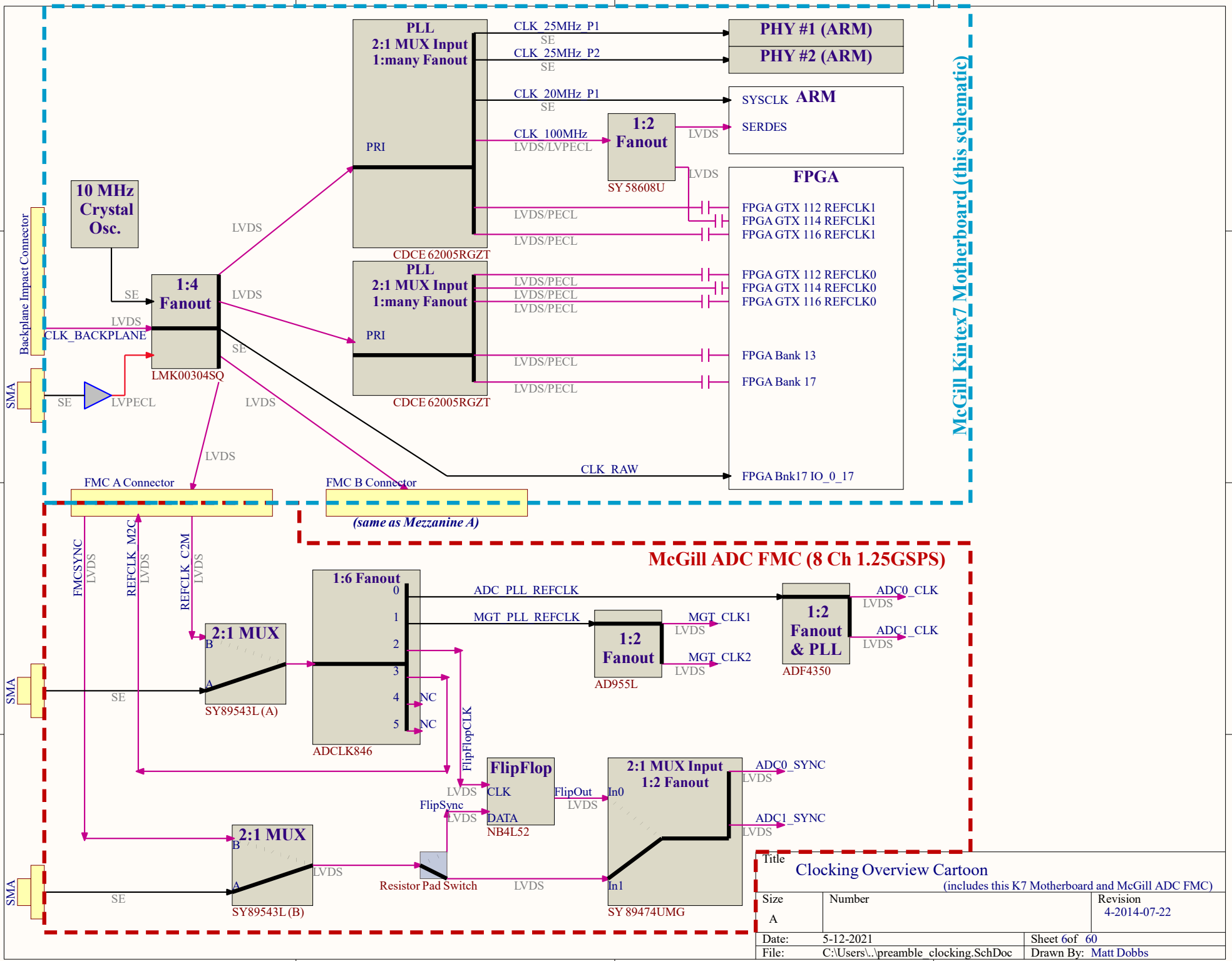
Low-speed FMC signals are controlled via the I2C matrix
FMCA has VREF signals attached to FPGA banks (e.g. for SSTL)

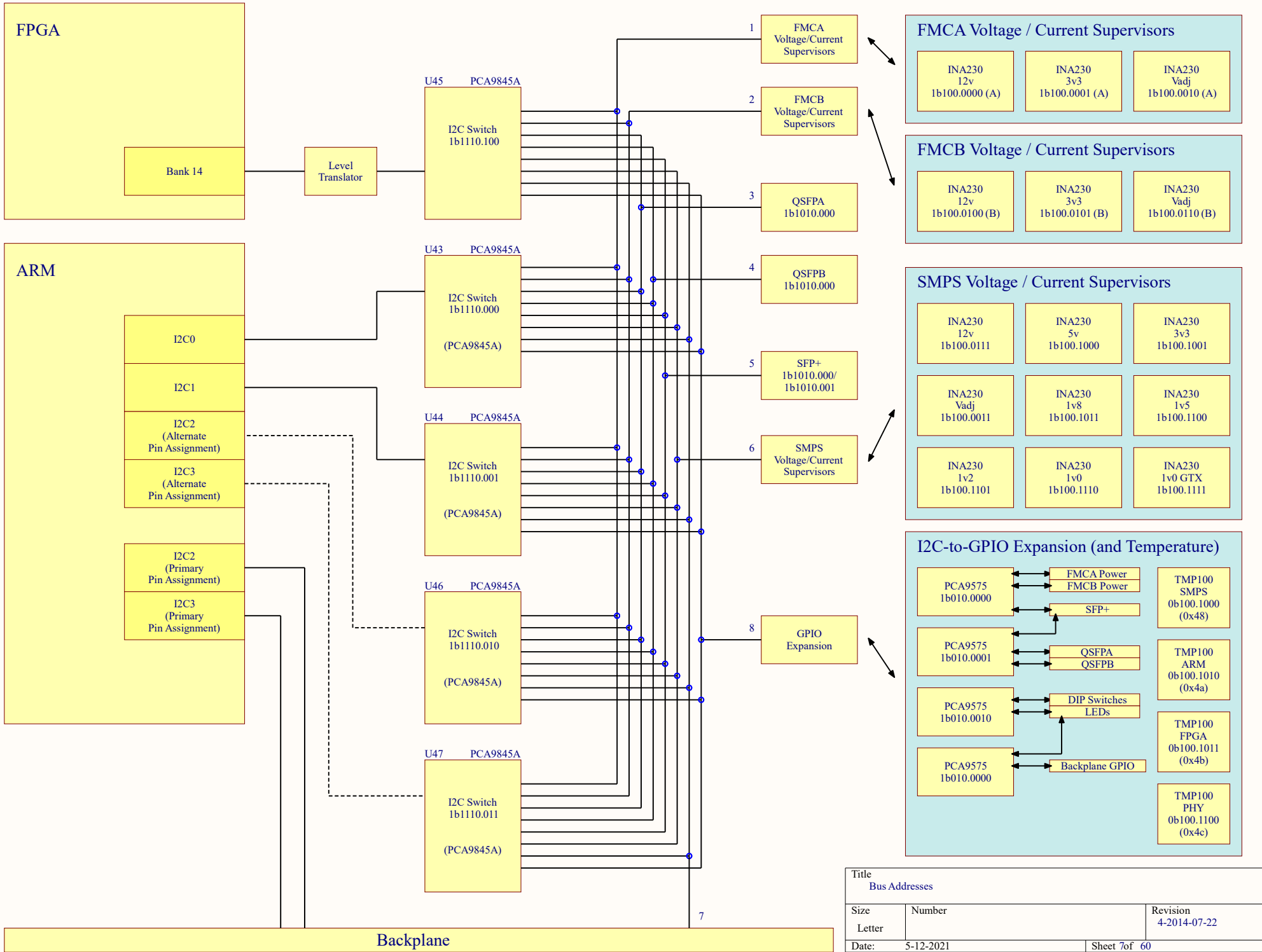
Title: FPGA Banking (Non-GTX)		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 4of 60
File: C:\Users\..\preamble_banking.SchDoc		Drawn By: Graeme Smecher



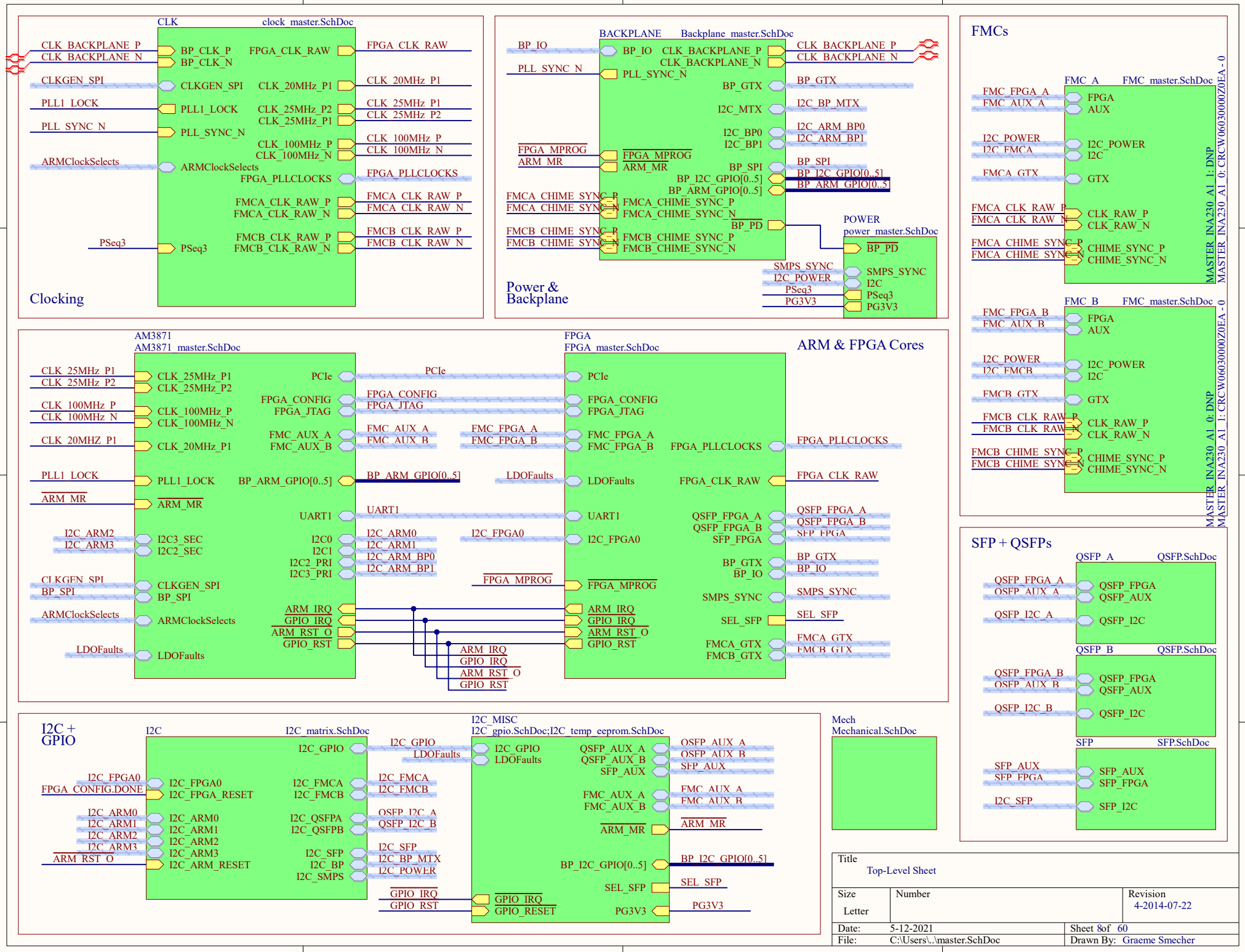
GTX clock interconnects. Each GTX clock can feed the 3 adjacent GTXes.

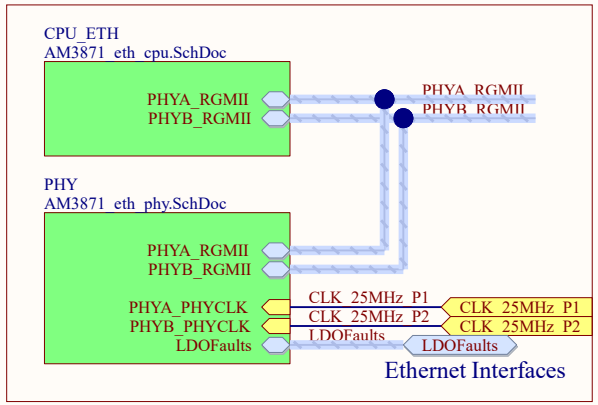
Title: Gigabit Transceivers: Clocking and Routing		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 5of 60
File: C:\Users\..\preamble_gtx.SchDoc		Drawn By: Graeme Smecher

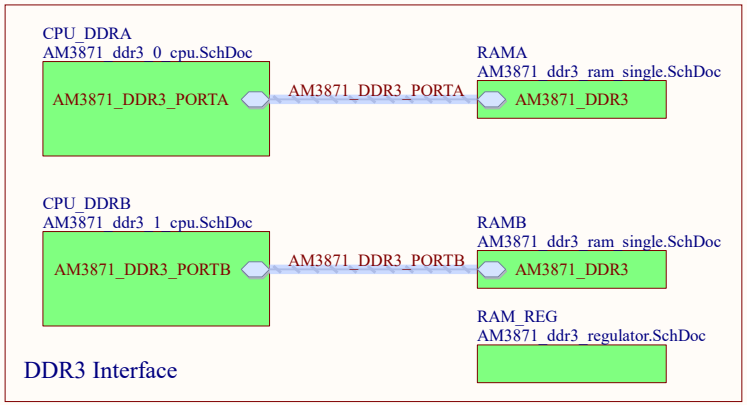

McGill Kintex7 Motherboard (this schematic)
PLL
2:1 MUX Input
1:many Fanout
PRI
CDCE 62005RGZT
PLL
2:1 MUX Input
1:many Fanout
PRI
CDCE 62005RGZT
10 MHz
Crystal
Osc.
1:4
Fanout
LMK00304SQ
Backplane Impact Connector
CLK_BACKPLANE
SMA
SE
LVPECL
LVDS
SE
CLK_25MHz_P1
CLK_25MHz_P2
PHY #1 (ARM)
PHY #2 (ARM)
CLK_20MHz_P1
SYSCLK ARM
SERDES
CLK_100MHz
LVDS/LVPECL
1:2
Fanout
SY 58608U
LVDS
FPGA
FPGA GTX 112 REFCLK1
FPGA GTX 114 REFCLK1
FPGA GTX 116 REFCLK1
FPGA GTX 112 REFCLK0
FPGA GTX 114 REFCLK0
FPGA GTX 116 REFCLK0
FPGA Bank 13
FPGA Bank 17
FPGA Bnk17 IO_0_17
LVDS/PECL
CLK_RAW
FMC A Connector
FMC B Connector
(same as Mezzanine A)
McGill ADC FMC (8 Ch 1.25GSPS)
FMCSYNC
REFCLK_M2C
REFCLK_C2M
1:6 Fanout
ADC_PLL_REFCLK
MGT_PLL_REFCLK
1:2
Fanout
AD955L
MGT_CLK1
MGT_CLK2
1:2
Fanout
& PLL
ADF4350
ADC0_CLK
ADC1_CLK
2:1 MUX
SY89543L (A)
ADCLK846
NC
FlipFlopCLK
FlipFlop
CLK
DATA
NB4L52
FlipSync
FlipOut
2:1 MUX Input
1:2 Fanout
In0
In1
SY 89474UMG
ADC0_SYNC
ADC1_SYNC
2:1 MUX
SY89543L (B)
Resistor Pad Switch
Title
Clocking Overview Cartoon
(includes this K7 Motherboard and McGill ADC FMC)
Size
A
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 6of 60
File: C:\Users\..\preamble_clocking.SchDoc
Drawn By: Matt Dobbs

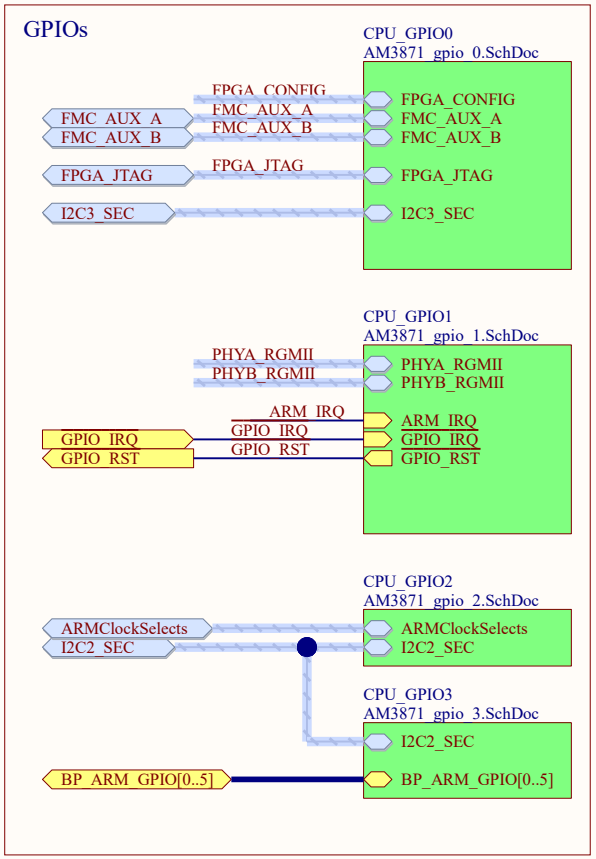


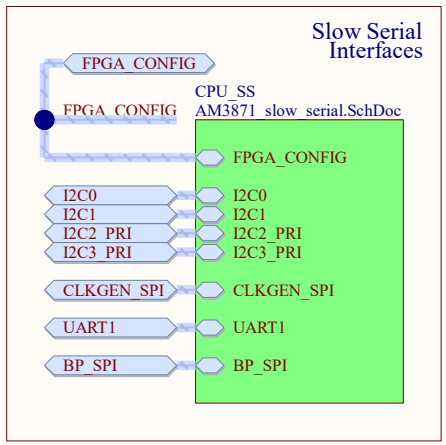
Title: Bus Addresses		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 7of 60
File: C:\Users\..\preamble_addresses.SchDoc		Drawn By: Graeme Smecher

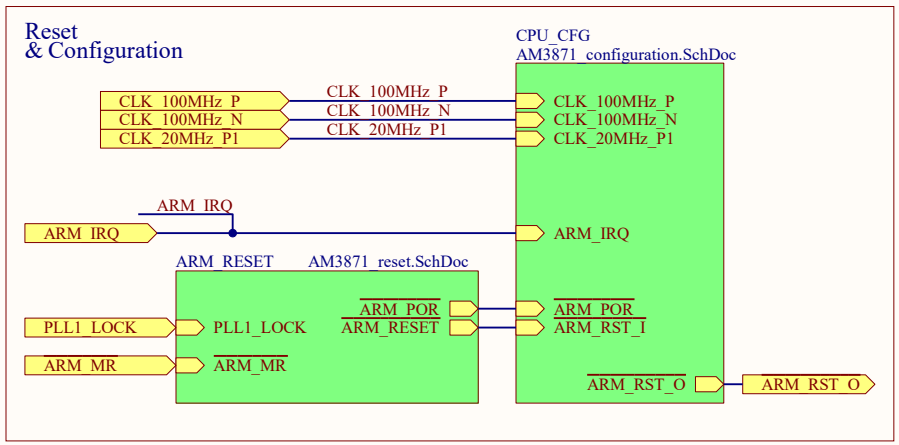

CLK
clock_master.SchDoc
CLK_BACKPLANE_P
CLK_BACKPLANE_N
BP_CLK_P
BP_CLK_N
FPGA_CLK_RAW
CLKGEN_SPI
CLK_20MHz_P1
PLL1_LOCK
CLK_25MHz_P2
CLK_25MHz_P1
PLL_SYNC_N
CLK_100MHz_P
CLK_100MHz_N
ARMClockSelects
FPGA_PLLCLOCKS
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
PSeq3
Clocking
BACKPLANE
Backplane_master.SchDoc
BP_IO
PLL_SYNC_N
BP_GTX
I2C_MTX
I2C_BP_MTX
I2C_BP0
I2C_BP1
I2C_ARM_BP0
I2C_ARM_BP1
FPGA_MPROG
ARM_MR
BP_SPI
BP_I2C_GPIO[0..5]
BP_ARM_GPIO[0..5]
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
BP_PD
POWER
power_master.SchDoc
SMPS_SYNC
I2C_POWER
I2C
Power &
Backplane
FMCs
FMC_A
FMC_master.SchDoc
FMC_FPGA_A
FMC_AUX_A
FPGA
AUX
I2C_FMCA
GTX
MASTER_INA230_A1_1: DNP
MASTER_INA230_A1_0: CRCW060300000Z0EA - 0
FMC_B
FMC_FPGA_B
FMC_AUX_B
I2C_FMCB
FMCB_GTX
AM3871
AM3871_master.SchDoc
FPGA
FPGA_master.SchDoc
ARM & FPGA Cores
CLK_25MHz_P1
CLK_25MHz_P2
CLK_100MHz_P
CLK_100MHz_N
CLK_20MHZ_P1
PCIe
FPGA_CONFIG
FPGA_JTAG
FMC_AUX_A
FMC_AUX_B
FMC_FPGA_A
FMC_FPGA_B
BP_ARM_GPIO[0..5]
LDOFaults
UART1
I2C_FPGA0
I2C3_SEC
I2C2_SEC
I2C0
I2C1
I2C2_PRI
I2C3_PRI
I2C_ARM0
I2C_ARM1
I2C_ARM_BP0
I2C_ARM_BP1
CLKGEN_SPI
BP_SPI
ARM_IRQ
GPIO_IRQ
ARM_RST_O
GPIO_RST
FPGA_PLLCLOCKS
FPGA_CLK_RAW
QSFP_FPGA_A
QSFP_FPGA_B
SFP_FPGA
BP_GTX
BP_IO
SMPS_SYNC
SEL_SFP
FMCA_GTX
FMCB_GTX
SFP + QSFPs
QSFP_A
QSFP.SchDoc
QSFP_FPGA
QSFP_AUX
QSFP_I2C
QSFP_B
SFP
SFP.SchDoc
SFP_AUX
SFP_I2C
I2C +
GPIO
I2C
I2C_matrix.SchDoc
I2C_GPIO
I2C_FPGA0
FPGA_CONFIG.DONE
I2C_FPGA_RESET
I2C_FMCA
I2C_FMCB
I2C_ARM0
I2C_ARM1
I2C_ARM2
I2C_ARM3
ARM_RST_O
I2C_ARM_RESET
I2C_QSFPA
I2C_QSFPB
QSFP_I2C_A
QSFP_I2C_B
I2C_SFP
I2C_BP
I2C_SMPS
I2C_BP_MTX
I2C_POWER
I2C_MISC
I2C_gpio.SchDoc;I2C_temp_eeprom.SchDoc
LDOFaults
QSFP_AUX_A
QSFP_AUX_B
SFP_AUX
FMC_AUX_A
FMC_AUX_B
ARM_MR
BP_I2C_GPIO[0..5]
SEL_SFP
GPIO_IRQ
GPIO_RESET
PG3V3
Mech
Mechanical.SchDoc
Title
Top-Level Sheet
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 8of 60
File:
C:\Users\..\master.SchDoc
Drawn By:
Graeme Smecher

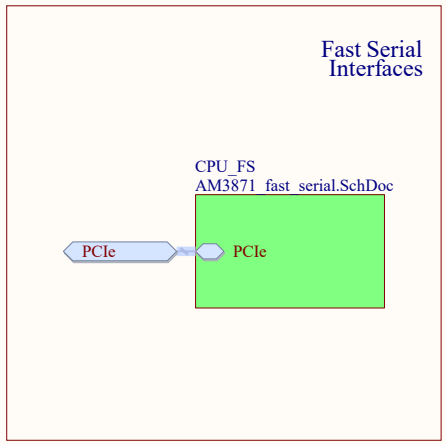




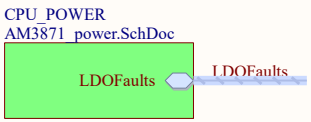


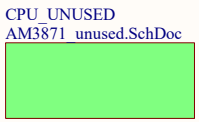




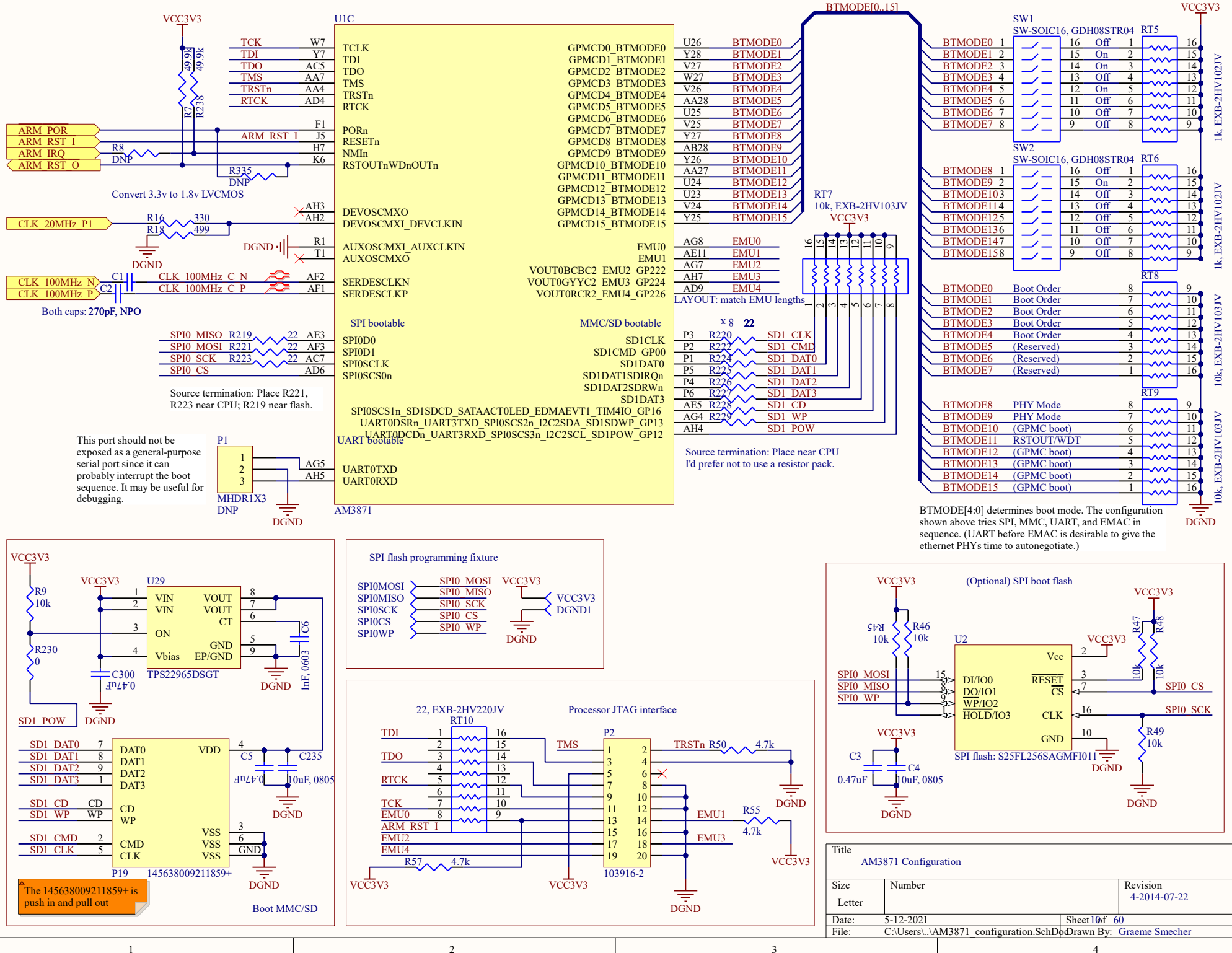


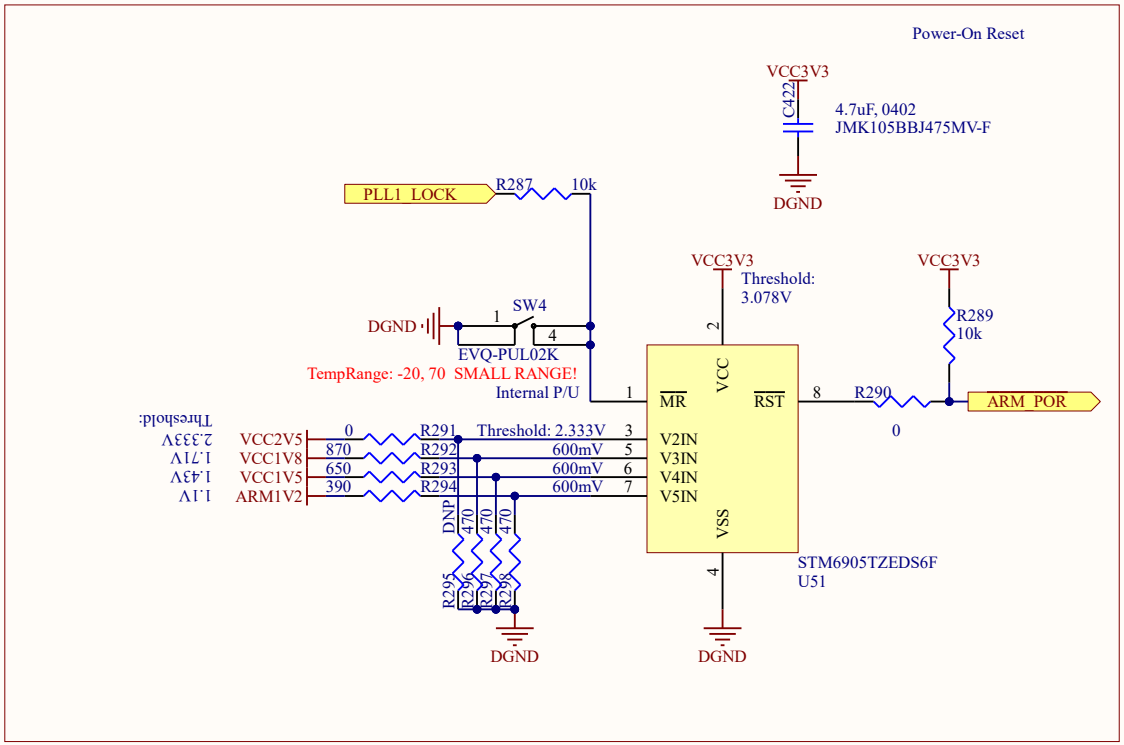


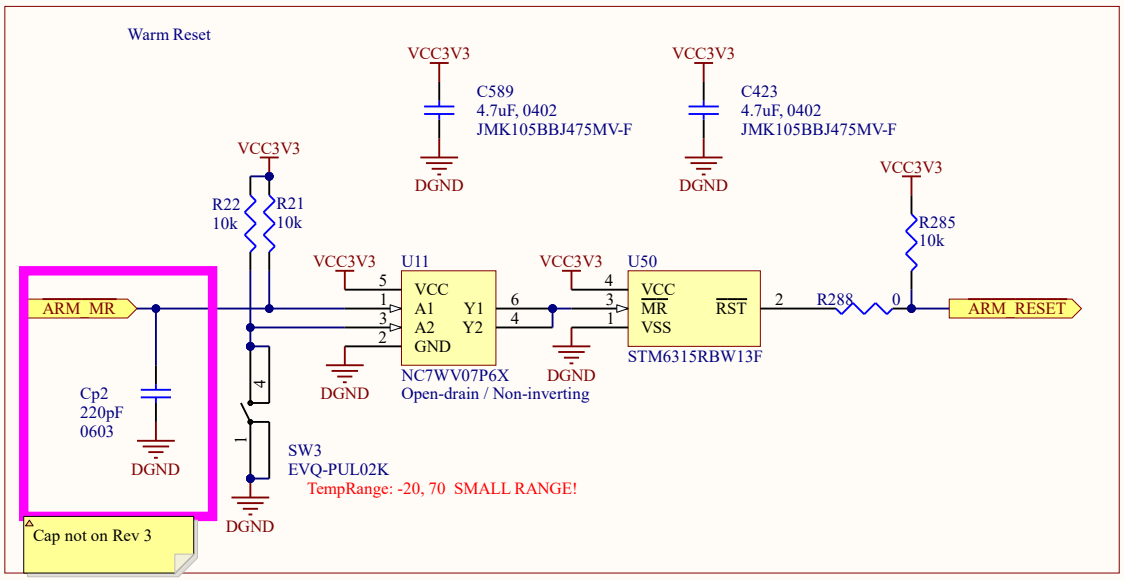




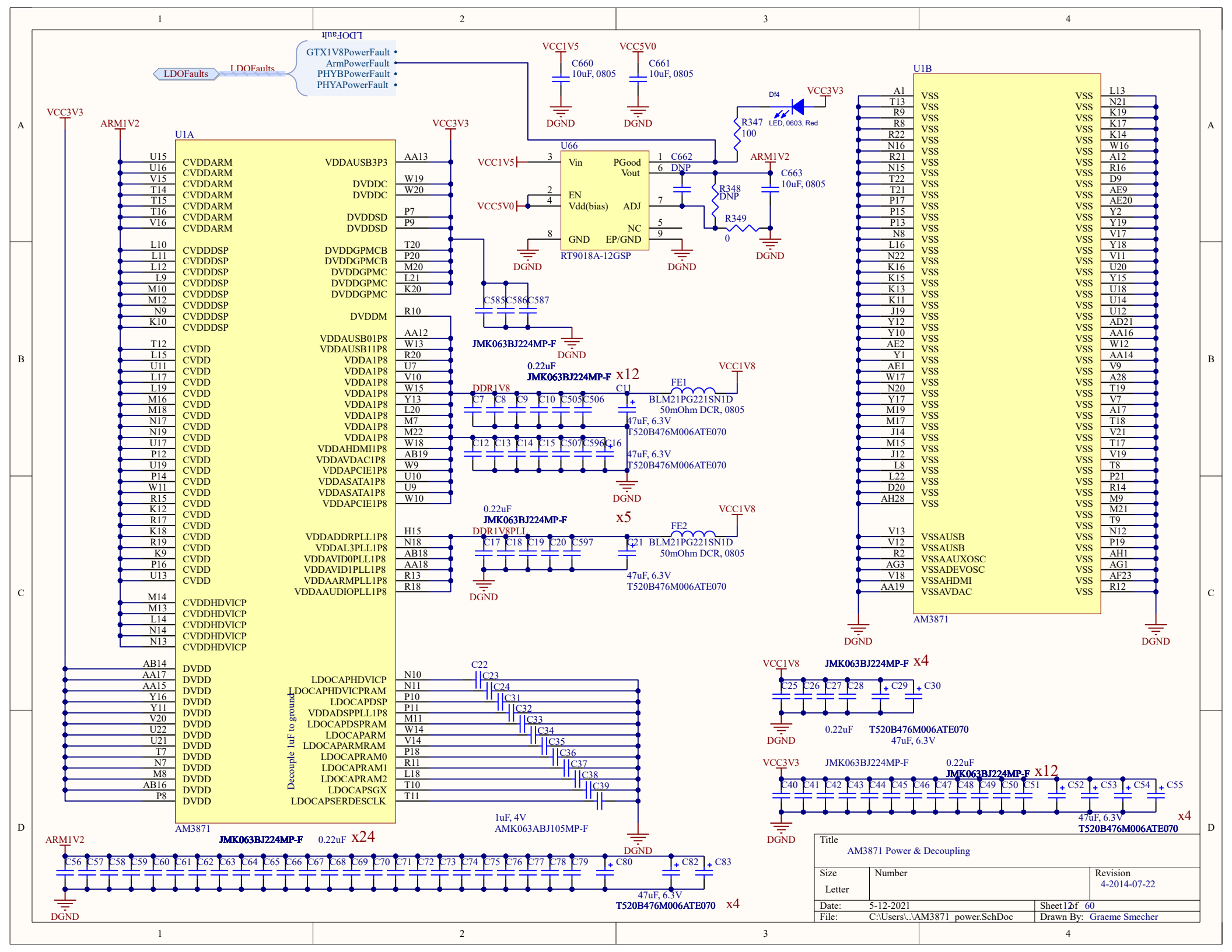
Title: AM3871 Master Sheet		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 9of 60
File: C:\Users\..\AM3871_master.SchDoc		Drawn By: Graeme Smecher


U1C
AM3871
VCC3V3
TCK W7
TDI Y7
TDO AC5
TMS AA7
TRSTn AA4
RTCK AD4
R7 49.9k
R238 49.9k
ARM_POR
ARM_RST_I
ARM_IRQ
ARM_RST_O
R8 DNP
R335 DNP
ARM_RST_I J5
F1
H7
K6
Convert 3.3v to 1.8v LVCMOS
CLK_20MHz_P1
R16 330
R18 499
DGND
CLK_100MHz_N
CLK_100MHz_P
C1
C2
CLK_100MHz_C_N
CLK_100MHz_C_P
AF2
AF1
Both caps: 270pF, NPO
SPI0_MISO R219 22 AE3
SPI0_MOSI R221 22 AF3
SPI0_SCK R223 22 AC7
SPI0_CS AD6
Source termination: Place R221, R223 near CPU; R219 near flash.
This port should not be exposed as a general-purpose serial port since it can probably interrupt the boot sequence. It may be useful for debugging.
P1
MHDR1X3
DNP
AG5
AH5
UART bootable
UART0TXD
UART0RXD
SPI bootable
SPI0D0
SPI0D1
SPI0SCLK
SPI0SCS0n
MMC/SD bootable
SD1CLK
SD1CMD_GP00
SD1DAT0
SD1DAT1SDIRQn
SD1DAT2SDRWn
SD1DAT3
SPI0SCS1n_SD1SDCD_SATAACT0LED_EDMAEVT1_TIM4IO_GP16
UART0DSRn_UART3TXD_SPI0SCS2n_I2C2SDA_SD1SDWP_GP13
UART0DCDn_UART3RXD_SPI0SCS3n_I2C2SCL_SD1POW_GP12
DEVOSCMXO
DEVOSCMXI_DEVCLKIN
AUXOSCMXI_AUXCLKIN
AUXOSCMXO
SERDESCLKN
SERDESCLKP
PORn
RESETn
NMIn
RSTOUTnWDnOUTn
GPMCD0_BTMODE0
GPMCD1_BTMODE1
GPMCD2_BTMODE2
GPMCD3_BTMODE3
GPMCD4_BTMODE4
GPMCD5_BTMODE5
GPMCD6_BTMODE6
GPMCD7_BTMODE7
GPMCD8_BTMODE8
GPMCD9_BTMODE9
GPMCD10_BTMODE10
GPMCD11_BTMODE11
GPMCD12_BTMODE12
GPMCD13_BTMODE13
GPMCD14_BTMODE14
GPMCD15_BTMODE15
EMU0
EMU1
VOUT0BCBC2_EMU2_GP222
VOUT0GYYC2_EMU3_GP224
VOUT0RCR2_EMU4_GP226
BTMODE[0..15]
U26 BTMODE0
Y28 BTMODE1
V27 BTMODE2
W27 BTMODE3
V26 BTMODE4
AA28 BTMODE5
U25 BTMODE6
V25 BTMODE7
Y27 BTMODE8
AB28 BTMODE9
Y26 BTMODE10
AA27 BTMODE11
U24 BTMODE12
U23 BTMODE13
V24 BTMODE14
Y25 BTMODE15
AG8 EMU0
AE11 EMU1
AG7 EMU2
AH7 EMU3
AD9 EMU4
LAYOUT: match EMU lengths
RT7
10k, EXB-2HV103JV
VCC3V3
x 8 22
P3 R220 SD1_CLK
P2 R222 SD1_CMD
P1 R224 SD1_DAT0
P5 R225 SD1_DAT1
P4 R226 SD1_DAT2
P6 R227 SD1_DAT3
AE5 R228 SD1_CD
AG4 R229 SD1_WP
AH4 SD1_POW
Source termination: Place near CPU
I'd prefer not to use a resistor pack.
SW1
SW-SOIC16, GDH08STR04
RT5
SW2
RT6
RT8
RT9
BTMODE0 Boot Order
BTMODE1 Boot Order
BTMODE2 Boot Order
BTMODE3 Boot Order
BTMODE4 Boot Order
BTMODE5 (Reserved)
BTMODE6 (Reserved)
BTMODE7 (Reserved)
BTMODE8 PHY Mode
BTMODE9 PHY Mode
BTMODE10 (GPMC boot)
BTMODE11 RSTOUT/WDT
BTMODE12 (GPMC boot)
BTMODE13 (GPMC boot)
BTMODE14 (GPMC boot)
BTMODE15 (GPMC boot)
1k, EXB-2HV102JV
10k, EXB-2HV103JV
BTMODE[4:0] determines boot mode. The configuration shown above tries SPI, MMC, UART, and EMAC in sequence. (UART before EMAC is desirable to give the ethernet PHYs time to autonegotiate.)
U29
TPS22965DSGT
R9 10k
R230 0
C300 0.47uF
1nF, 0603
C6
SD1_POW
SD1_DAT0
SD1_DAT1
SD1_DAT2
SD1_DAT3
SD1_CD
SD1_WP
SD1_CMD
SD1_CLK
P19 145638009211859+
C5 0.47uF
C235 10uF, 0805
The 145638009211859+ is push in and pull out
Boot MMC/SD
SPI flash programming fixture
SPI0MOSI
SPI0MISO
SPI0SCK
SPI0CS
SPI0WP
22, EXB-2HV220JV
RT10
Processor JTAG interface
P2
103916-2
TDI
TDO
RTCK
TCK
EMU0
ARM_RST_I
EMU2
EMU4
TMS
TRSTn R50 4.7k
EMU1
EMU3
R55 4.7k
R57 4.7k
(Optional) SPI boot flash
R45 10k
R46 10k
U2
SPI flash: S25FL256SAGMFI011
DI/IO0
DO/IO1
WP/IO2
HOLD/IO3
C3 0.47uF
C4 10uF, 0805
R47 10k
R48 10k
R49 10k
Title
AM3871 Configuration
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 10 of 60
File: C:\Users\..\AM3871_configuration.SchDoc
Drawn By: Graeme Smecher





Title: AM3871 Reset		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 1 bf 60	
File: C:\Users\..\AM3871_reset.SchDoc	Drawn By: Graeme Smecher	


LDOFaults
GTX1V8PowerFault
ArmPowerFault
PHYBPowerFault
PHYAPowerFault
U1A
AM3871
U1B
AM3871
U66
RT9018A-12GSP
ARM1V2
VCC3V3
VCC1V5
VCC5V0
VCC1V8
DGND
JMK063BJ224MP-F
T520B476M006ATE070
AMK063ABJ105MP-F
BLM21PG221SN1D
50mOhm DCR, 0805
LED, 0603, Red
Decouple 1uF to ground
Title
AM3871 Power & Decoupling
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
File: C:\Users\..\AM3871_power.SchDoc
Sheet 12 of 60
Drawn By: Graeme Smecher

Datasheet (http://www.ti.com/lit/ds/symlink/am3871.pdf) contains extensive stackup/layout directions. Here are pointers:

* Stackup: See 8.13.4.2.4.4.
* Placement: See 8.13.4.2.4.5
* Keepout: See 8.13.4.2.4.6
* Bypass caps: See 8.13.4.2.4.8

* VREFSSTL: Use 20-mil trace; narrower is permitted if needed.
* VTT: Routed like a power sub-plane; bypassed near terminator resistors

DDR0_D[0..31]
DDR0_A[0..14]
VCC1V5
G16 E21 E20 J18 J17 J16 J15 H17 H16
DVDDDDR0

VREF_DDR
R58 49.9
C102 JMK063BJ224MP-F 0.22uF
DGND DGND

U1F
AM3871

DDR0_A0	B19	DDR0A0	
DDR0_A1	F19	DDR0A1	
DDR0_A2	D18	DDR0A2	
DDR0_A3	E15	DDR0A3	
DDR0_A4	D15	DDR0A4	
DDR0_A5	B17	DDR0A5	
DDR0_A6	A15	DDR0A6	
DDR0_A7	E18	DDR0A7	
DDR0_A8	B15	DDR0A8	
DDR0_A9	C15	DDR0A9	
DDR0_A10	A19	DDR0A10	
DDR0_A11	D17	DDR0A11	
DDR0_A12	E17	DDR0A12	
DDR0_A13	F17	DDR0A13	
DDR0_A14	F16	DDR0A14	
DDR0_D0	E27	DDR0D0	
DDR0_D1	F26	DDR0D1	
DDR0_D2	E26	DDR0D2	
DDR0_D3	F25	DDR0D3	
DDR0_D4	F24	DDR0D4	
DDR0_D5	E25	DDR0D5	
DDR0_D6	D26	DDR0D6	
DDR0_D7	C28	DDR0D7	
DDR0_D8	C27	DDR0D8	
DDR0_D9	C25	DDR0D9	
DDR0_D10	C26	DDR0D10	
DDR0_D11	A27	DDR0D11	
DDR0_D12	B25	DDR0D12	
DDR0_D13	D24	DDR0D13	
DDR0_D14	A25	DDR0D14	
DDR0_D15	A24	DDR0D15	
DDR0_D16	B24	DDR0D16	
DDR0_D17	F22	DDR0D17	
DDR0_D18	H21	DDR0D18	
DDR0_D19	G21	DDR0D19	
DDR0_D20	D23	DDR0D20	
DDR0_D21	E23	DDR0D21	
DDR0_D22	C23	DDR0D22	
DDR0_D23	B22	DDR0D23	
DDR0_D24	H20	DDR0D24	
DDR0_D25	F21	DDR0D25	
DDR0_D26	G20	DDR0D26	
DDR0_D27	A22	DDR0D27	
DDR0_D28	C20	DDR0D28	
DDR0_D29	C21	DDR0D29	
DDR0_D30	D21	DDR0D30	
DDR0_D31	B20	DDR0D31	

DDR0CLK	B16	DDR0_CLK_P	
DDR0CLKn	A16	DDR0_CLK_N	
DDR0WEn	C17	DDR0_WEn	
DDR0CASn	C18	DDR0_CASn	
DDR0RASn	B18	DDR0_RASn	
DDR0CKE	H18	DDR0_CKE	
DDR0CS0n	F18	DDR0_CS0n	
DDR0CS1n	G17	DDR0_CS1n	
DDR0RST	G19	DDR0_RST	
DDR0DQM0	E28	DDR0_DQM0	
DDR0DQS0	D28	DDR0_DQS0_P	
DDR0DQS0n	D27	DDR0_DQS0_N	
DDR0DQM1	B28	DDR0_DQM1	
DDR0DQS1	B26	DDR0_DQS1_P	
DDR0DQS1n	A26	DDR0_DQS1_N	
DDR0DQM2	C24	DDR0_DQM2	
DDR0DQS2	B23	DDR0_DQS2_P	
DDR0DQS2n	A23	DDR0_DQS2_N	
DDR0DQM3	F20	DDR0_DQM3	
DDR0DQS3	B21	DDR0_DQS3_P	
DDR0DQS3n	A21	DDR0_DQS3_N	
DDR0BA0	F15	DDR0_BA0	
DDR0BA1	A20	DDR0_BA1	
DDR0BA2	A18	DDR0_BA2	
DDR0ODT0	G18	DDR0_ODT0	
DDR0ODT1	H19	DDR0_ODT1	

VCC1V5
R59 10k

AM3871_DDR3

DDR0_D[0..31]	D[0..31]
DDR0_A[0..14]	A[0..14]
DDR0_CLK_P	CLK_P
DDR0_CLK_N	CLK_N
DDR0_WEN	WEn
DDR0_CASN	CASn
DDR0_RASN	RASn
DDR0_CKE	CKE
DDR0_CS0N	CS0n
DDR0_CS1N	CS1n
DDR0_RST	RST
DDR0_DQM0	DQM0
DDR0_DQS0_P	DQS0_P
DDR0_DQS0_N	DQS0_N
DDR0_DQM1	DQM1
DDR0_DQS1_P	DQS1_P
DDR0_DQS1_N	DQS1_N
DDR0_DQM2	DQM2
DDR0_DQS2_P	DQS2_P
DDR0_DQS2_N	DQS2_N
DDR0_DQM3	DQM3
DDR0_DQS3_P	DQS3_P
DDR0_DQS3_N	DQS3_N
DDR0_ODT0	ODT0
DDR0_ODT1	ODT1
DDR0_BA0	BA0
DDR0_BA1	BA1
DDR0_BA2	BA2

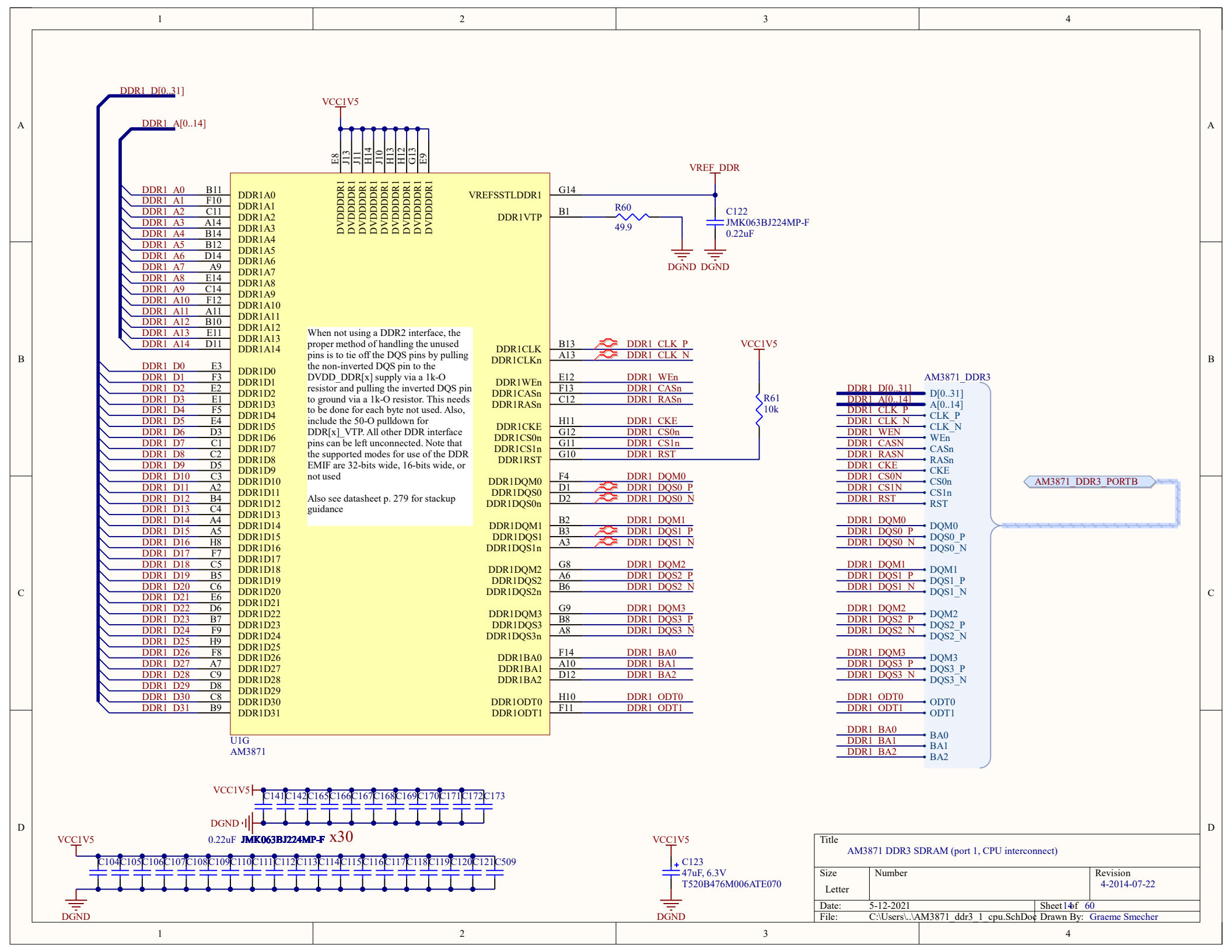
AM3871_DDR3_PORTA

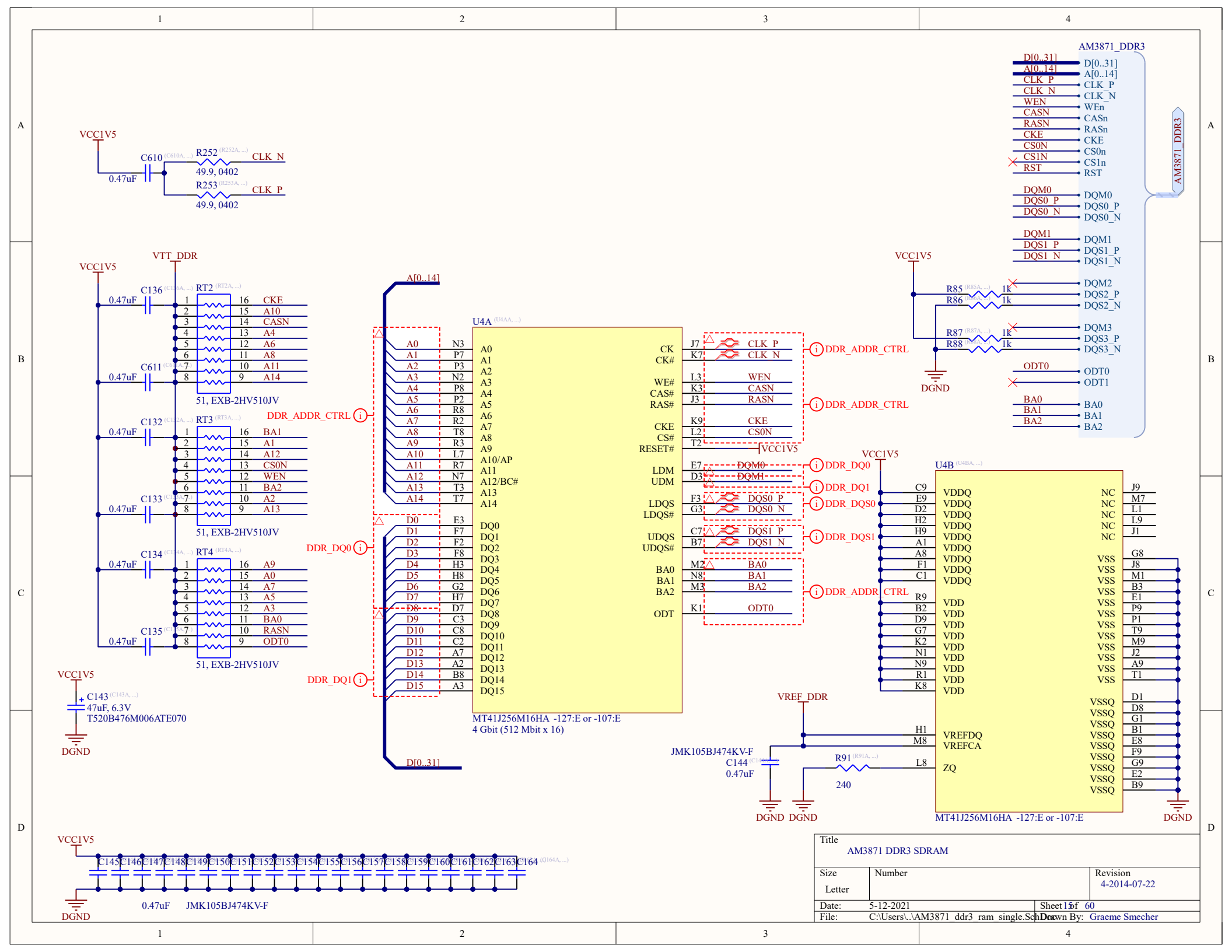
VCC1V5
C174 C175 C176 C580 C581 C582 C583 C584
0.22uF DGND
JMK063BJ224MP-F x30

VCC1V5
C84 C85 C86 C87 C88 C89 C90 C91 C92 C93 C94 C95 C96 C97 C98 C99 C100 C101 C508
DGND

VCC1V5
C103 47uF, 6.3V T520B476M006ATE070
DGND

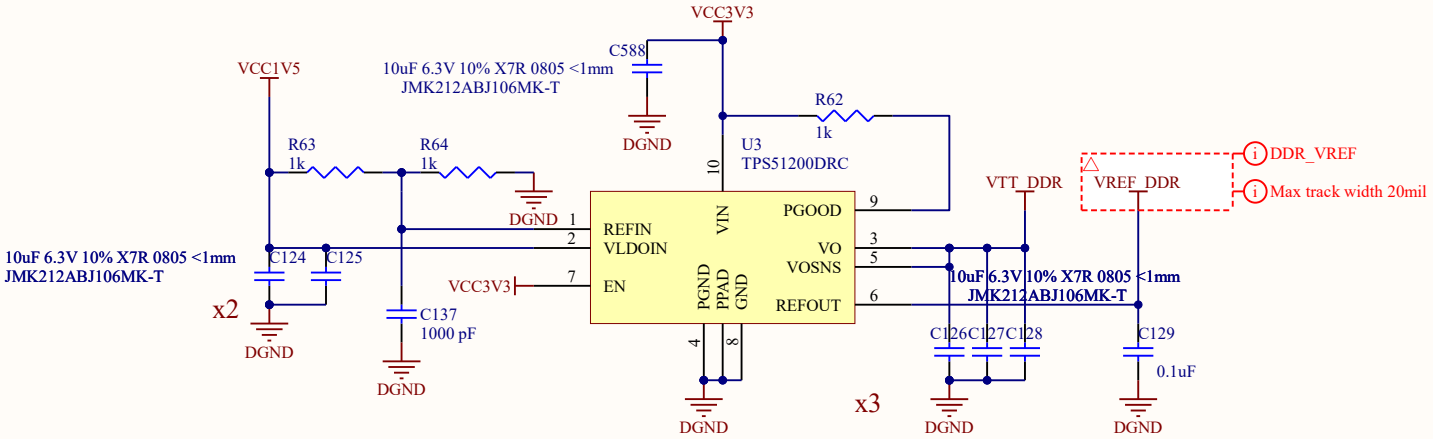
Title	AM3871 DDR3 SDRAM (port 0, CPU interconnect)		
Size	Number		Revision
Letter			4-2014-07-22
Date:	5-12-2021	Sheet 13 of 60	
File:	C:\Users\..\AM3871_ddr3_0_cpu.SchDoc	Drawn By:	Graeme Smecher


DDR1_D[0..31]
DDR1_A[0..14]
VCC1V5
E8 J13 J11 H14 J10 H13 H12 G13 E9
DVDDDDR1
VREF_DDR
VREFSSTLDDR1
DDR1VTP
G14
B1
R60
49.9
C122
JMK063BJ224MP-F
0.22uF
DGND
DDR1A0 DDR1A1 DDR1A2 DDR1A3 DDR1A4 DDR1A5 DDR1A6 DDR1A7 DDR1A8 DDR1A9 DDR1A10 DDR1A11 DDR1A12 DDR1A13 DDR1A14
DDR1D0 DDR1D1 DDR1D2 DDR1D3 DDR1D4 DDR1D5 DDR1D6 DDR1D7 DDR1D8 DDR1D9 DDR1D10 DDR1D11 DDR1D12 DDR1D13 DDR1D14 DDR1D15 DDR1D16 DDR1D17 DDR1D18 DDR1D19 DDR1D20 DDR1D21 DDR1D22 DDR1D23 DDR1D24 DDR1D25 DDR1D26 DDR1D27 DDR1D28 DDR1D29 DDR1D30 DDR1D31
When not using a DDR2 interface, the proper method of handling the unused pins is to tie off the DQS pins by pulling the non-inverted DQS pin to the DVDD_DDR[x] supply via a 1k-O resistor and pulling the inverted DQS pin to ground via a 1k-O resistor. This needs to be done for each byte not used. Also, include the 50-O pulldown for DDR[x]_VTP. All other DDR interface pins can be left unconnected. Note that the supported modes for use of the DDR EMIF are 32-bits wide, 16-bits wide, or not used
Also see datasheet p. 279 for stackup guidance
DDR1CLK DDR1CLKn
DDR1WEn DDR1CASn DDR1RASn
DDR1CKE DDR1CS0n DDR1CS1n DDR1RST
DDR1DQM0 DDR1DQS0 DDR1DQS0n
DDR1DQM1 DDR1DQS1 DDR1DQS1n
DDR1DQM2 DDR1DQS2 DDR1DQS2n
DDR1DQM3 DDR1DQS3 DDR1DQS3n
DDR1BA0 DDR1BA1 DDR1BA2
DDR1ODT0 DDR1ODT1
B13 A13 E12 F13 C12 H11 G12 G11 G10 F4 D1 D2 B2 B3 A3 G8 A6 B6 G9 B8 A8 F14 A10 D12 H10 F11
DDR1_CLK_P DDR1_CLK_N DDR1_WEn DDR1_CASn DDR1_RASn DDR1_CKE DDR1_CS0n DDR1_CS1n DDR1_RST DDR1_DQM0 DDR1_DQS0_P DDR1_DQS0_N DDR1_DQM1 DDR1_DQS1_P DDR1_DQS1_N DDR1_DQM2 DDR1_DQS2_P DDR1_DQS2_N DDR1_DQM3 DDR1_DQS3_P DDR1_DQS3_N DDR1_BA0 DDR1_BA1 DDR1_BA2 DDR1_ODT0 DDR1_ODT1
R61
10k
U1G
AM3871
AM3871_DDR3
DDR1_D[0..31] DDR1_A[0..14] DDR1_CLK_P DDR1_CLK_N DDR1_WEN DDR1_CASN DDR1_RASN DDR1_CKE DDR1_CS0N DDR1_CS1N DDR1_RST DDR1_DQM0 DDR1_DQS0_P DDR1_DQS0_N DDR1_DQM1 DDR1_DQS1_P DDR1_DQS1_N DDR1_DQM2 DDR1_DQS2_P DDR1_DQS2_N DDR1_DQM3 DDR1_DQS3_P DDR1_DQS3_N DDR1_ODT0 DDR1_ODT1 DDR1_BA0 DDR1_BA1 DDR1_BA2
D[0..31] A[0..14] CLK_P CLK_N WEn CASn RASn CKE CS0n CS1n RST DQM0 DQS0_P DQS0_N DQM1 DQS1_P DQS1_N DQM2 DQS2_P DQS2_N DQM3 DQS3_P DQS3_N ODT0 ODT1 BA0 BA1 BA2
AM3871_DDR3_PORTB
C141 C142 C165 C166 C167 C168 C169 C170 C171 C172 C173
0.22uF JMK063BJ224MP-F x30
C104 C105 C106 C107 C108 C109 C110 C111 C112 C113 C114 C115 C116 C117 C118 C119 C120 C121 C509
C123
47uF, 6.3V
T520B476M006ATE070
Title
AM3871 DDR3 SDRAM (port 1, CPU interconnect)
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 14 of 60
File: C:\Users\..\AM3871_ddr3_1_cpu.SchDoc
Drawn By: Graeme Smecher



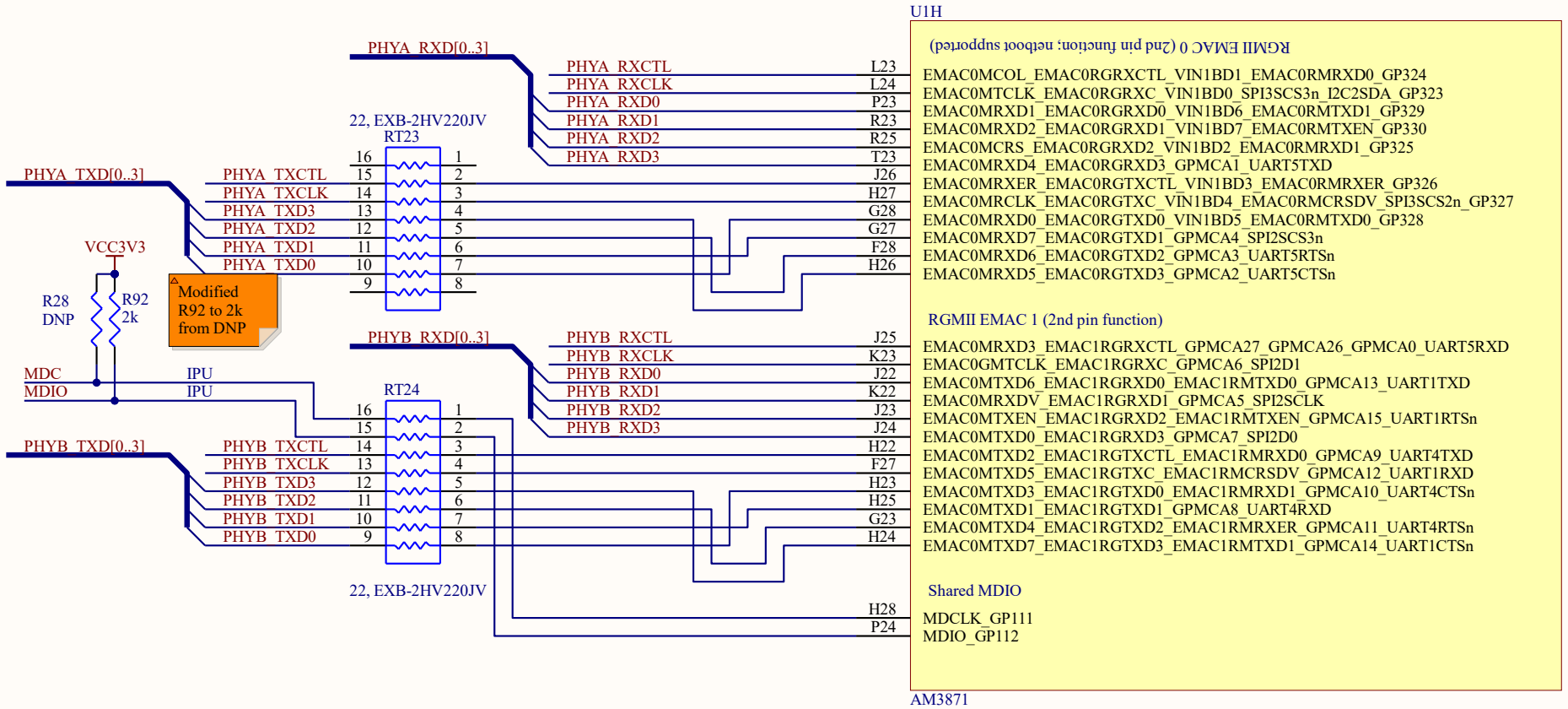
Title	AM3871 DDR3 SDRAM	
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 15 of 60	
File: C:\Users\..\AM3871_ddr3_ram_single.SchDoc	Drawn By: Graeme Smecher	

VTT_DDR supply is 0.75v (1.5v/2)
Current could be 2 Amps

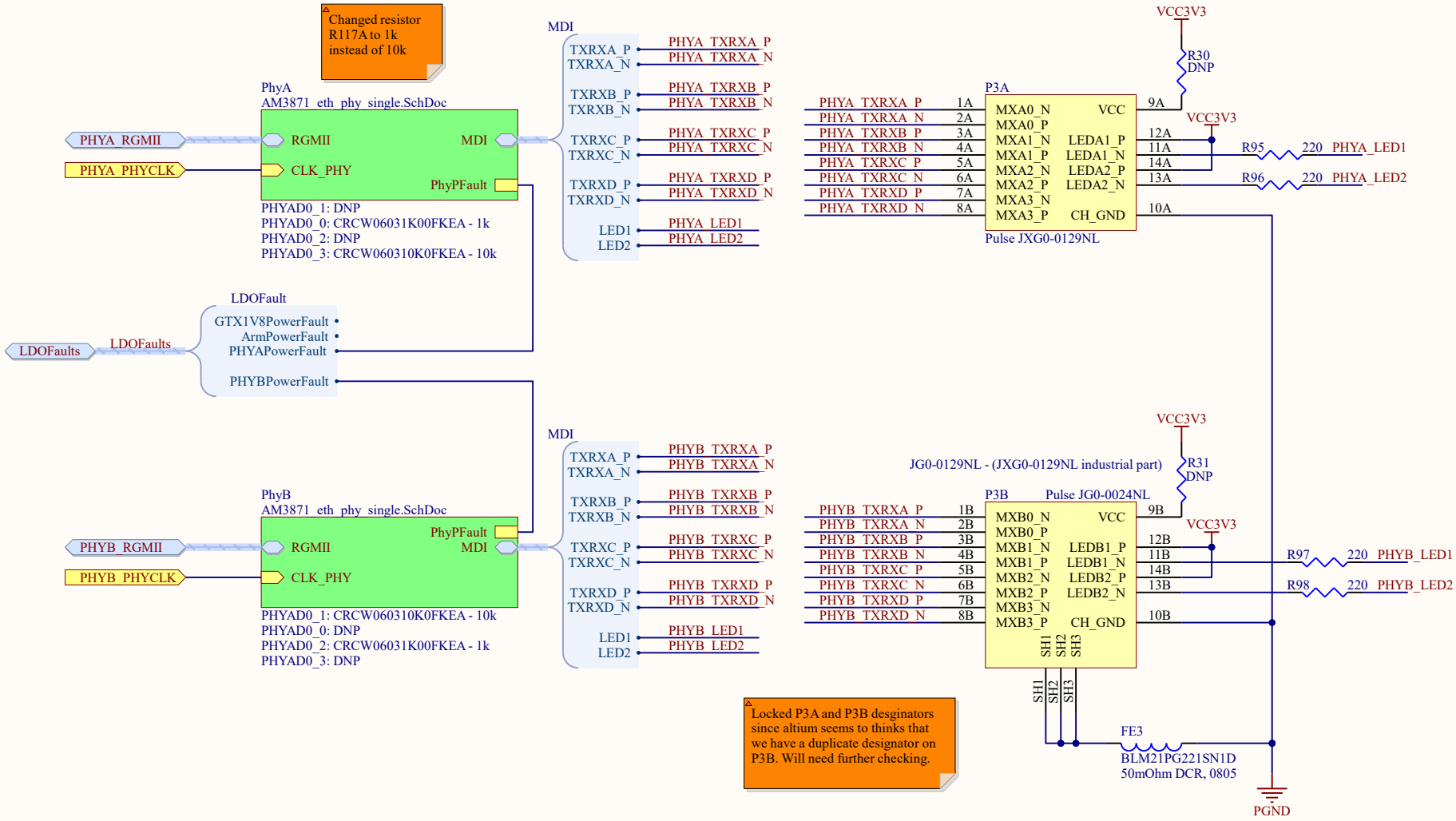


Layout considerations at http://www.ti.com/lit/ds/symlink/tps51200.pdf#page=17

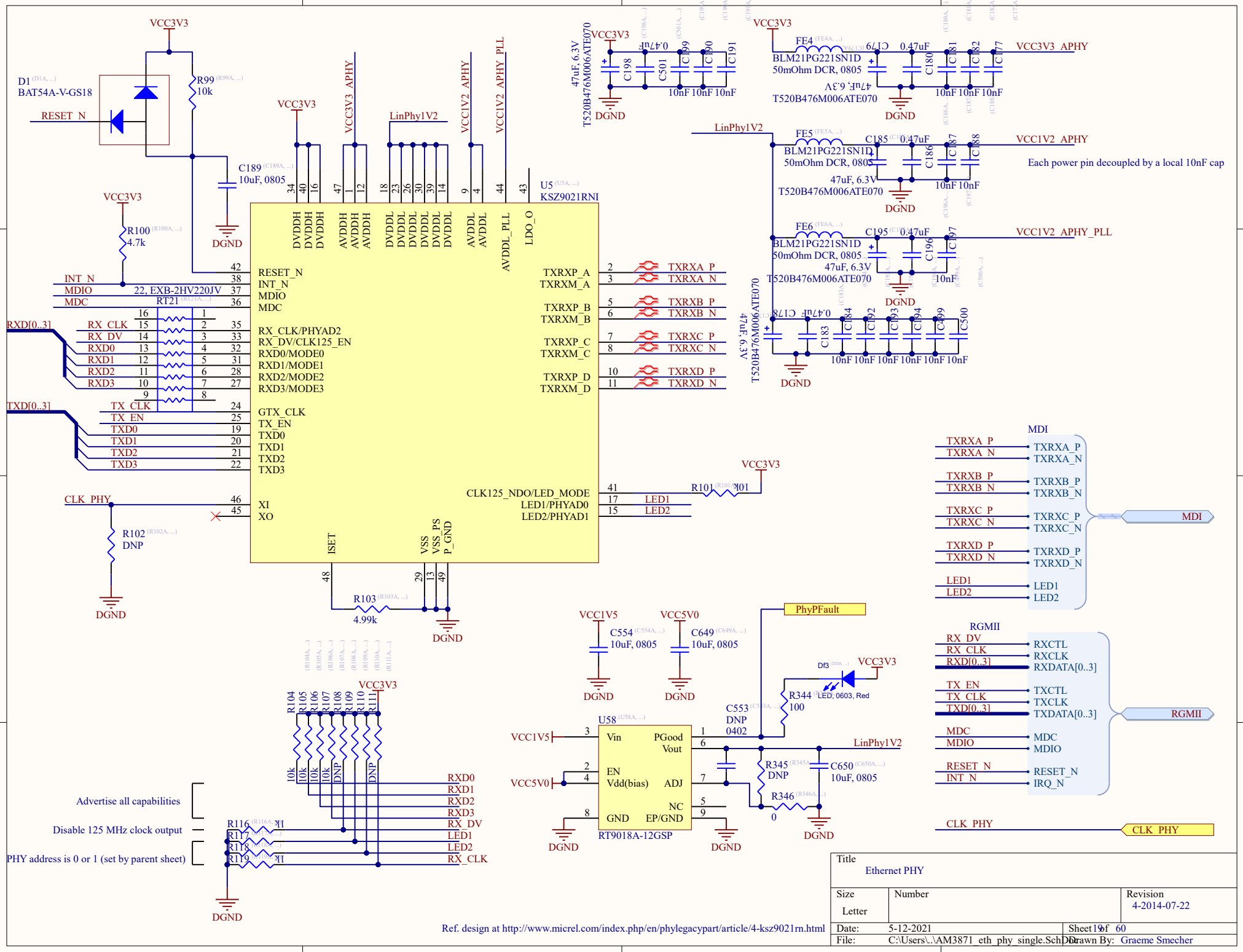
Title: AM3871 DDR3 Termination Regulator		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 16 of 60	
File: C:\Users\..\AM3871_ddr3_regulator.SchDoc	Drawn By: Graeme Smecher	

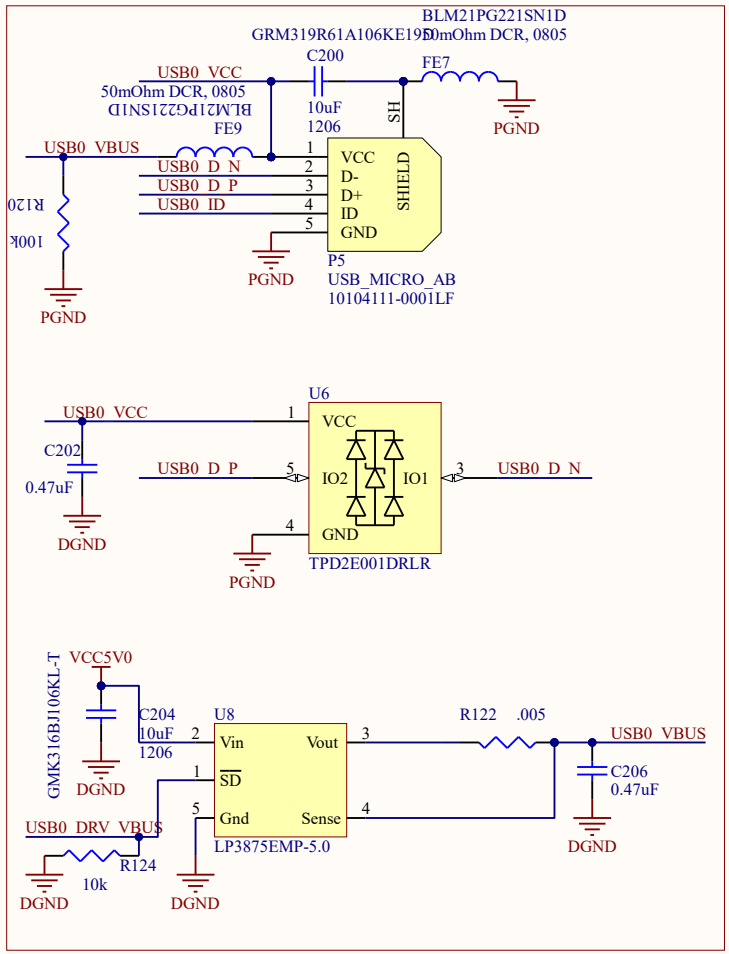


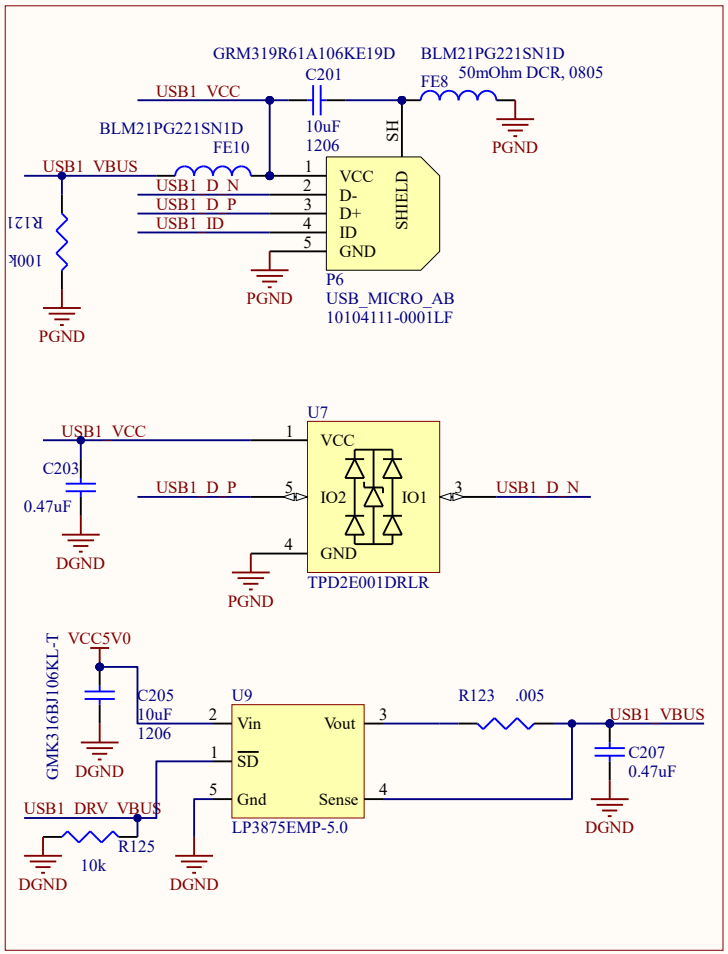
Title: Ethernet PHY		
Size: A	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 17 of 60
File: C:\Users\..\AM3871_eth_cpu.SchDoc		Drawn By: Graeme Smecher

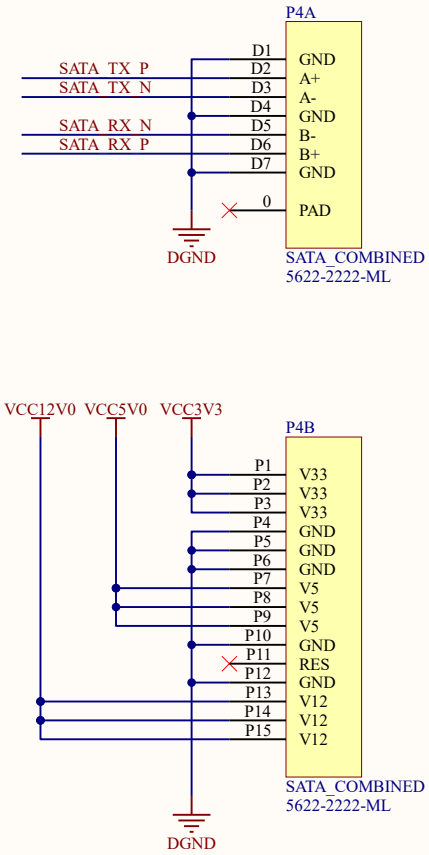


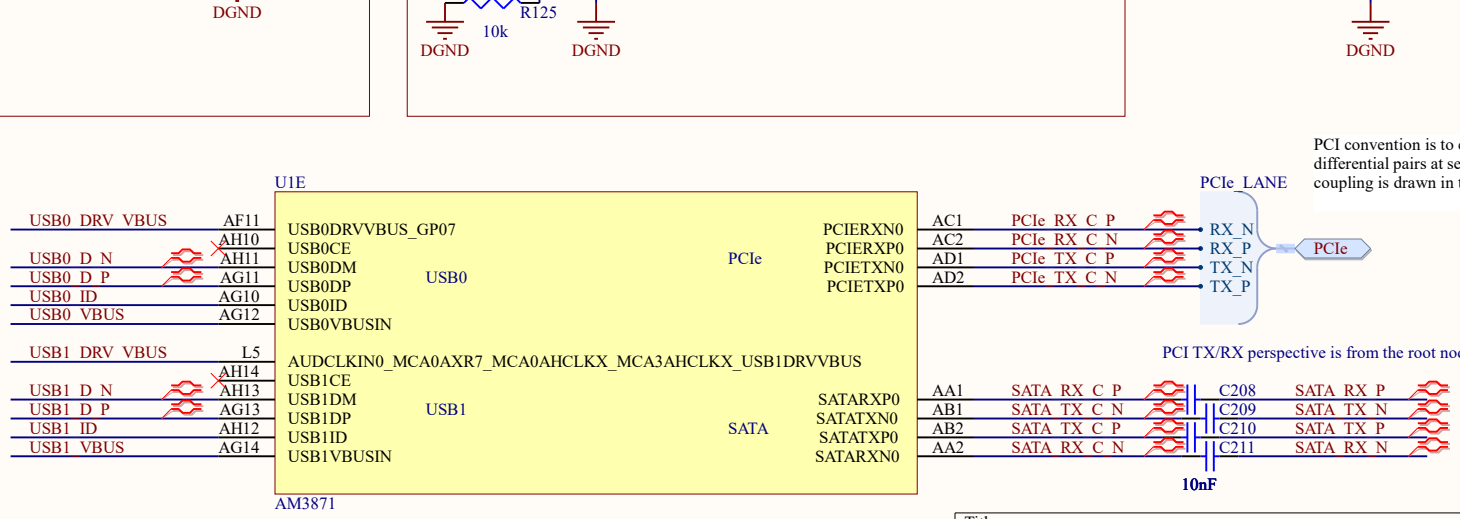
Title: Ethernet PHY		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 18 of 60
File: C:\Users\..\AM3871_eth_phy.SchDoc		Drawn By: Graeme Smecher


VCC3V3
D1 BAT54A-V-GS18
RESET_N
R99 10k
C189 10uF, 0805
DGND
R100 4.7k
INT_N
MDIO
MDC
22, EXB-2HV220JV
RT21
RXD[0..3]
RX_CLK
RX_DV
RXD0
RXD1
RXD2
RXD3
TXD[0..3]
TX_CLK
TX_EN
TXD0
TXD1
TXD2
TXD3
CLK_PHY
R102 DNP
VCC3V3_APHY
VCC1V2_APHY
VCC1V2_APHY_PLL
LinPhy1V2
U5 KSZ9021RNI
DVDDH
AVDDH
DVDDL
AVDDL
AVDDL_PLL
LDO_O
RESET_N
INT_N
MDIO
MDC
RX_CLK/PHYAD2
RX_DV/CLK125_EN
RXD0/MODE0
RXD1/MODE1
RXD2/MODE2
RXD3/MODE3
GTX_CLK
TX_EN
TXD0
TXD1
TXD2
TXD3
XI
XO
ISET
VSS
VSS_PS
P_GND
TXRXP_A
TXRXM_A
TXRXP_B
TXRXM_B
TXRXP_C
TXRXM_C
TXRXP_D
TXRXM_D
TXRXA_P
TXRXA_N
TXRXB_P
TXRXB_N
TXRXC_P
TXRXC_N
TXRXD_P
TXRXD_N
CLK125_NDO/LED_MODE
LED1/PHYAD0
LED2/PHYAD1
LED1
LED2
R101 10k
R103 4.99k
47uF, 6.3V T520B476M006ATE070
0.47uF
10nF
FE4 BLM21PG221SN1D 50mOhm DCR, 0805
FE5 BLM21PG221SN1D 50mOhm DCR, 0805
FE6 BLM21PG221SN1D 50mOhm DCR, 0805
VCC3V3_APHY
VCC1V2_APHY
VCC1V2_APHY_PLL
Each power pin decoupled by a local 10nF cap
MDI
RGMII
RXCTL
RXCLK
RXDATA[0..3]
TXCTL
TXCLK
TXDATA[0..3]
IRQ_N
PhyPFault
VCC1V5
VCC5V0
C554 10uF, 0805
C649 10uF, 0805
C553 DNP 0402
R344 100
LED, 0603, Red
U58 RT9018A-12GSP
Vin
PGood
Vout
EN
Vdd(bias)
ADJ
NC
GND
EP/GND
R345 DNP
C650 10uF, 0805
R346 0
R104 R105 R106 R107 R108 R109 R110 R111
10k
DNP
Advertise all capabilities
Disable 125 MHz clock output
PHY address is 0 or 1 (set by parent sheet)
R116 1k
R117
R118
R119 1k
Ref. design at http://www.micrel.com/index.php/en/phylegacypart/article/4-ksz9021rn.html
Title
Ethernet PHY
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 19 of 60
File: C:\Users\..\AM3871_eth_phy_single.SchDoc
Drawn By: Graeme Smecher







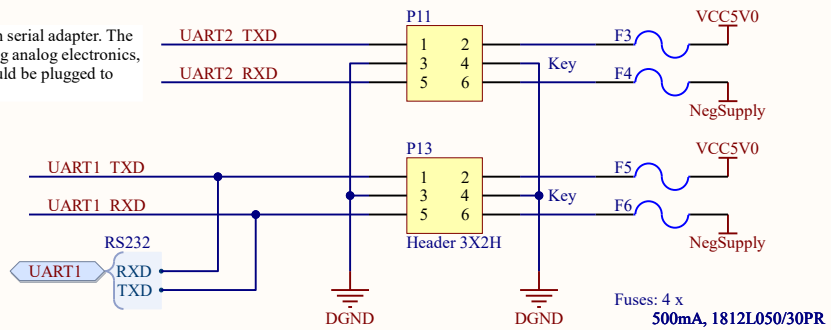


Title	AM3871 Fast Serial Interfaces	
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 20 of 60	
File: C:\Users\..\AM3871_fast_serial.SchDoc	Drawn By: Graeme Smecher	

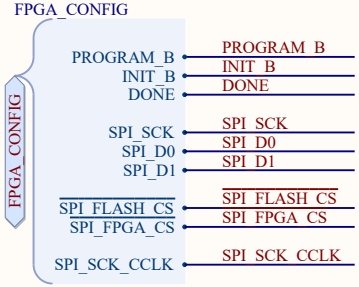
RS-232 Headers

These ports allow a standard, 3-pin serial adapter. The even pins are reserved for powering analog electronics, e.g. in an active dongle. Pin 4 should be plugged to prevent accidents.

UART1 also passes through to the FPGA. Note that TXD/RXD are reversed here.



FPGA_CONFIG

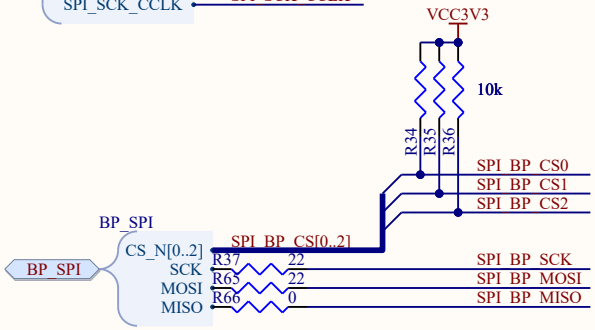


FPGA configuration status pins

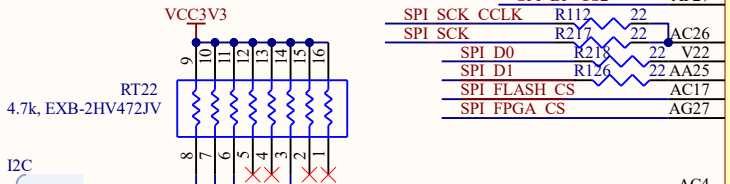
Programming (SS + SPI flash) and data bus

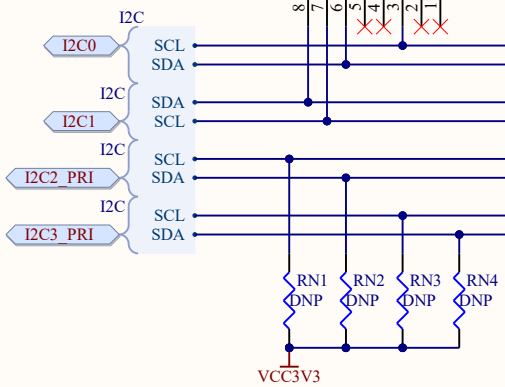
Roles cannot be separated since the FPGA pins are combined

SPI chip-selects



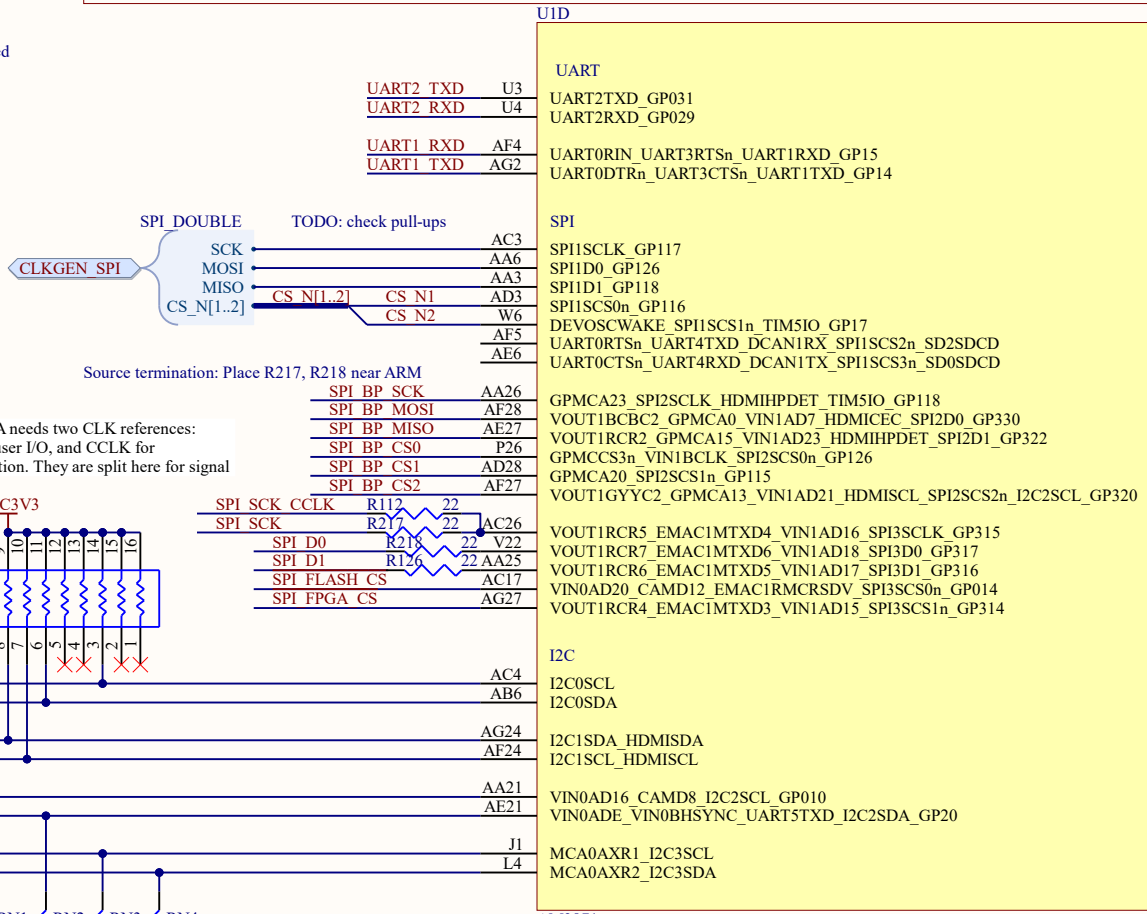
The FPGA needs two CLK references: SCK for user I/O, and CCLK for configuration. They are split here for signal





I2C2 and I2C3 have two pin-mux possibilities at the ARM: they can connect to neighbouring backplane slots (primary), or the I2C interconnect matrix (secondary).

U1D



AM3871

Title: AM3871 Slow Serial Interfaces		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 2 of 60
File: C:\Users\..\AM3871_slow_serial.SchDoc		Drawn By: Graeme Smecher

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.

FMC_AUX

FMC_AUX_A
- EN_12V0
- EN_3V3
- EN_VADJ
- TMS — FMCA_TMS
- TCK — FMCA_TCK
- TDO — FMCA_TDO
- TDI — FMCA_TDI
- TRST — FMCA_TRST
- PG_M2C
- PG_C2M
- PRSNT_M2C_L — Unused pins: see I2C GPIO
- CLK_DIR

FMC_AUX

FMC_AUX_B
- EN_12V0
- EN_3V3
- EN_VADJ
- TMS — FMCB_TMS
- TCK — FMCB_TCK
- TDO — FMCB_TDO
- TDI — FMCB_TDI
- TRST — FMCB_TRST
- PG_M2C
- PG_C2M
- PRSNT_M2C_L — Unused pins: see I2C GPIO
- CLK_DIR

JTAG

FPGA_JTAG
- TMS — FPGA_TMS
- TCK — FPGA_TCK
- TDO — FPGA_TDO
- TDI — FPGA_TDI
- TRST

FPGA_CONFIG

FPGA_CONFIG
- PROGRAM_B — PROGRAM_B
- INIT_B — INIT_B
- DONE — DONE
- SPI_SCK — (See ARM slow serial sheet)
- SPI_D0
- SPI_D1
- SPI_FLASH_CS — (See ARM slow serial sheet)
- SPI_FPGA_CS
- SPI_SCK_CCLK

VCC3V3

R27 4.7k, R29 4.7k

I2C: SCL, SDA — I2C3_SEC

I2C3's primary connection is to the backplane neighbour. Alternately, it connects to the I2C interconnect matrix.

FPGA Configuration: 1 PROGRAM_B, 2 INIT_B, 3 DONE

FMCA: 8 FMCA_TMS, 9 FMCA_TCK, 10 FMCA_TDO, 11 FMCA_TDI

GP014: SPI3 bit-banging

15 FMCA_TRST

FMCB: 16 FMCB_TMS, 17 FMCB_TCK, 18 FMCB_TDO, 19 FMCB_TDI, 20 FMCB_TRST

FPGA JTAG: 24 FPGA_TMS, 25 FPGA_TCK, 26 FPGA_TDO, 27 FPGA_TDI

FPGA_TMS	R73	22	J4
FPGA_TCK	R244	22	AA22
FPGA_TDO	R245	22	H5
FPGA_TDI	R246	22	AD23

C615 FPGA_TMS, C616 FPGA_TCK, C617 FPGA_TDO, C618 FPGA_TDI — DNP — DGND

CC0402KRX7R9BB221

U1I

AM3871

Y6	SD0CLK_GP01
N1	SD0CMD_SD1CMD_GP02
R7	SD0DAT0_SD1DAT4_GP03
Y5	SD0DAT1SDIRQn_SD1DAT5_GP04
Y3	SD0DAT2SDRWn_SD1DAT6_GP05
Y4	SD0DAT3_SD1DAT7_GP06
R5	AUDCLKIN1_MCA0AXR8_MCA1AHCLKX_MCA4AHCLKX_EDMAEVT3_TIM2IO_GP08
H1	AUDCLKIN2_MCA0AXR9_MCA2AHCLKX_MCA5AHCLKX_EDMAEVT2_TIM3IO_GP09
U6	MCA2ACLKX_GP010
AA5	MCA2AFSX_GP011
AB21	VIN0AD17_CAMD9_EMAC1RMRXER_GP011
AF20	VIN0AD18_CAMD10_EMAC1RMRXD1_I2C3SCL_GP012
N2	MCA2AXR0_SD0DAT6_UART5RXD_GP012
AF21	VIN0AD19_CAMD11_EMAC1RMRXD0_I2C3SDA_GP013
V6	MCA2AXR1_SD0DAT7_UART5TXD_GP013
V5	MCA2AXR2_MCA1AXR6_TIM2IO_GP014
H2	MCA2AXR3_MCA1AXR7_TIM3IO_GP015
AE18	VIN0AD21_CAMD13_EMAC1RMTXD0_SPI3SCLK_GP015
G6	MCA3ACLKX_GP016
AC21	VIN0AD22_CAMD14_EMAC1RMTXD1_SPI3D1_GP016
AC16	VIN0AD23_CAMD15_EMAC1RMTXEN_SPI3D0_GP017
H4	MCA3AFSX_GP017
AB17	VIN0ADE_CAMD7_GP018
G1	MCA3AXR0_TIM4IO_GP018
G2	MCA3AXR1_TIM5IO_GP019
AC15	VIN0BDE_CAMD6_GP019
AC22	VIN0AFLD_CAMD5_GP020
F2	MCA3AXR2_MCA1AXR8_GP020
AD17	VIN0BFLD_CAMD4_GP021
K7	MCA4ACLKX_GP021
AD18	VOUT1GYYC1_CAMD3_GPMCA5_UART4RXD_GP022
H3	MCA4AFSX_GP022
AC18	VOUT1GYYC0_CAMD2_GPMCA6_UART4TXD_GP023
H6	MCA4AXR0_GP023
AC19	MCA4AXR1_TIM6IO_GP024
J3	VOUT1RCR1_CAMD1_GPMCA7_UART4CTSn_GP024
AE23	VOUT1RCR0_CAMD0_GPMCA8_UART4RTSn_GP025
L7	MCA5ACLKX_GP025
L6	MCA5AFSX_GP026
AB23	VOUT1BCBC1_CAMHS_GPMCA9_UART2RXD_GP026
T2	VOUT1BCBC0_CAMVS_GPMCA10_UART2TXD_GP027
	MCA5AXR0_MCA4AXR2_GP027
	MCA5AXR1_MCA4AXR3_TIM7IO_GP028
	VOUT1FLD_CAMFLD_CAMWEn_GPMCA11_UART2CTSn_GP028
	TCLKIN_GP030

GPIO bank 0

Title	AM3871 GPIO		
Size	Letter	Number	Revision 4-2014-07-22
Date:	5-12-2021	Sheet 22 of 60	
File:	C:\Users\..\AM3871_gpio_0.SchDoc	Drawn By: Graeme Smecher	

RGMII

PHYA_RGMII
- RXCTL
- RXCLK
- RXDATA[0..3]
- TXCTL
- TXCLK
- TXDATA[0..3]
- MDC
- MDIO
- RESET_N
- IRQ_N

(See ARM Ethernet sheet)
(See ARM Ethernet sheet)
(See ARM Ethernet sheet)
PHYA_RESET_N
PHYA_IRQ_N

RGMII

PHYB_RGMII
- RXCTL
- RXCLK
- RXDATA[0..3]
- TXCTL
- TXCLK
- TXDATA[0..3]
- MDC
- MDIO
- RESET_N
- IRQ_N

(See ARM Ethernet sheet)
(See ARM Ethernet sheet)
(See ARM Ethernet sheet)
PHYB_RESET_N
PHYB_IRQ_N

Share PHY IRQs by default

PHYA_IRQ_N R130 0
PHYB_IRQ_N

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.

U1J

PHY IRQ/Reset

PHYA_IRQ_N	AH6	DCAN0TX_UART2TXD_I2C3SDA_GP10
PHYA_RESET_N	AG6	DCAN0RX_UART2RXD_I2C3SCL_GP11
PHYB_IRQ_N	W1	GP17
PHYB_RESET_N	W2	GP18
	P25	GPMCCS4n_SD2CMD_GP18
ARM_IRQ	AE17	VIN0BCLK_CLKOUT0_GP19
	V1	GP19
GPIO_IRQ	V2	GP110
	J27	EMACRMREFCLK_TIM2IO_GP110
GPIO_RST	AF9	VIN0AD0_GP111
	AB11	VIN0AD1_GP112
	AE28	GPMCA18_TIM2IO_GP113
	M24	SD2DAT1SDIRQn_GPMCA3_GP113
	L26	SD2DAT0_GPMCA4_GP114
	AC27	GPMCA19_TIM3IO_GP114
	M23	SD2SCLK_GP115
	AC28	GPMCA21_SPI2D0_GP116
	AB27	GPMCA22_SPI2D1_HDMICEC_TIM4IO_GP117
	L25	SD2DAT7_GPMCA24_GPMCA20_UART2RXD_GP119
	N23	SD2DAT6_GPMCA25_GPMCA21_UART2TXD_GP120
	P22	SD2DAT5_GPMCA26_GPMCA22_TIM6IO_GP121
	R24	SD2DAT4_GPMCA27_GPMCA23_GPMCCS7n_EDMAEVT0_TIM7IO_GP122
	T28	GPMCCS0n_GP123
	K28	GPMCCS1n_GPMCA25_GP124
	M25	GPMCCS2n_GPMCA24_GP125
	R26	GPMCCLK_GPMCCS5n_GPMCWAIT1_CLKOUT1_EDMAEVT3_TIM4IO_GP127
	M26	GPMCADVnALE_GPMCCS6n_TIM5IO_GP128
	U27	GPMCBE0nCLE_GPMCA25_EDMAEVT2_TIM6IO_GP129
	V28	GPMCBE1n_GPMCA24_EDMAEVT1_TIM7IO_GP130
	W28	GPMCWAIT0_GPMCA26_EDMAEVT0_GP131

GP115 is reserved for manual SPI_BP_CS1

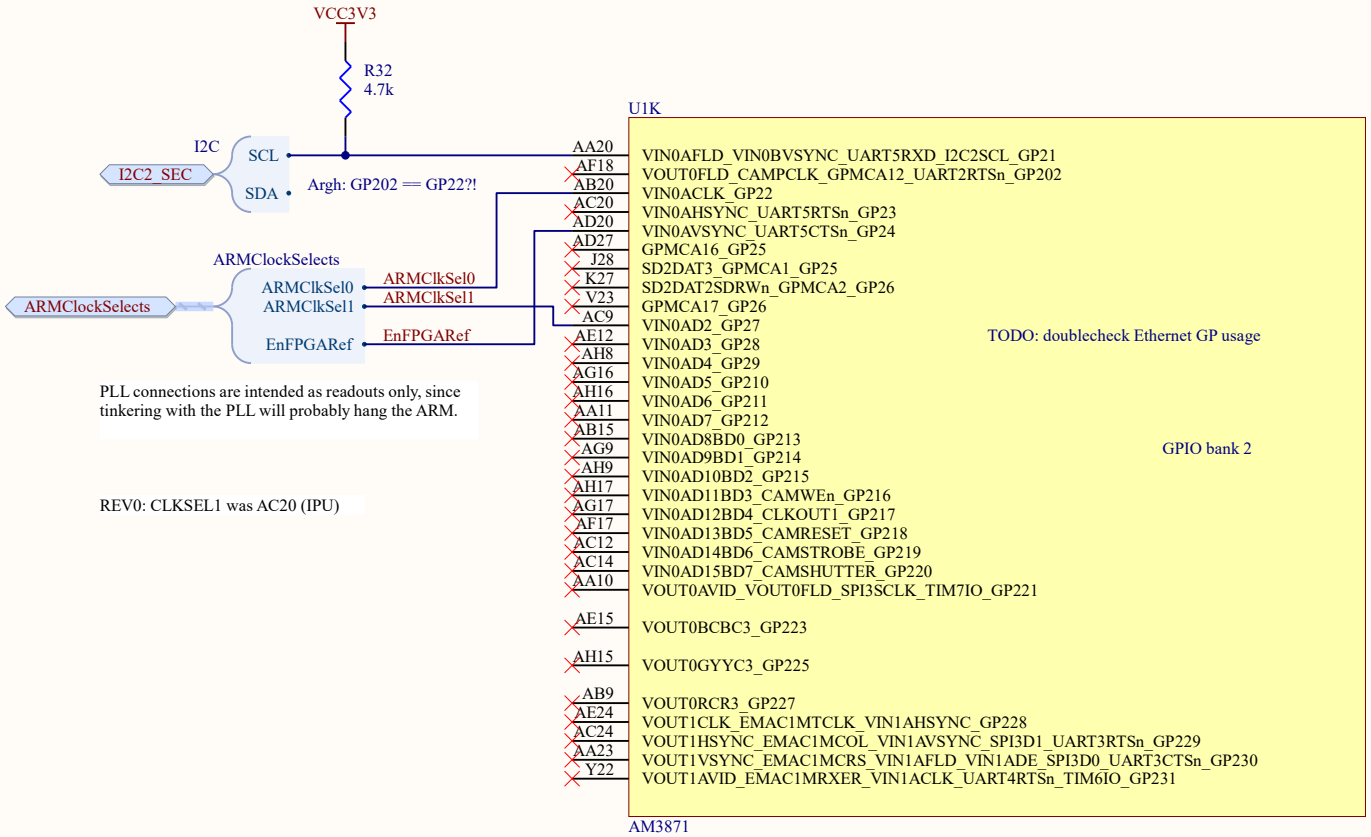
GPIO bank 1

AM3871

Title: AM3871 GPIO		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 23 of 60	
File: C:\Users\..\AM3871_gpio_1.SchDoc	Drawn By: Graeme Smecher	

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.



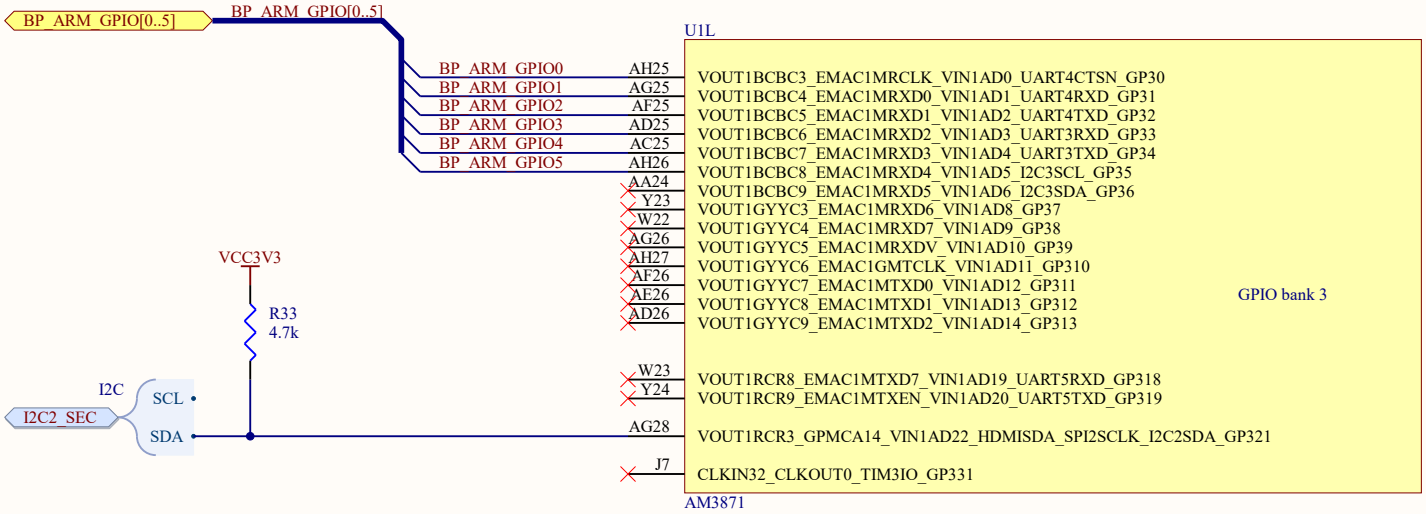
PLL connections are intended as readouts only, since tinkering with the PLL will probably hang the ARM.

REV0: CLKSEL1 was AC20 (IPU)

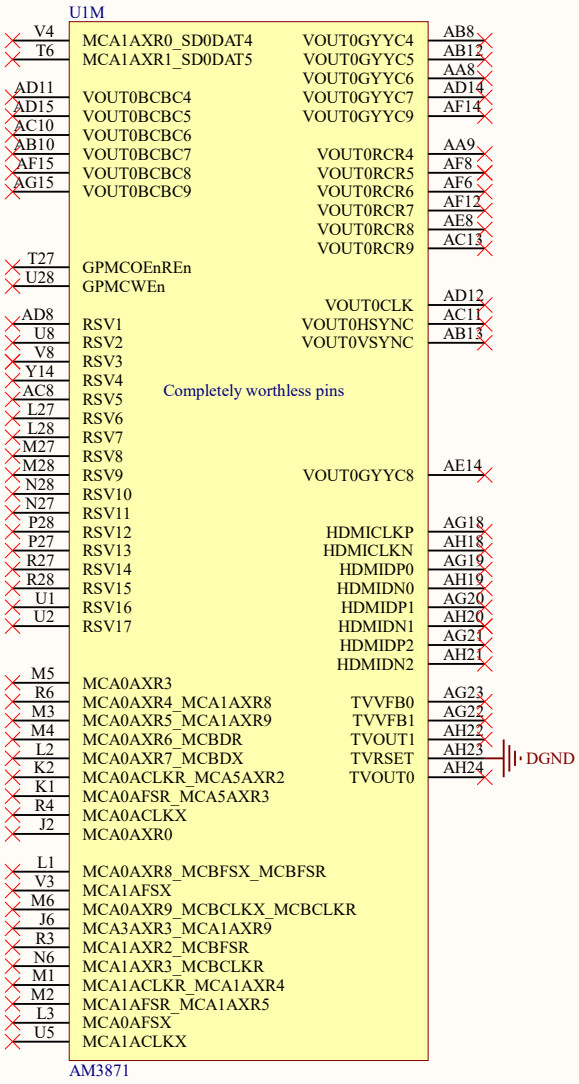
Title: AM3871 GPIO		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 24 of 60
File: C:\Users\..\AM3871_gpio_2.SchDoc		Drawn By: Graeme Smecher

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

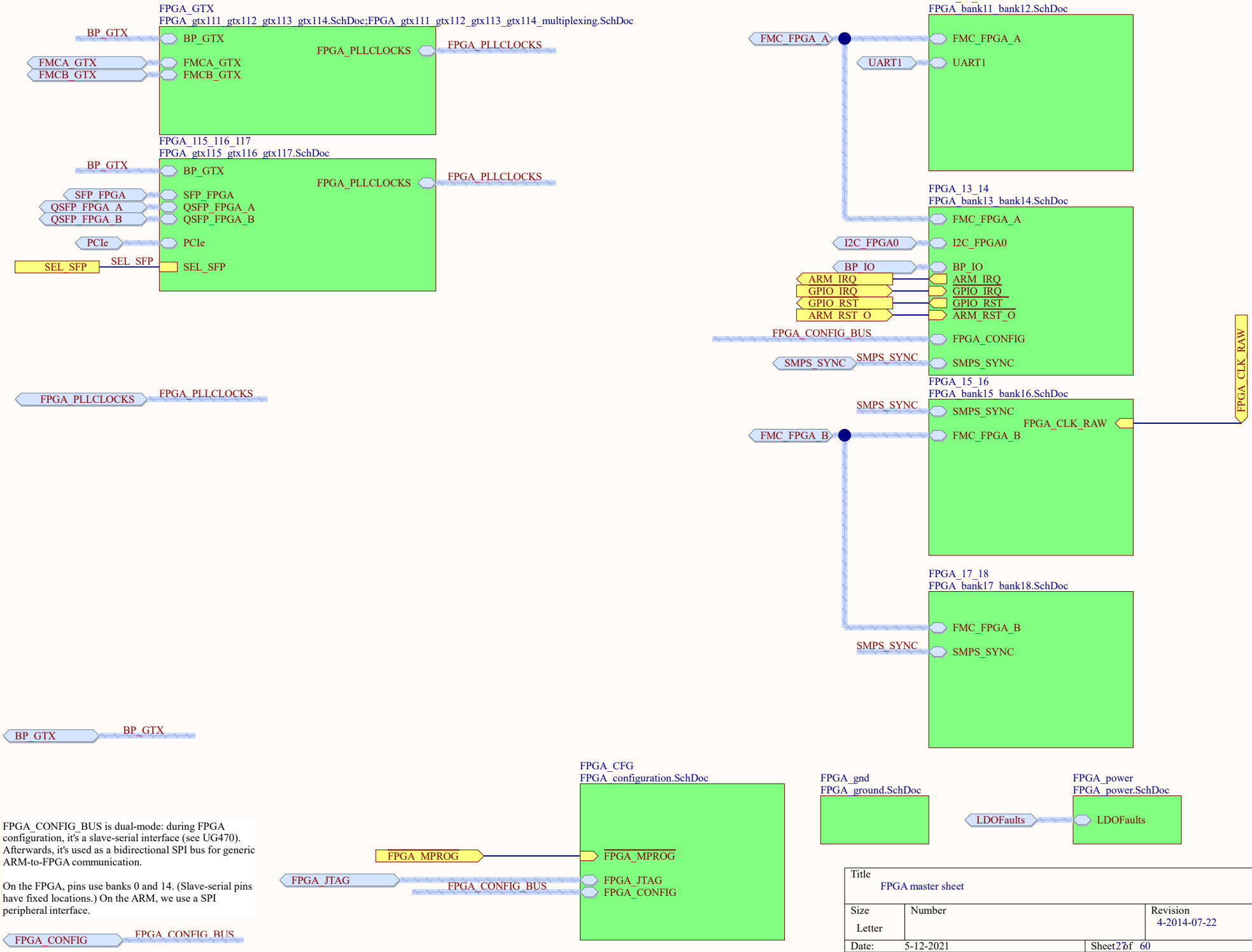
LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.



Title: AM3871 GPIO		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 25 of 60	
File: C:\Users\..\AM3871_gpio_3.SchDoc	Drawn By: Graeme Smecher	



Title: AM3871 Useless Pins		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 26 of 60	
File: C:\Users\..\AM3871_unused.SchDoc	Drawn By: Graeme Smecher	


FPGA_GTX
FPGA_gtx111_gtx112_gtx113_gtx114.SchDoc;FPGA_gtx111_gtx112_gtx113_gtx114_multiplexing.SchDoc
BP_GTX
FMCA_GTX
FMCB_GTX
FPGA_PLLCLOCKS
FPGA_115_116_117
FPGA_gtx115_gtx116_gtx117.SchDoc
SFP_FPGA
QSFP_FPGA_A
QSFP_FPGA_B
PCIe
SEL_SFP
FPGA_11_12
FPGA_bank11_bank12.SchDoc
FMC_FPGA_A
UART1
FPGA_13_14
FPGA_bank13_bank14.SchDoc
I2C_FPGA0
BP_IO
ARM_IRQ
GPIO_IRQ
GPIO_RST
ARM_RST_O
FPGA_CONFIG_BUS
FPGA_CONFIG
SMPS_SYNC
FPGA_15_16
FPGA_bank15_bank16.SchDoc
FPGA_CLK_RAW
FMC_FPGA_B
FPGA_17_18
FPGA_bank17_bank18.SchDoc
FPGA_CFG
FPGA_configuration.SchDoc
FPGA_MPROG
FPGA_JTAG
FPGA_gnd
FPGA_ground.SchDoc
FPGA_power
FPGA_power.SchDoc
LDOFaults
FPGA_CONFIG_BUS is dual-mode: during FPGA configuration, it's a slave-serial interface (see UG470). Afterwards, it's used as a bidirectional SPI bus for generic ARM-to-FPGA communication.
On the FPGA, pins use banks 0 and 14. (Slave-serial pins have fixed locations.) On the ARM, we use a SPI peripheral interface.
Title
FPGA master sheet
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 2 of 60
File: C:\Users\..\FPGA_master.SchDoc
Drawn By: Graeme Smecher

The configuration bus is muddled amongst three regimes:
* FPGA is configured from the SPI flash
* FPGA is configured from the ARM, via slave serial (SS)
* ARM communicates with the FPGA, using standard SPI

The SPI flash is a fallback we don't intend to use. Note that it's effectively off-limits after configuration, unless we figure out multi-mastering (to keep the ARM and FPGA from competing for the bus).

U27B XC7K420T-2FFG901I

JTAG

FPGA_CONFIG

FPGA Reset

Triggers an FPGA reset. In SS mode, the ARM needs to notice that PROGRAM_B has fallen and initiate its own programming procedure.

Cap not on Rev 3

Configuration Mode Select

M[2:0]=000: Master serial
M[2:0]=001: Master SPI
M[2:0]=010: Master BPI
M[2:0]=011:
M[2:0]=100: Master SelectMAP
M[2:0]=101: JTAG
M[2:0]=110: Slave SelectMAP
M[2:0]=111: Slave Serial

JTAG

For TRST_N, to match FMC JTAG

CC0402KRX7R9BB221

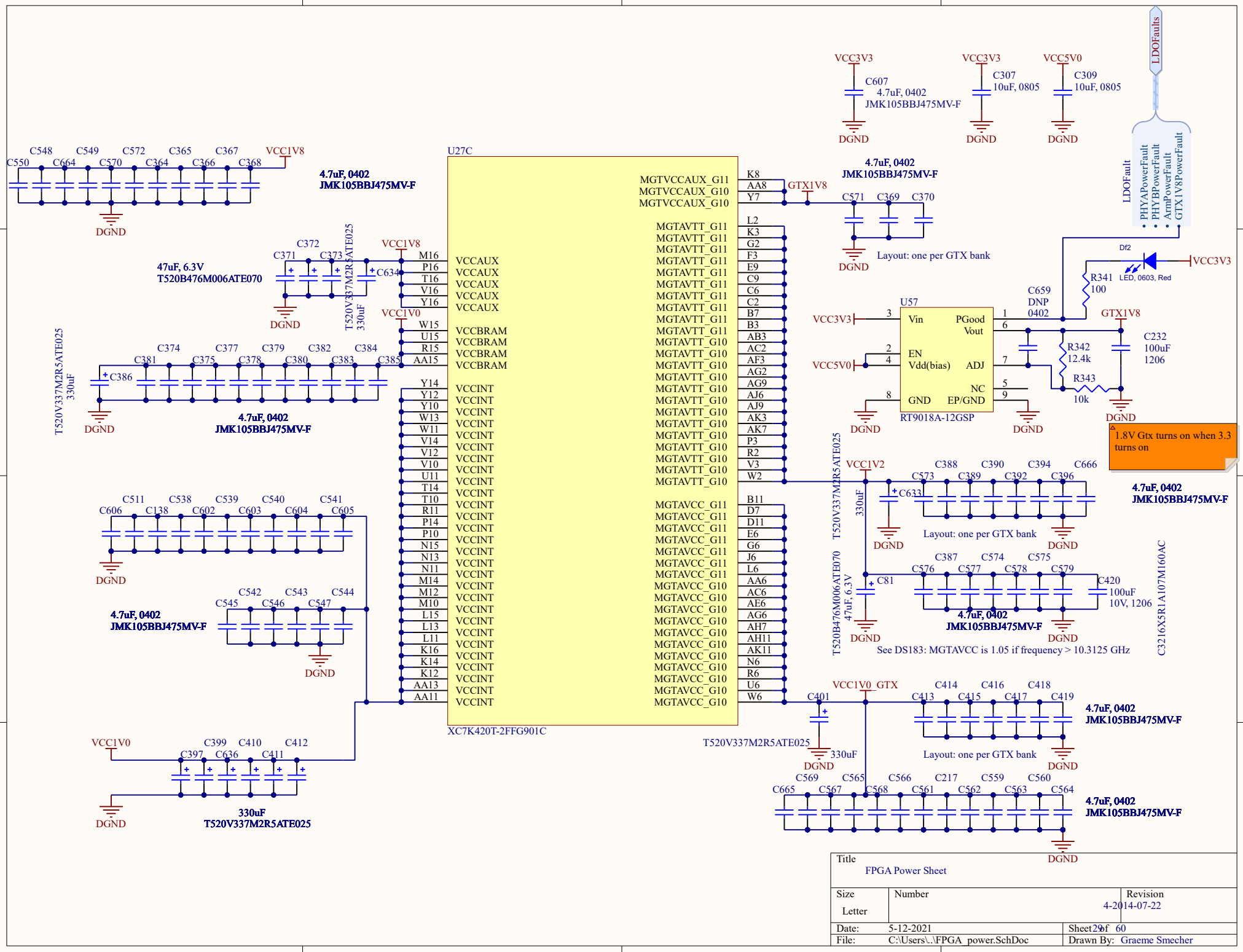
(Optional) SPI bitstream flash

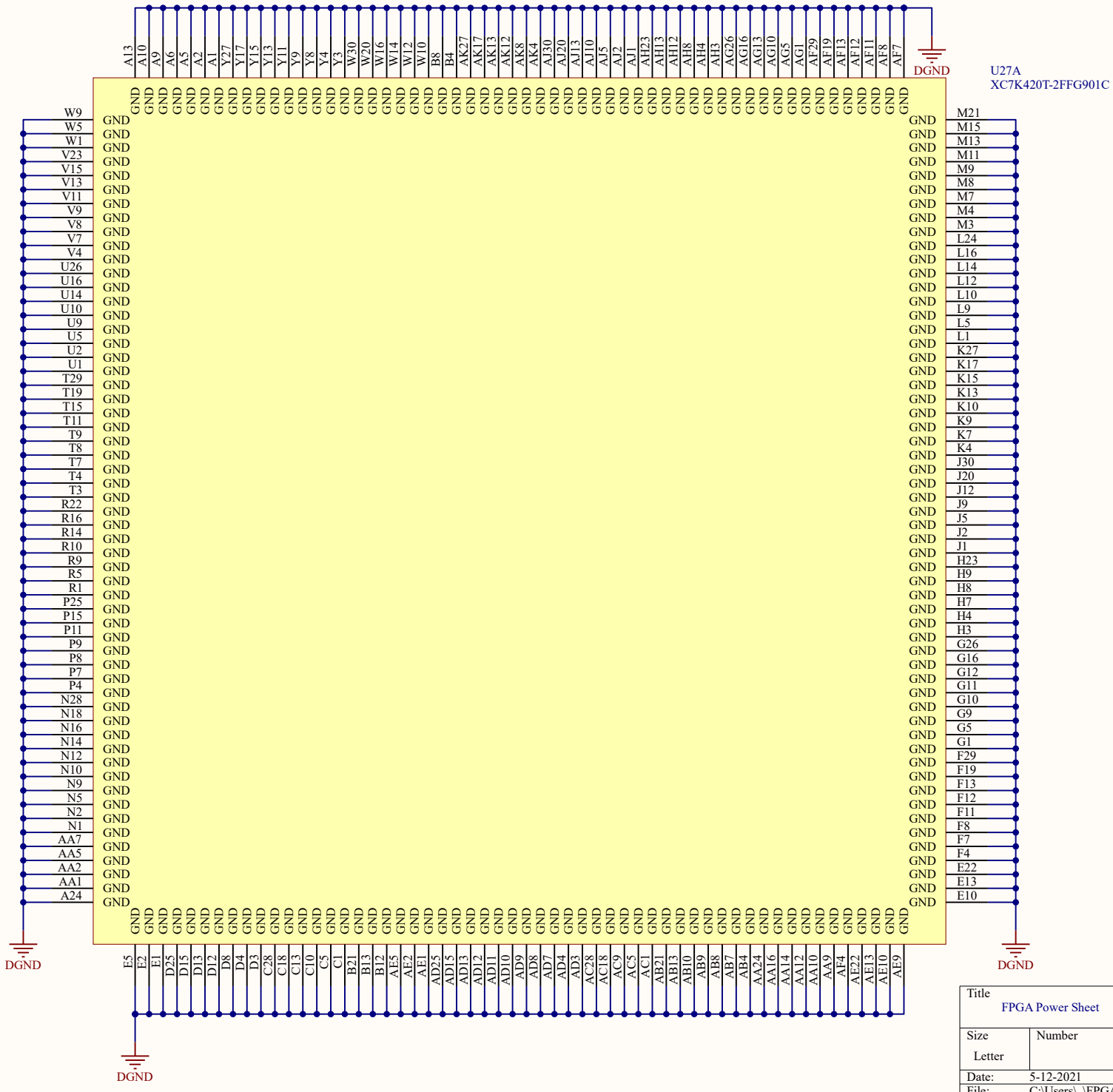
10k pull-up baked into TXS0102

SPI flash: S25FL256SAGMFI011

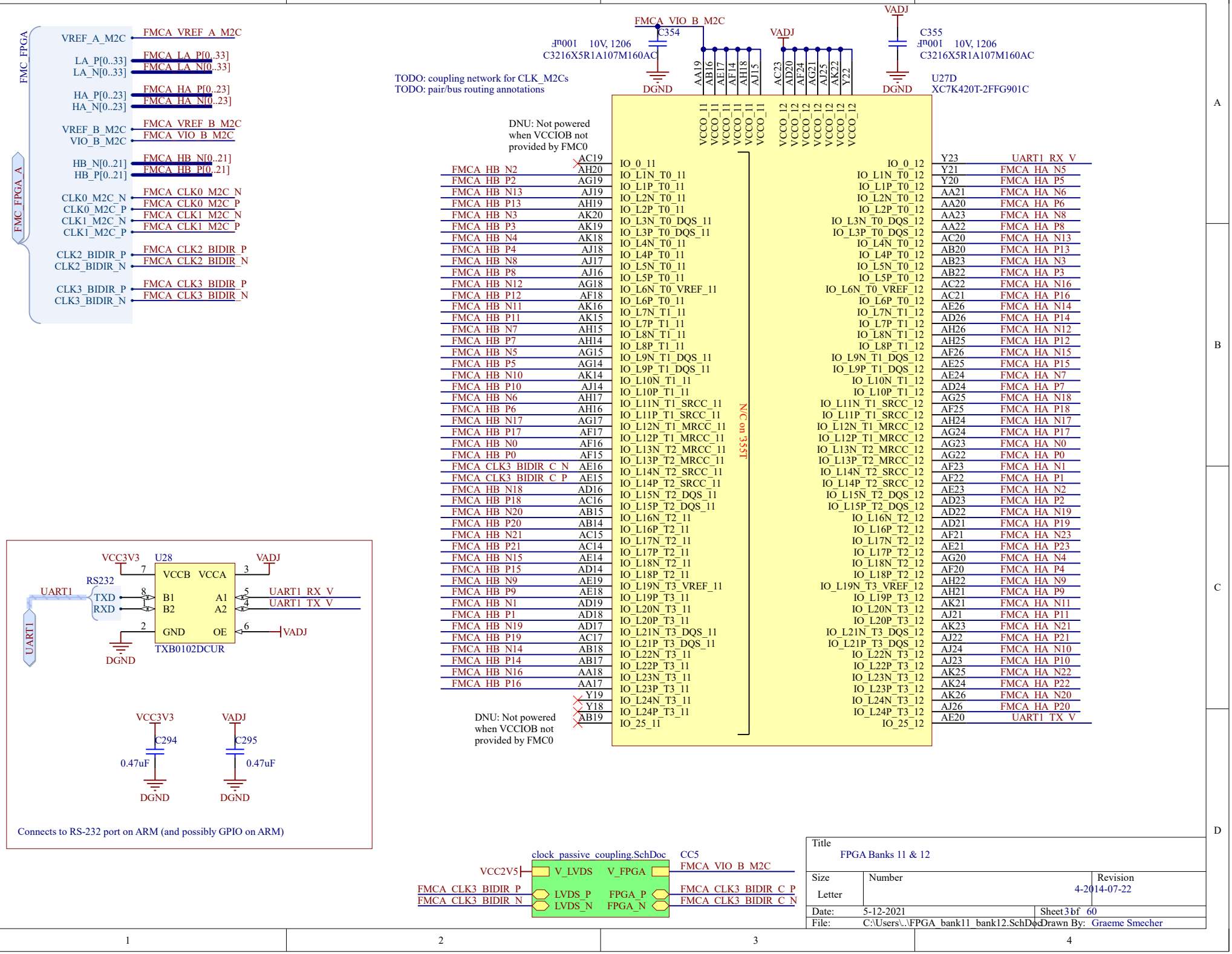
Test points and J1 allow flash programming, even while board isn't powered.

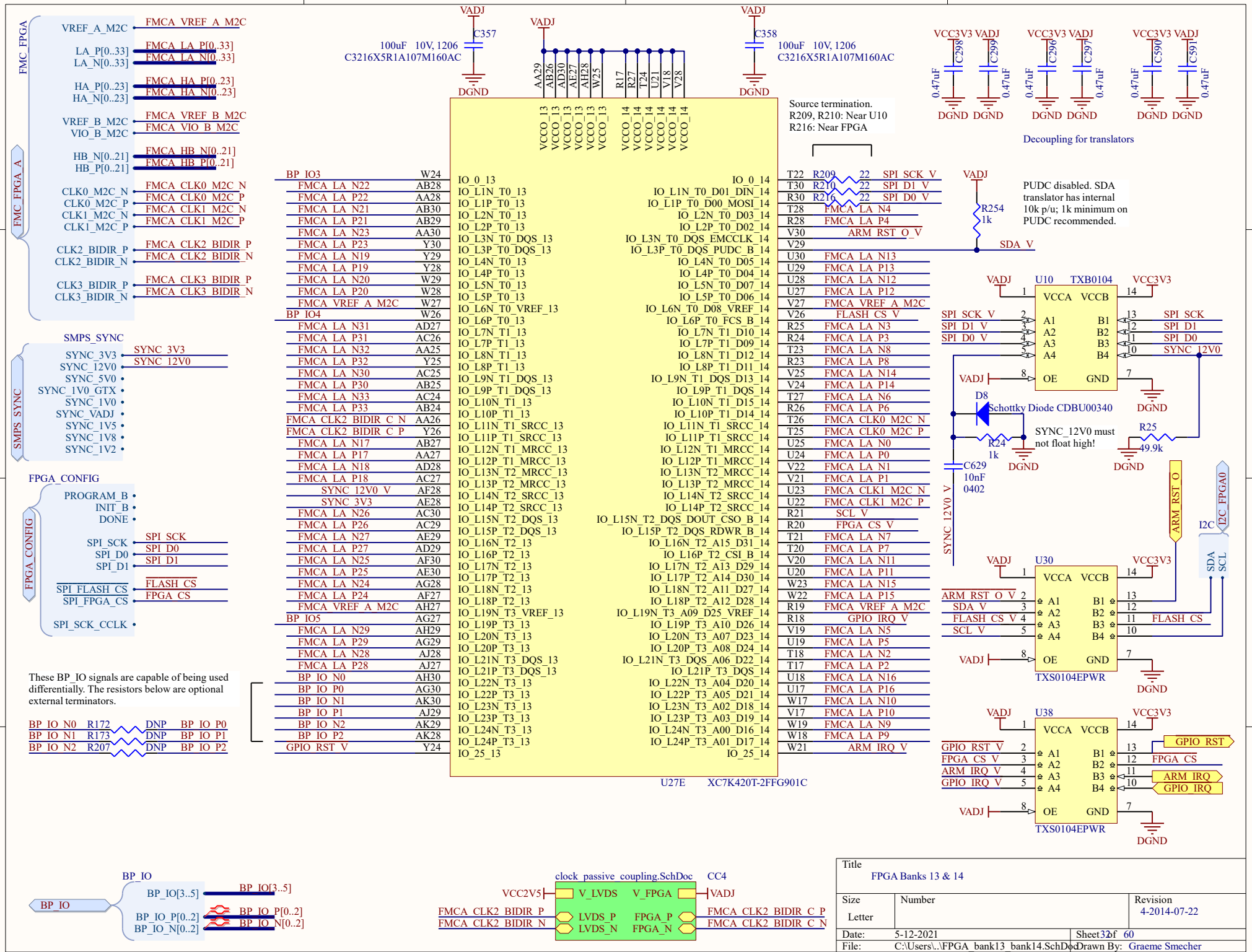
Title: FPGA configuration sheet		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 28 of 60	
File: C:\Users\..\FPGA_configuration.SchDoc	Drawn By: Graeme Smecher	

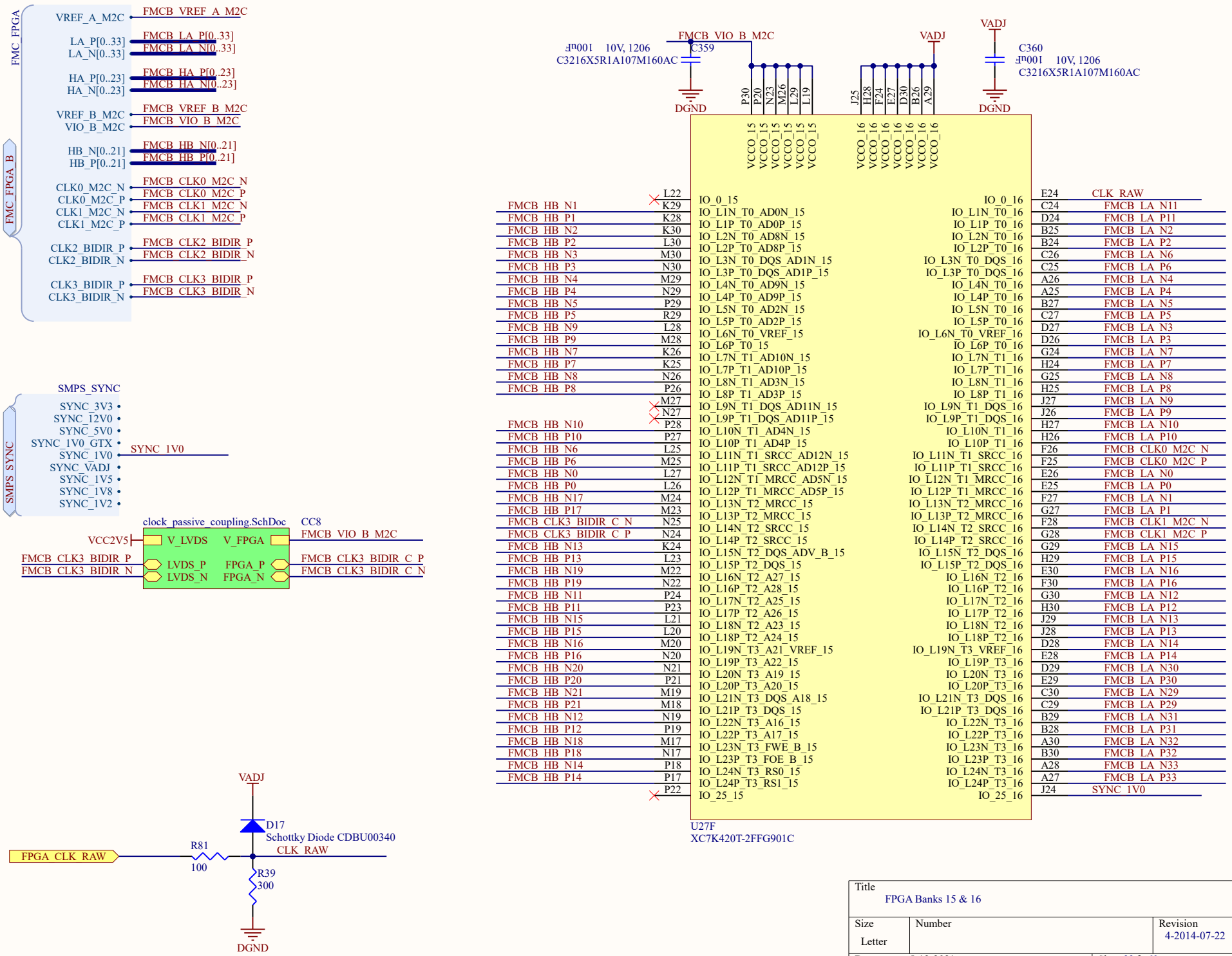

U27C
XC7K420T-2FFG901C
VCCAUX
VCCBRAM
VCCINT
MGTVCCAUX_G11
MGTVCCAUX_G10
MGTAVTT_G11
MGTAVTT_G10
MGTAVCC_G11
MGTAVCC_G10
VCC1V8
VCC1V0
VCC1V2
VCC1V0_GTX
VCC3V3
VCC5V0
GTX1V8
DGND
4.7uF, 0402
JMK105BBJ475MV-F
47uF, 6.3V
T520B476M006ATE070
330uF
T520V337M2R5ATE025
Layout: one per GTX bank
U57
RT9018A-12GSP
Vin
PGood
Vout
EN
Vdd(bias)
ADJ
NC
GND
EP/GND
R341
100
LED, 0603, Red
R342
12.4k
R343
10k
C659
DNP
0402
C232
100uF
1206
LDOFaults
LDOFault
PHYAPowerFault
PHYBPowerFault
ArmPowerFault
GTX1V8PowerFault
1.8V Gtx turns on when 3.3 turns on
C81
47uF, 6.3V
T520B476M006ATE070
C420
100uF
10V, 1206
C3216X5R1A107M160AC
See DS183: MGTAVCC is 1.05 if frequency > 10.3125 GHz
Title
FPGA Power Sheet
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 2 of 60
File:
C:\Users\..\FPGA_power.SchDoc
Drawn By:
Graeme Smecher

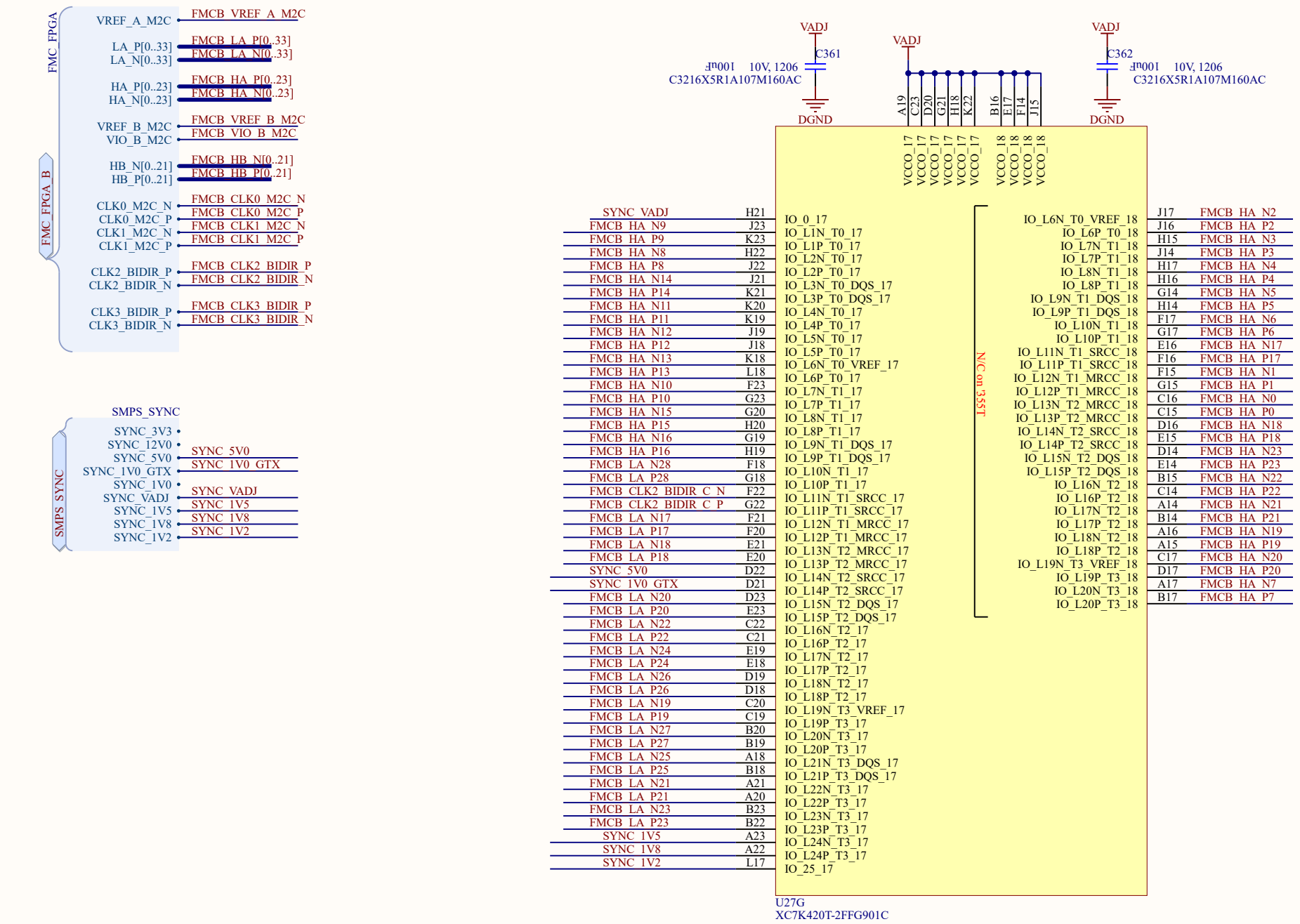


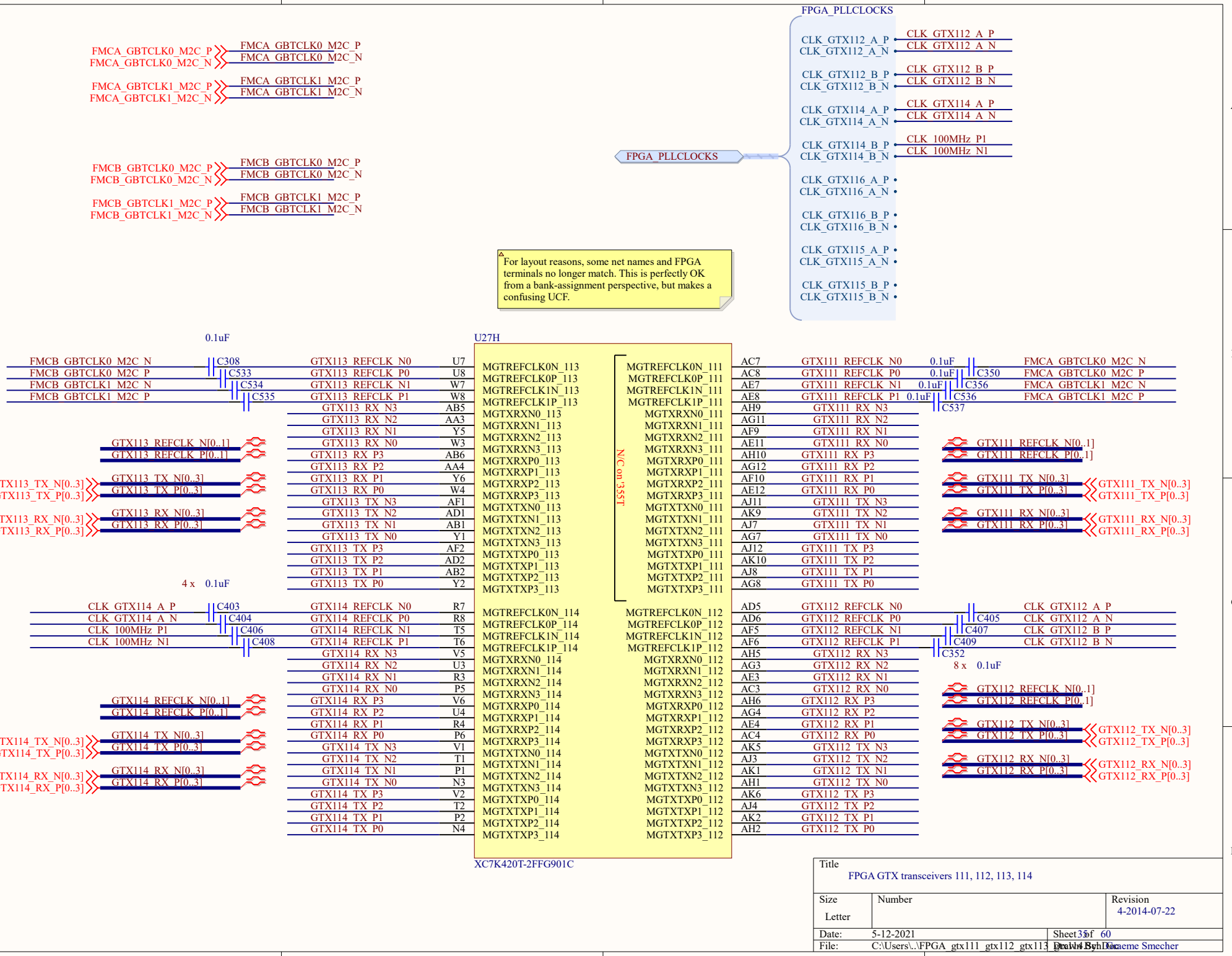
Title: FPGA Power Sheet		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 30 of 60
File: C:\Users\..\FPGA_ground.SchDoc		Drawn By: Graeme Smecher


FMC_FPGA
VREF_A_M2C
FMCA_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCA_LA_P[0..33]
FMCA_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCA_HA_P[0..23]
FMCA_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCA_VREF_B_M2C
FMCA_VIO_B_M2C
HB_N[0..21]
HB_P[0..21]
FMCA_HB_N[0..21]
FMCA_HB_P[0..21]
FMC_FPGA_A
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCA_CLK0_M2C_N
FMCA_CLK0_M2C_P
FMCA_CLK1_M2C_N
FMCA_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCA_CLK2_BIDIR_P
FMCA_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCA_CLK3_BIDIR_P
FMCA_CLK3_BIDIR_N
TODO: coupling network for CLK_M2Cs
TODO: pair/bus routing annotations
FMCA_VIO_B_M2C
C354
0.001uF 10V, 1206
C3216X5R1A107M160AC
DGND
VADJ
C355
0.001uF 10V, 1206
C3216X5R1A107M160AC
DGND
U27D
XC7K420T-2FFG901C
AA19
AB16
AE17
AF14
AH18
AJ15
AC23
AD20
AF24
AG21
AJ25
AK22
Y22
VCCO_11
VCCO_12
DNU: Not powered when VCCIOB not provided by FMC0
DNU: Not powered when VCCIOB not provided by FMC0
N/C on 355T
AC19 IO_0_11
AH20 FMCA_HB_N2 IO_L1N_T0_11
AG19 FMCA_HB_P2 IO_L1P_T0_11
AJ19 FMCA_HB_N13 IO_L2N_T0_11
AH19 FMCA_HB_P13 IO_L2P_T0_11
AK20 FMCA_HB_N3 IO_L3N_T0_DQS_11
AK19 FMCA_HB_P3 IO_L3P_T0_DQS_11
AK18 FMCA_HB_N4 IO_L4N_T0_11
AJ18 FMCA_HB_P4 IO_L4P_T0_11
AJ17 FMCA_HB_N8 IO_L5N_T0_11
AJ16 FMCA_HB_P8 IO_L5P_T0_11
AG18 FMCA_HB_N12 IO_L6N_T0_VREF_11
AF18 FMCA_HB_P12 IO_L6P_T0_11
AK16 FMCA_HB_N11 IO_L7N_T1_11
AK15 FMCA_HB_P11 IO_L7P_T1_11
AH15 FMCA_HB_N7 IO_L8N_T1_11
AH14 FMCA_HB_P7 IO_L8P_T1_11
AG15 FMCA_HB_N5 IO_L9N_T1_DQS_11
AG14 FMCA_HB_P5 IO_L9P_T1_DQS_11
AK14 FMCA_HB_N10 IO_L10N_T1_11
AJ14 FMCA_HB_P10 IO_L10P_T1_11
AH17 FMCA_HB_N6 IO_L11N_T1_SRCC_11
AH16 FMCA_HB_P6 IO_L11P_T1_SRCC_11
AG17 FMCA_HB_N17 IO_L12N_T1_MRCC_11
AF17 FMCA_HB_P17 IO_L12P_T1_MRCC_11
AF16 FMCA_HB_N0 IO_L13N_T2_MRCC_11
AF15 FMCA_HB_P0 IO_L13P_T2_MRCC_11
AE16 FMCA_CLK3_BIDIR_C_N IO_L14N_T2_SRCC_11
AE15 FMCA_CLK3_BIDIR_C_P IO_L14P_T2_SRCC_11
AD16 FMCA_HB_N18 IO_L15N_T2_DQS_11
AC16 FMCA_HB_P18 IO_L15P_T2_DQS_11
AB15 FMCA_HB_N20 IO_L16N_T2_11
AB14 FMCA_HB_P20 IO_L16P_T2_11
AC15 FMCA_HB_N21 IO_L17N_T2_11
AC14 FMCA_HB_P21 IO_L17P_T2_11
AE14 FMCA_HB_N15 IO_L18N_T2_11
AD14 FMCA_HB_P15 IO_L18P_T2_11
AE19 FMCA_HB_N9 IO_L19N_T3_VREF_11
AE18 FMCA_HB_P9 IO_L19P_T3_11
AD19 FMCA_HB_N1 IO_L20N_T3_11
AD18 FMCA_HB_P1 IO_L20P_T3_11
AD17 FMCA_HB_N19 IO_L21N_T3_DQS_11
AC17 FMCA_HB_P19 IO_L21P_T3_DQS_11
AB18 FMCA_HB_N14 IO_L22N_T3_11
AB17 FMCA_HB_P14 IO_L22P_T3_11
AA18 FMCA_HB_N16 IO_L23N_T3_11
AA17 FMCA_HB_P16 IO_L23P_T3_11
Y19 IO_L24N_T3_11
Y18 IO_L24P_T3_11
AB19 IO_25_11
IO_0_12 Y23 UART1_RX_V
IO_L1N_T0_12 Y21 FMCA_HA_N5
IO_L1P_T0_12 Y20 FMCA_HA_P5
IO_L2N_T0_12 AA21 FMCA_HA_N6
IO_L2P_T0_12 AA20 FMCA_HA_P6
IO_L3N_T0_DQS_12 AA23 FMCA_HA_N8
IO_L3P_T0_DQS_12 AA22 FMCA_HA_P8
IO_L4N_T0_12 AC20 FMCA_HA_N13
IO_L4P_T0_12 AB20 FMCA_HA_P13
IO_L5N_T0_12 AB23 FMCA_HA_N3
IO_L5P_T0_12 AB22 FMCA_HA_P3
IO_L6N_T0_VREF_12 AC22 FMCA_HA_N16
IO_L6P_T0_12 AC21 FMCA_HA_P16
IO_L7N_T1_12 AE26 FMCA_HA_N14
IO_L7P_T1_12 AD26 FMCA_HA_P14
IO_L8N_T1_12 AH26 FMCA_HA_N12
IO_L8P_T1_12 AH25 FMCA_HA_P12
IO_L9N_T1_DQS_12 AF26 FMCA_HA_N15
IO_L9P_T1_DQS_12 AE25 FMCA_HA_P15
IO_L10N_T1_12 AE24 FMCA_HA_N7
IO_L10P_T1_12 AD24 FMCA_HA_P7
IO_L11N_T1_SRCC_12 AG25 FMCA_HA_N18
IO_L11P_T1_SRCC_12 AF25 FMCA_HA_P18
IO_L12N_T1_MRCC_12 AH24 FMCA_HA_N17
IO_L12P_T1_MRCC_12 AG24 FMCA_HA_P17
IO_L13N_T2_MRCC_12 AG23 FMCA_HA_N0
IO_L13P_T2_MRCC_12 AG22 FMCA_HA_P0
IO_L14N_T2_SRCC_12 AF23 FMCA_HA_N1
IO_L14P_T2_SRCC_12 AF22 FMCA_HA_P1
IO_L15N_T2_DQS_12 AE23 FMCA_HA_N2
IO_L15P_T2_DQS_12 AD23 FMCA_HA_P2
IO_L16N_T2_12 AD22 FMCA_HA_N19
IO_L16P_T2_12 AD21 FMCA_HA_P19
IO_L17N_T2_12 AF21 FMCA_HA_N23
IO_L17P_T2_12 AE21 FMCA_HA_P23
IO_L18N_T2_12 AG20 FMCA_HA_N4
IO_L18P_T2_12 AF20 FMCA_HA_P4
IO_L19N_T3_VREF_12 AH22 FMCA_HA_N9
IO_L19P_T3_12 AH21 FMCA_HA_P9
IO_L20N_T3_12 AK21 FMCA_HA_N11
IO_L20P_T3_12 AJ21 FMCA_HA_P11
IO_L21N_T3_DQS_12 AK23 FMCA_HA_N21
IO_L21P_T3_DQS_12 AJ22 FMCA_HA_P21
IO_L22N_T3_12 AJ24 FMCA_HA_N10
IO_L22P_T3_12 AJ23 FMCA_HA_P10
IO_L23N_T3_12 AK25 FMCA_HA_N22
IO_L23P_T3_12 AK24 FMCA_HA_P22
IO_L24N_T3_12 AK26 FMCA_HA_N20
IO_L24P_T3_12 AJ26 FMCA_HA_P20
IO_25_12 AE20 UART1_TX_V
VCC3V3
U28
VADJ
RS232
UART1
TXD
RXD
VCCB
VCCA
B1
B2
A1
A2
GND
OE
UART1_RX_V
UART1_TX_V
VADJ
TXB0102DCUR
DGND
VCC3V3
VADJ
C294
0.47uF
C295
0.47uF
DGND
DGND
Connects to RS-232 port on ARM (and possibly GPIO on ARM)
clock_passive_coupling.SchDoc
CC5
FMCA_VIO_B_M2C
VCC2V5
V_LVDS
V_FPGA
FMCA_CLK3_BIDIR_P
FMCA_CLK3_BIDIR_N
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCA_CLK3_BIDIR_C_P
FMCA_CLK3_BIDIR_C_N
Title
FPGA Banks 11 & 12
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 3 of 60
File: C:\Users\..\FPGA_bank11_bank12.SchDoc
Drawn By: Graeme Smecher

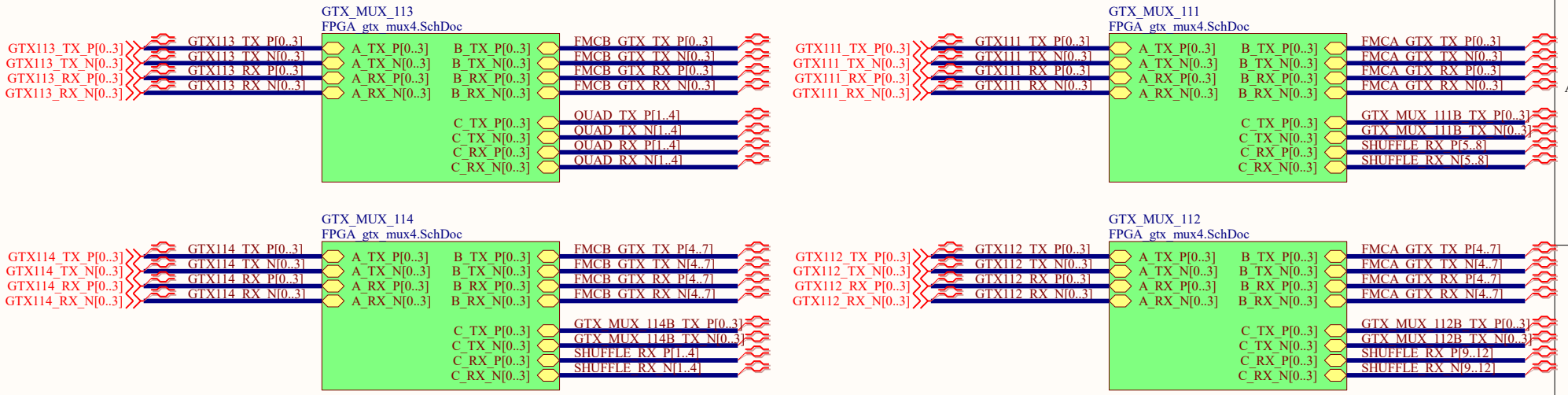

FMC_FPGA
VREF_A_M2C
FMCA_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCA_LA_P[0..33]
FMCA_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCA_HA_P[0..23]
FMCA_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCA_VREF_B_M2C
FMCA_VIO_B_M2C
HB_N[0..21]
HB_P[0..21]
FMCA_HB_N[0..21]
FMCA_HB_P[0..21]
FMC_FPGA_A
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCA_CLK0_M2C_N
FMCA_CLK0_M2C_P
FMCA_CLK1_M2C_N
FMCA_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCA_CLK2_BIDIR_P
FMCA_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCA_CLK3_BIDIR_P
FMCA_CLK3_BIDIR_N
SMPS_SYNC
SYNC_3V3
SYNC_12V0
SYNC_5V0
SYNC_1V0_GTX
SYNC_1V0
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
SMPS_SYNC
FPGA_CONFIG
PROGRAM_B
INIT_B
DONE
SPI_SCK
SPI_D0
SPI_D1
SPI_FLASH_CS
SPI_FPGA_CS
FLASH_CS
FPGA_CS
SPI_SCK_CCLK
FPGA_CONFIG
These BP_IO signals are capable of being used differentially. The resistors below are optional external terminators.
BP_IO_N0 R172 DNP BP_IO_P0
BP_IO_N1 R173 DNP BP_IO_P1
BP_IO_N2 R207 DNP BP_IO_P2
VADJ
C357
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
C358
100uF 10V, 1206
C3216X5R1A107M160AC
AA29 AB26 AD30 AE27 AH28 W25 R17 R27 T24 U21 V18 V28
VCCO_13 VCCO_13 VCCO_13 VCCO_13 VCCO_13 VCCO_13
VCCO_14 VCCO_14 VCCO_14 VCCO_14 VCCO_14 VCCO_14
Source termination.
R209, R210: Near U10
R216: Near FPGA
BP_IO3 W24
FMCA_LA_N22 AB28
FMCA_LA_P22 AA28
FMCA_LA_N21 AB30
FMCA_LA_P21 AB29
FMCA_LA_N23 AA30
FMCA_LA_P23 Y30
FMCA_LA_N19 Y29
FMCA_LA_P19 Y28
FMCA_LA_N20 W29
FMCA_LA_P20 W28
FMCA_VREF_A_M2C W27
BP_IO4 W26
FMCA_LA_N31 AD27
FMCA_LA_P31 AC26
FMCA_LA_N32 AA25
FMCA_LA_P32 Y25
FMCA_LA_N30 AC25
FMCA_LA_P30 AB25
FMCA_LA_N33 AC24
FMCA_LA_P33 AB24
FMCA_CLK2_BIDIR_C_N AA26
FMCA_CLK2_BIDIR_C_P Y26
FMCA_LA_N17 AB27
FMCA_LA_P17 AA27
FMCA_LA_N18 AD28
FMCA_LA_P18 AC27
SYNC_12V0_V AF28
SYNC_3V3 AE28
FMCA_LA_N26 AC30
FMCA_LA_P26 AC29
FMCA_LA_N27 AE29
FMCA_LA_P27 AD29
FMCA_LA_N25 AF30
FMCA_LA_P25 AE30
FMCA_LA_N24 AG28
FMCA_LA_P24 AF27
FMCA_VREF_A_M2C AH27
BP_IO5 AG27
FMCA_LA_N29 AH29
FMCA_LA_P29 AG29
FMCA_LA_N28 AJ28
FMCA_LA_P28 AJ27
BP_IO_N0 AH30
BP_IO_P0 AG30
BP_IO_N1 AK30
BP_IO_P1 AJ29
BP_IO_N2 AK29
BP_IO_P2 AK28
GPIO_RST_V Y24
IO_0_13 IO_L1N_T0_13 IO_L1P_T0_13 IO_L2N_T0_13 IO_L2P_T0_13 IO_L3N_T0_DQS_13 IO_L3P_T0_DQS_13 IO_L4N_T0_13 IO_L4P_T0_13 IO_L5N_T0_13 IO_L5P_T0_13 IO_L6N_T0_VREF_13 IO_L6P_T0_13 IO_L7N_T1_13 IO_L7P_T1_13 IO_L8N_T1_13 IO_L8P_T1_13 IO_L9N_T1_DQS_13 IO_L9P_T1_DQS_13 IO_L10N_T1_13 IO_L10P_T1_13 IO_L11N_T1_SRCC_13 IO_L11P_T1_SRCC_13 IO_L12N_T1_MRCC_13 IO_L12P_T1_MRCC_13 IO_L13N_T2_MRCC_13 IO_L13P_T2_MRCC_13 IO_L14N_T2_SRCC_13 IO_L14P_T2_SRCC_13 IO_L15N_T2_DQS_13 IO_L15P_T2_DQS_13 IO_L16N_T2_13 IO_L16P_T2_13 IO_L17N_T2_13 IO_L17P_T2_13 IO_L18N_T2_13 IO_L18P_T2_13 IO_L19N_T3_VREF_13 IO_L19P_T3_13 IO_L20N_T3_13 IO_L20P_T3_13 IO_L21N_T3_DQS_13 IO_L21P_T3_DQS_13 IO_L22N_T3_13 IO_L22P_T3_13 IO_L23N_T3_13 IO_L23P_T3_13 IO_L24N_T3_13 IO_L24P_T3_13 IO_25_13
IO_0_14 IO_L1N_T0_D01_DIN_14 IO_L1P_T0_D00_MOSI_14 IO_L2N_T0_D03_14 IO_L2P_T0_D02_14 IO_L3N_T0_DQS_EMCCLK_14 IO_L3P_T0_DQS_PUDC_B_14 IO_L4N_T0_D05_14 IO_L4P_T0_D04_14 IO_L5N_T0_D07_14 IO_L5P_T0_D06_14 IO_L6N_T0_D08_VREF_14 IO_L6P_T0_FCS_B_14 IO_L7N_T1_D10_14 IO_L7P_T1_D09_14 IO_L8N_T1_D12_14 IO_L8P_T1_D11_14 IO_L9N_T1_DQS_D13_14 IO_L9P_T1_DQS_14 IO_L10N_T1_D15_14 IO_L10P_T1_D14_14 IO_L11N_T1_SRCC_14 IO_L11P_T1_SRCC_14 IO_L12N_T1_MRCC_14 IO_L12P_T1_MRCC_14 IO_L13N_T2_MRCC_14 IO_L13P_T2_MRCC_14 IO_L14N_T2_SRCC_14 IO_L14P_T2_SRCC_14 IO_L15N_T2_DQS_DOUT_CSO_B_14 IO_L15P_T2_DQS_RDWR_B_14 IO_L16N_T2_A15_D31_14 IO_L16P_T2_CSI_B_14 IO_L17N_T2_A13_D29_14 IO_L17P_T2_A14_D30_14 IO_L18N_T2_A11_D27_14 IO_L18P_T2_A12_D28_14 IO_L19N_T3_A09_D25_VREF_14 IO_L19P_T3_A10_D26_14 IO_L20N_T3_A07_D23_14 IO_L20P_T3_A08_D24_14 IO_L21N_T3_DQS_A06_D22_14 IO_L21P_T3_DQS_14 IO_L22N_T3_A04_D20_14 IO_L22P_T3_A05_D21_14 IO_L23N_T3_A02_D18_14 IO_L23P_T3_A03_D19_14 IO_L24N_T3_A00_D16_14 IO_L24P_T3_A01_D17_14 IO_25_14
U27E XC7K420T-2FFG901C
T22 R209 22 SPI_SCK_V
T30 R210 22 SPI_D1_V
R30 R216 22 SPI_D0_V
T28 FMCA_LA_N4
R28 FMCA_LA_P4
V30 ARM_RST_O_V
V29
U30 FMCA_LA_N13
U29 FMCA_LA_P13
U28 FMCA_LA_N12
U27 FMCA_LA_P12
V27 FMCA_VREF_A_M2C
V26 FLASH_CS_V
R25 FMCA_LA_N3
R24 FMCA_LA_P3
T23 FMCA_LA_N8
R23 FMCA_LA_P8
V25 FMCA_LA_N14
V24 FMCA_LA_P14
T27 FMCA_LA_N6
R26 FMCA_LA_P6
T26 FMCA_CLK0_M2C_N
T25 FMCA_CLK0_M2C_P
U25 FMCA_LA_N0
U24 FMCA_LA_P0
V22 FMCA_LA_N1
V21 FMCA_LA_P1
U23 FMCA_CLK1_M2C_N
U22 FMCA_CLK1_M2C_P
R21 SCL_V
R20 FPGA_CS_V
T21 FMCA_LA_N7
T20 FMCA_LA_P7
V20 FMCA_LA_N11
U20 FMCA_LA_P11
W23 FMCA_LA_N15
W22 FMCA_LA_P15
R19 FMCA_VREF_A_M2C
R18 GPIO_IRQ_V
V19 FMCA_LA_N5
U19 FMCA_LA_P5
T18 FMCA_LA_N2
T17 FMCA_LA_P2
U18 FMCA_LA_N16
U17 FMCA_LA_P16
W17 FMCA_LA_N10
V17 FMCA_LA_P10
W19 FMCA_LA_N9
W18 FMCA_LA_P9
W21 ARM_IRQ_V
VCC3V3 VADJ
C298 C299 C296 C297 C590 C591
0.47uF
DGND
Decoupling for translators
VADJ
R254
1k
SDA_V
PUDC disabled. SDA translator has internal 10k p/u; 1k minimum on PUDC recommended.
U10 TXB0104
VCCA VCCB
A1 A2 A3 A4
B1 B2 B3 B4
OE GND
SPI_SCK_V
SPI_D1_V
SPI_D0_V
SPI_SCK
SPI_D1
SPI_D0
SYNC_12V0
D8
Schottky Diode CDBU00340
R24
1k
SYNC_12V0 must not float high!
R25
49.9k
C629
10nF
0402
SYNC_12V0_V
ARM_RST_O
I2C_FPGA0
I2C
SDA
SCL
U30
ARM_RST_O_V
SDA_V
FLASH_CS_V
SCL_V
FLASH_CS
TXS0104EPWR
U38
GPIO_RST_V
FPGA_CS_V
ARM_IRQ_V
GPIO_IRQ_V
GPIO_RST
FPGA_CS
ARM_IRQ
GPIO_IRQ
BP_IO
BP_IO[3..5]
BP_IO_P[0..2]
BP_IO_N[0..2]
clock_passive_coupling.SchDoc
CC4
VCC2V5
V_LVDS
V_FPGA
VADJ
FMCA_CLK2_BIDIR_P
FMCA_CLK2_BIDIR_N
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCA_CLK2_BIDIR_C_P
FMCA_CLK2_BIDIR_C_N
Title
FPGA Banks 13 & 14
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
Sheet 32 of 60
File: C:\Users\..\FPGA_bank13_bank14.SchDoc
Drawn By: Graeme Smecher

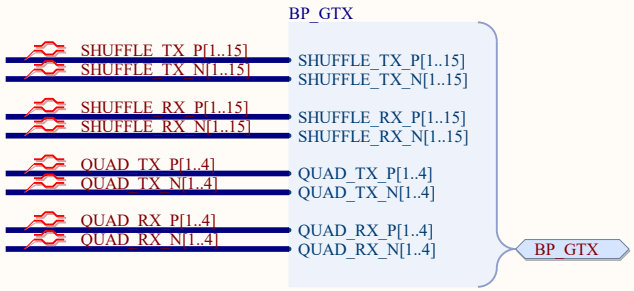

FMC_FPGA
VREF_A_M2C
FMCB_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCB_LA_P[0..33]
FMCB_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCB_HA_P[0..23]
FMCB_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCB_VREF_B_M2C
FMCB_VIO_B_M2C
FMC_FPGA_B
HB_N[0..21]
HB_P[0..21]
FMCB_HB_N[0..21]
FMCB_HB_P[0..21]
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCB_CLK0_M2C_N
FMCB_CLK0_M2C_P
FMCB_CLK1_M2C_N
FMCB_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCB_CLK3_BIDIR_P
FMCB_CLK3_BIDIR_N
SMPS_SYNC
SYNC_3V3
SYNC_12V0
SYNC_5V0
SYNC_1V0_GTX
SYNC_1V0
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
SYNC_1V0
clock_passive_coupling.SchDoc
CC8
VCC2V5
V_LVDS
V_FPGA
FMCB_VIO_B_M2C
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCB_CLK3_BIDIR_C_P
FMCB_CLK3_BIDIR_C_N
VADJ
D17
Schottky Diode CDBU00340
FPGA_CLK_RAW
R81
100
CLK_RAW
R39
300
DGND
C359
100nF 10V, 1206
C3216X5R1A107M160AC
C360
100nF 10V, 1206
C3216X5R1A107M160AC
P30
P20
N23
M26
L29
L19
J25
H28
F24
E27
D30
B26
A29
VCCO_15
VCCO_16
IO_0_15
IO_L1N_T0_AD0N_15
IO_L1P_T0_AD0P_15
IO_L2N_T0_AD8N_15
IO_L2P_T0_AD8P_15
IO_L3N_T0_DQS_AD1N_15
IO_L3P_T0_DQS_AD1P_15
IO_L4N_T0_AD9N_15
IO_L4P_T0_AD9P_15
IO_L5N_T0_AD2N_15
IO_L5P_T0_AD2P_15
IO_L6N_T0_VREF_15
IO_L6P_T0_15
IO_L7N_T1_AD10N_15
IO_L7P_T1_AD10P_15
IO_L8N_T1_AD3N_15
IO_L8P_T1_AD3P_15
IO_L9N_T1_DQS_AD11N_15
IO_L9P_T1_DQS_AD11P_15
IO_L10N_T1_AD4N_15
IO_L10P_T1_AD4P_15
IO_L11N_T1_SRCC_AD12N_15
IO_L11P_T1_SRCC_AD12P_15
IO_L12N_T1_MRCC_AD5N_15
IO_L12P_T1_MRCC_AD5P_15
IO_L13N_T2_MRCC_15
IO_L13P_T2_MRCC_15
IO_L14N_T2_SRCC_15
IO_L14P_T2_SRCC_15
IO_L15N_T2_DQS_ADV_B_15
IO_L15P_T2_DQS_15
IO_L16N_T2_A27_15
IO_L16P_T2_A28_15
IO_L17N_T2_A25_15
IO_L17P_T2_A26_15
IO_L18N_T2_A23_15
IO_L18P_T2_A24_15
IO_L19N_T3_A21_VREF_15
IO_L19P_T3_A22_15
IO_L20N_T3_A19_15
IO_L20P_T3_A20_15
IO_L21N_T3_DQS_A18_15
IO_L21P_T3_DQS_15
IO_L22N_T3_A16_15
IO_L22P_T3_A17_15
IO_L23N_T3_FWE_B_15
IO_L23P_T3_FOE_B_15
IO_L24N_T3_RS0_15
IO_L24P_T3_RS1_15
IO_25_15
IO_0_16
IO_L1N_T0_16
IO_L1P_T0_16
IO_L2N_T0_16
IO_L2P_T0_16
IO_L3N_T0_DQS_16
IO_L3P_T0_DQS_16
IO_L4N_T0_16
IO_L4P_T0_16
IO_L5N_T0_16
IO_L5P_T0_16
IO_L6N_T0_VREF_16
IO_L6P_T0_16
IO_L7N_T1_16
IO_L7P_T1_16
IO_L8N_T1_16
IO_L8P_T1_16
IO_L9N_T1_DQS_16
IO_L9P_T1_DQS_16
IO_L10N_T1_16
IO_L10P_T1_16
IO_L11N_T1_SRCC_16
IO_L11P_T1_SRCC_16
IO_L12N_T1_MRCC_16
IO_L12P_T1_MRCC_16
IO_L13N_T2_MRCC_16
IO_L13P_T2_MRCC_16
IO_L14N_T2_SRCC_16
IO_L14P_T2_SRCC_16
IO_L15N_T2_DQS_16
IO_L15P_T2_DQS_16
IO_L16N_T2_16
IO_L16P_T2_16
IO_L17N_T2_16
IO_L17P_T2_16
IO_L18N_T2_16
IO_L18P_T2_16
IO_L19N_T3_VREF_16
IO_L19P_T3_16
IO_L20N_T3_16
IO_L20P_T3_16
IO_L21N_T3_DQS_16
IO_L21P_T3_DQS_16
IO_L22N_T3_16
IO_L22P_T3_16
IO_L23N_T3_16
IO_L23P_T3_16
IO_L24N_T3_16
IO_L24P_T3_16
IO_25_16
L22
K29
K28
K30
L30
M30
N30
M29
N29
P29
R29
L28
M28
K26
K25
N26
P26
M27
N27
P28
P27
L25
M25
L27
L26
M24
M23
N25
N24
K24
L23
M22
N22
P24
P23
L21
L20
M20
N20
N21
P21
M19
M18
N19
P19
M17
N17
P18
P17
P22
FMCB_HB_N1
FMCB_HB_P1
FMCB_HB_N2
FMCB_HB_P2
FMCB_HB_N3
FMCB_HB_P3
FMCB_HB_N4
FMCB_HB_P4
FMCB_HB_N5
FMCB_HB_P5
FMCB_HB_N9
FMCB_HB_P9
FMCB_HB_N7
FMCB_HB_P7
FMCB_HB_N8
FMCB_HB_P8
FMCB_HB_N10
FMCB_HB_P10
FMCB_HB_N6
FMCB_HB_P6
FMCB_HB_N0
FMCB_HB_P0
FMCB_HB_N17
FMCB_HB_P17
FMCB_CLK3_BIDIR_C_N
FMCB_CLK3_BIDIR_C_P
FMCB_HB_N13
FMCB_HB_P13
FMCB_HB_N19
FMCB_HB_P19
FMCB_HB_N11
FMCB_HB_P11
FMCB_HB_N15
FMCB_HB_P15
FMCB_HB_N16
FMCB_HB_P16
FMCB_HB_N20
FMCB_HB_P20
FMCB_HB_N21
FMCB_HB_P21
FMCB_HB_N12
FMCB_HB_P12
FMCB_HB_N18
FMCB_HB_P18
FMCB_HB_N14
FMCB_HB_P14
U27F
XC7K420T-2FFG901C
E24 CLK_RAW
C24 FMCB_LA_N11
D24 FMCB_LA_P11
B25 FMCB_LA_N2
B24 FMCB_LA_P2
C26 FMCB_LA_N6
C25 FMCB_LA_P6
A26 FMCB_LA_N4
A25 FMCB_LA_P4
B27 FMCB_LA_N5
C27 FMCB_LA_P5
D27 FMCB_LA_N3
D26 FMCB_LA_P3
G24 FMCB_LA_N7
H24 FMCB_LA_P7
G25 FMCB_LA_N8
H25 FMCB_LA_P8
J27 FMCB_LA_N9
J26 FMCB_LA_P9
H27 FMCB_LA_N10
H26 FMCB_LA_P10
F26 FMCB_CLK0_M2C_N
F25 FMCB_CLK0_M2C_P
E26 FMCB_LA_N0
E25 FMCB_LA_P0
F27 FMCB_LA_N1
G27 FMCB_LA_P1
F28 FMCB_CLK1_M2C_N
G28 FMCB_CLK1_M2C_P
G29 FMCB_LA_N15
H29 FMCB_LA_P15
E30 FMCB_LA_N16
F30 FMCB_LA_P16
G30 FMCB_LA_N12
H30 FMCB_LA_P12
J29 FMCB_LA_N13
J28 FMCB_LA_P13
D28 FMCB_LA_N14
E28 FMCB_LA_P14
D29 FMCB_LA_N30
E29 FMCB_LA_P30
C30 FMCB_LA_N29
C29 FMCB_LA_P29
B29 FMCB_LA_N31
B28 FMCB_LA_P31
A30 FMCB_LA_N32
B30 FMCB_LA_P32
A28 FMCB_LA_N33
A27 FMCB_LA_P33
J24 SYNC_1V0
Title
FPGA Banks 15 & 16
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Letter
Number
Revision
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Date: 5-12-2021
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File: C:\Users\..\FPGA_bank15_bank16.SchDoc
Drawn By: Graeme Smecher

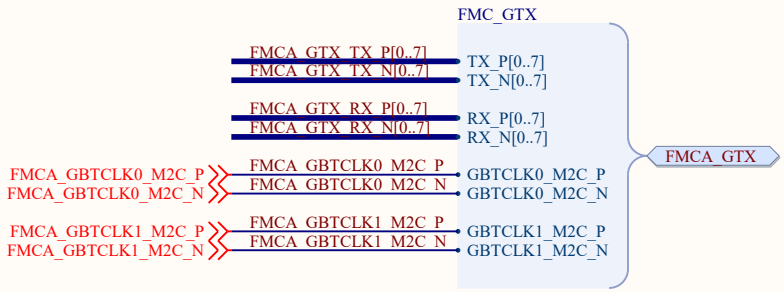

FMC_FPGA
VREF_A_M2C
FMCB_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCB_LA_P[0..33]
FMCB_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCB_HA_P[0..23]
FMCB_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCB_VREF_B_M2C
FMCB_VIO_B_M2C
HB_N[0..21]
HB_P[0..21]
FMCB_HB_N[0..21]
FMCB_HB_P[0..21]
FMC_FPGA_B
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCB_CLK0_M2C_N
FMCB_CLK0_M2C_P
FMCB_CLK1_M2C_N
FMCB_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCB_CLK3_BIDIR_P
FMCB_CLK3_BIDIR_N
SMPS_SYNC
SYNC_3V3
SYNC_12V0
SYNC_5V0
SYNC_1V0_GTX
SYNC_1V0
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
SMPS_SYNC
SYNC_5V0
SYNC_1V0_GTX
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
VADJ
C361
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
VADJ
A19
C23
D20
G21
H18
K22
B16
E17
F14
J15
VADJ
C362
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_18
VCCO_18
VCCO_18
VCCO_18
SYNC_VADJ H21
FMCB_HA_N9 J23
FMCB_HA_P9 K23
FMCB_HA_N8 H22
FMCB_HA_P8 J22
FMCB_HA_N14 J21
FMCB_HA_P14 K21
FMCB_HA_N11 K20
FMCB_HA_P11 K19
FMCB_HA_N12 J19
FMCB_HA_P12 J18
FMCB_HA_N13 K18
FMCB_HA_P13 L18
FMCB_HA_N10 F23
FMCB_HA_P10 G23
FMCB_HA_N15 G20
FMCB_HA_P15 H20
FMCB_HA_N16 G19
FMCB_HA_P16 H19
FMCB_LA_N28 F18
FMCB_LA_P28 G18
FMCB_CLK2_BIDIR_C_N F22
FMCB_CLK2_BIDIR_C_P G22
FMCB_LA_N17 F21
FMCB_LA_P17 F20
FMCB_LA_N18 E21
FMCB_LA_P18 E20
SYNC_5V0 D22
SYNC_1V0_GTX D21
FMCB_LA_N20 D23
FMCB_LA_P20 E23
FMCB_LA_N22 C22
FMCB_LA_P22 C21
FMCB_LA_N24 E19
FMCB_LA_P24 E18
FMCB_LA_N26 D19
FMCB_LA_P26 D18
FMCB_LA_N19 C20
FMCB_LA_P19 C19
FMCB_LA_N27 B20
FMCB_LA_P27 B19
FMCB_LA_N25 A18
FMCB_LA_P25 B18
FMCB_LA_N21 A21
FMCB_LA_P21 A20
FMCB_LA_N23 B23
FMCB_LA_P23 B22
SYNC_1V5 A23
SYNC_1V8 A22
SYNC_1V2 L17
IO_0_17
IO_L1N_T0_17
IO_L1P_T0_17
IO_L2N_T0_17
IO_L2P_T0_17
IO_L3N_T0_DQS_17
IO_L3P_T0_DQS_17
IO_L4N_T0_17
IO_L4P_T0_17
IO_L5N_T0_17
IO_L5P_T0_17
IO_L6N_T0_VREF_17
IO_L6P_T0_17
IO_L7N_T1_17
IO_L7P_T1_17
IO_L8N_T1_17
IO_L8P_T1_17
IO_L9N_T1_DQS_17
IO_L9P_T1_DQS_17
IO_L10N_T1_17
IO_L10P_T1_17
IO_L11N_T1_SRCC_17
IO_L11P_T1_SRCC_17
IO_L12N_T1_MRCC_17
IO_L12P_T1_MRCC_17
IO_L13N_T2_MRCC_17
IO_L13P_T2_MRCC_17
IO_L14N_T2_SRCC_17
IO_L14P_T2_SRCC_17
IO_L15N_T2_DQS_17
IO_L15P_T2_DQS_17
IO_L16N_T2_17
IO_L16P_T2_17
IO_L17N_T2_17
IO_L17P_T2_17
IO_L18N_T2_17
IO_L18P_T2_17
IO_L19N_T3_VREF_17
IO_L19P_T3_17
IO_L20N_T3_17
IO_L20P_T3_17
IO_L21N_T3_DQS_17
IO_L21P_T3_DQS_17
IO_L22N_T3_17
IO_L22P_T3_17
IO_L23N_T3_17
IO_L23P_T3_17
IO_L24N_T3_17
IO_L24P_T3_17
IO_25_17
N/C on 355T
IO_L6N_T0_VREF_18
IO_L6P_T0_18
IO_L7N_T1_18
IO_L7P_T1_18
IO_L8N_T1_18
IO_L8P_T1_18
IO_L9N_T1_DQS_18
IO_L9P_T1_DQS_18
IO_L10N_T1_18
IO_L10P_T1_18
IO_L11N_T1_SRCC_18
IO_L11P_T1_SRCC_18
IO_L12N_T1_MRCC_18
IO_L12P_T1_MRCC_18
IO_L13N_T2_MRCC_18
IO_L13P_T2_MRCC_18
IO_L14N_T2_SRCC_18
IO_L14P_T2_SRCC_18
IO_L15N_T2_DQS_18
IO_L15P_T2_DQS_18
IO_L16N_T2_18
IO_L16P_T2_18
IO_L17N_T2_18
IO_L17P_T2_18
IO_L18N_T2_18
IO_L18P_T2_18
IO_L19N_T3_VREF_18
IO_L19P_T3_18
IO_L20N_T3_18
IO_L20P_T3_18
J17 FMCB_HA_N2
J16 FMCB_HA_P2
H15 FMCB_HA_N3
J14 FMCB_HA_P3
H17 FMCB_HA_N4
H16 FMCB_HA_P4
G14 FMCB_HA_N5
H14 FMCB_HA_P5
F17 FMCB_HA_N6
G17 FMCB_HA_P6
E16 FMCB_HA_N17
F16 FMCB_HA_P17
F15 FMCB_HA_N1
G15 FMCB_HA_P1
C16 FMCB_HA_N0
C15 FMCB_HA_P0
D16 FMCB_HA_N18
E15 FMCB_HA_P18
D14 FMCB_HA_N23
E14 FMCB_HA_P23
B15 FMCB_HA_N22
C14 FMCB_HA_P22
A14 FMCB_HA_N21
B14 FMCB_HA_P21
A16 FMCB_HA_N19
A15 FMCB_HA_P19
C17 FMCB_HA_N20
D17 FMCB_HA_P20
A17 FMCB_HA_N7
B17 FMCB_HA_P7
U27G
XC7K420T-2FFG901C
clock_passive_coupling.SchDoc
CC10
VCC2V5
V_LVDS
V_FPGA
VADJ
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCB_CLK2_BIDIR_C_P
FMCB_CLK2_BIDIR_C_N
Title
FPGA Banks 17 & 18
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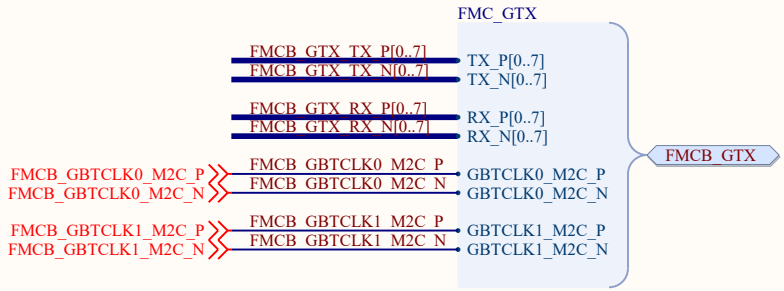


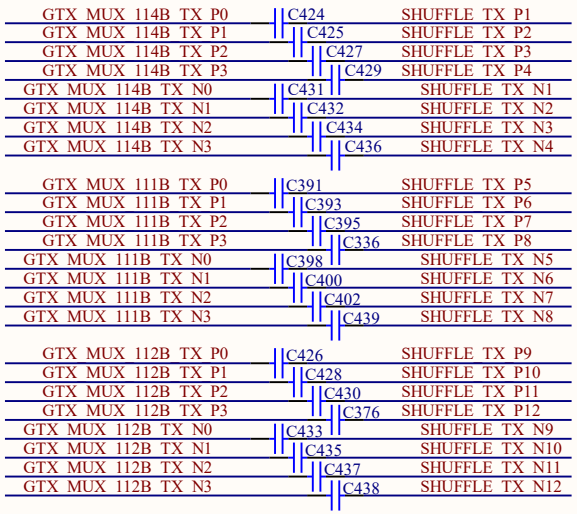
Title		
FPGA GTX transceivers 111, 112, 113, 114		
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File: C:\Users\..\FPGA_gtx111_gtx112_gtx113_gtx114.SchDoc	Drawn By: Dimeme Smecher	











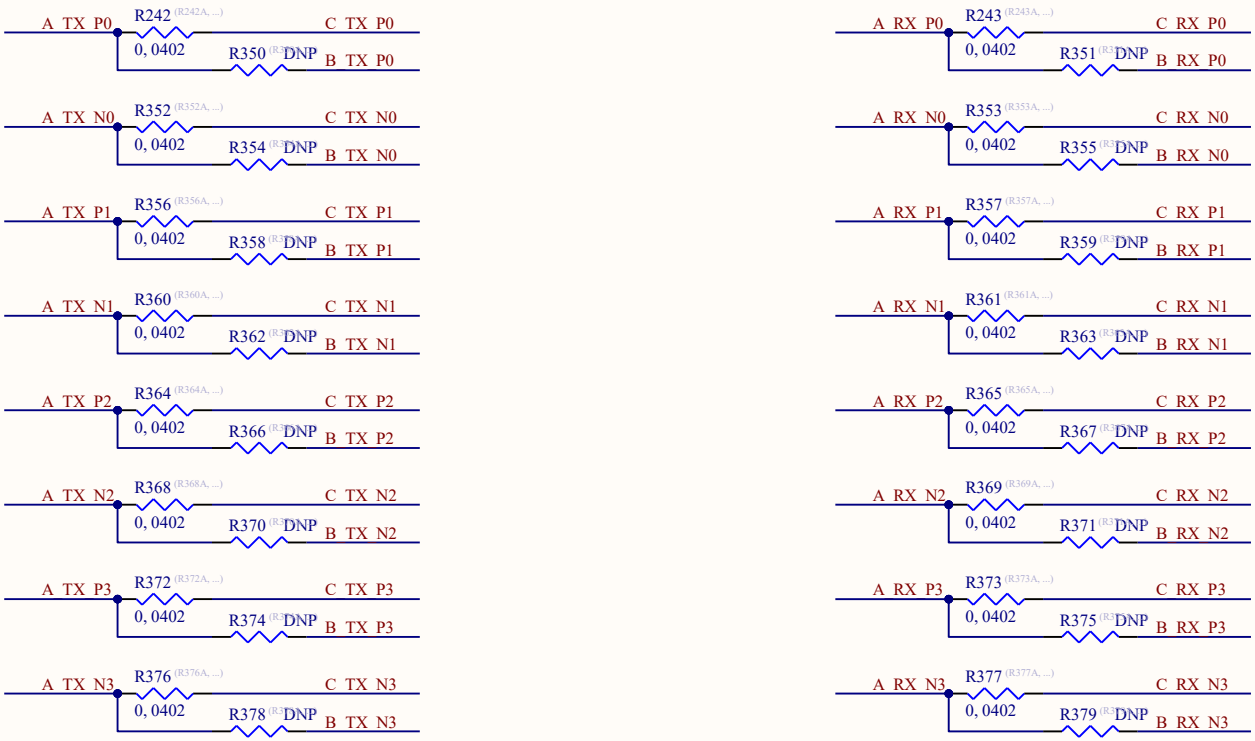
24 x 0.1uF

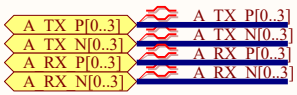
There are several conventions for coupling caps:

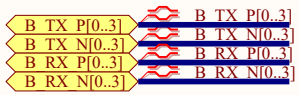
* For PCIe, coupling caps are placed wherever there's a transceiver.
* For QSFP/SFP, coupling caps are placed in the module.
* For FMC, caps are placed (if needed) on the mezzanine.
* For backplane shuffles, we follow PCIe convention.

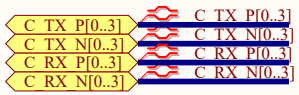
Title: FPGA GTX transceivers 111, 112, 113, 114 (Multiplexing)		
Size: Letter	Number:	Revision: 4-2014-07-22
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We connect A <-> C by default. This connects GTXes to the backplane.









Title: GTX Multiplexer		
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CLK_BANK_14 now goes to GTX clock 115A
CLK_BANK_17 now goes to GTX clock 115B

U27I

CLK_GTX115_A_N — 0.1uF — GTX115_REFCLK_N0 J7
CLK_GTX115_A_P C646 — 0.1uF — C645 — GTX115_REFCLK_P0 J8
CLK_GTX115_B_N — C648 — C647 — GTX115_REFCLK_N1 L7
CLK_GTX115_B_P — GTX115_REFCLK_P1 L8

SHUFFLE_RX_N13 — 0.1uF — M5
SHUFFLE_RX_N14 — 0.1uF — L3
SHUFFLE_RX_N15 — K5
PCI_SFP_RX_N — J3
SHUFFLE_RX_P13 — M6
SHUFFLE_RX_P14 — L4
SHUFFLE_RX_P15 — K6
PCI_SFP_RX_P — J4

SHUFFLE_TX_N13 — C327 — GTX115_TX_N0 M1
SHUFFLE_TX_N14 — C329 — GTX115_TX_N1 K1
SHUFFLE_TX_N15 — C331 — GTX115_TX_N2 H1
PCI_SFP_TX_N — F1
SHUFFLE_TX_P13 — C335 — GTX115_TX_P0 M2
SHUFFLE_TX_P14 — C337 — GTX115_TX_P1 K2
SHUFFLE_TX_P15 — C339 — GTX115_TX_P2 H2
PCI_SFP_TX_P — F2

0.1uF x10
CLK_GTX116_A_P — C344 — GTX116_REFCLK_N0 F5
CLK_GTX116_A_N — C345 — GTX116_REFCLK_P0 F6
CLK_GTX116_B_P — C347 — GTX116_REFCLK_N1 H5
CLK_GTX116_B_N — C349 — GTX116_REFCLK_P1 H6

QSFPA_RX_N1 G3
QSFPA_RX_N2 E3
QSFPA_RX_N3 D5
QSFPA_RX_N4 C3
QSFPA_RX_P1 G4
QSFPA_RX_P2 E4
QSFPA_RX_P3 D6
QSFPA_RX_P4 C4
QSFPA_TX_N1 D1
QSFPA_TX_N2 B1
QSFPA_TX_N3 A3
QSFPA_TX_N4 B5
QSFPA_TX_P1 D2
QSFPA_TX_P2 B2
QSFPA_TX_P3 A4
QSFPA_TX_P4 B6

MGTREFCLK0N_115 MGTREFCLK0P_115 MGTREFCLK1N_115 MGTREFCLK1P_115
MGTXRXN0_115 MGTXRXN1_115 MGTXRXN2_115 MGTXRXN3_115
MGTXRXP0_115 MGTXRXP1_115 MGTXRXP2_115 MGTXRXP3_115
MGTXTXN0_115 MGTXTXN1_115 MGTXTXN2_115 MGTXTXN3_115
MGTXTXP0_115 MGTXTXP1_115 MGTXTXP2_115 MGTXTXP3_115

MGTREFCLK0N_116 MGTREFCLK0P_116 MGTREFCLK1N_116 MGTREFCLK1P_116
MGTXRXN0_116 MGTXRXN1_116 MGTXRXN2_116 MGTXRXN3_116
MGTXRXP0_116 MGTXRXP1_116 MGTXRXP2_116 MGTXRXP3_116
MGTXTXN0_116 MGTXTXN1_116 MGTXTXN2_116 MGTXTXN3_116
MGTXTXP0_116 MGTXTXP1_116 MGTXTXP2_116 MGTXTXP3_116

MGTREFCLK0N_117 E7
MGTREFCLK0P_117 E8
MGTREFCLK1N_117 G7
MGTREFCLK1P_117 G8
MGTXRXN0_117 F9
MGTXRXN1_117 D9
MGTXRXN2_117 E11
MGTXRXN3_117 C11
MGTXRXP0_117 F10
MGTXRXP1_117 D10
MGTXRXP2_117 E12
MGTXRXP3_117 C12
MGTXTXN0_117 C7
MGTXTXN1_117 A7
MGTXTXN2_117 B9
MGTXTXN3_117 A11
MGTXTXP0_117 C8
MGTXTXP1_117 A8
MGTXTXP2_117 B10
MGTXTXP3_117 A12

MGTAVTTRCAL_115 N8
MGTRREF_115 N7

QSFPB_RX_N1 QSFPB_RX_N2 QSFPB_RX_N3 QSFPB_RX_N4
QSFPB_RX_P1 QSFPB_RX_P2 QSFPB_RX_P3 QSFPB_RX_P4
QSFPB_TX_N1 QSFPB_TX_N2 QSFPB_TX_N3 QSFPB_TX_N4
QSFPB_TX_P1 QSFPB_TX_P2 QSFPB_TX_P3 QSFPB_TX_P4

XC7K420T-2FFG901C

GTX115_TX_N[0..2] GTX115_TX_P[0..2]
GTX115_REFCLK_N[0..1] GTX115_REFCLK_P[0..1]
GTX116_REFCLK_N[0..1] GTX116_REFCLK_P[0..1]

VCC1V2 — R214 100

Layout guidelines in UG476, fig. 5-3

SEL_SFP

BP_GTX
SHUFFLE_TX_P[1..15] SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15] SHUFFLE_RX_N[1..15]
• QUAD_TX_P[1..4] • QUAD_TX_N[1..4]
• QUAD_RX_P[1..4] • QUAD_RX_N[1..4]

FPGA_PLLCLOCKS
CLK_GTX112_A_P CLK_GTX112_A_N
CLK_GTX112_B_P CLK_GTX112_B_N
CLK_GTX114_A_P CLK_GTX114_A_N
CLK_GTX114_B_P CLK_GTX114_B_N
CLK_GTX116_A_P CLK_GTX116_A_N
CLK_GTX116_B_P CLK_GTX116_B_N
CLK_GTX115_A_P CLK_GTX115_A_N
CLK_GTX115_B_P CLK_GTX115_B_N

VCC3V3 — 4.7k R15 — 4.7k R166 DNP R194 — DGND

U40
CBTL02043A

SEL_SFP → XSD, SEL
PCI_SFP_TX_P PCI_SFP_TX_N PCI_SFP_RX_P PCI_SFP_RX_N → A0_P A0_N A1_P A1_N
B0_P B0_N B1_P B1_N ← PCIe_RX_P PCIe_RX_N PCIe_TX_P PCIe_TX_N
C0_P C0_N C1_P C1_N ← SFP_TX_P SFP_TX_N SFP_RX_P SFP_RX_N
VDD VDD VDD
GND GND GND

PCI TX/RX perspective is from the ARM core

0.1uF x4
PCIe_RX_P → C317 → PCIe_RX_C_P
PCIe_RX_N → C325 → PCIe_RX_C_N
PCIe_TX_P → C333 → PCIe_TX_C_P
PCIe_TX_N → C341 → PCIe_TX_C_N

PCIe_LANE: RX_P RX_N TX_P TX_N
PCIe

QSFP_FPGA_A
QSFPA_TX_N[1..4] QSFPA_TX_P[1..4] → TX_N[1..4] TX_P[1..4]
QSFPA_RX_N[1..4] QSFPA_RX_P[1..4] → RX_N[1..4] RX_P[1..4]

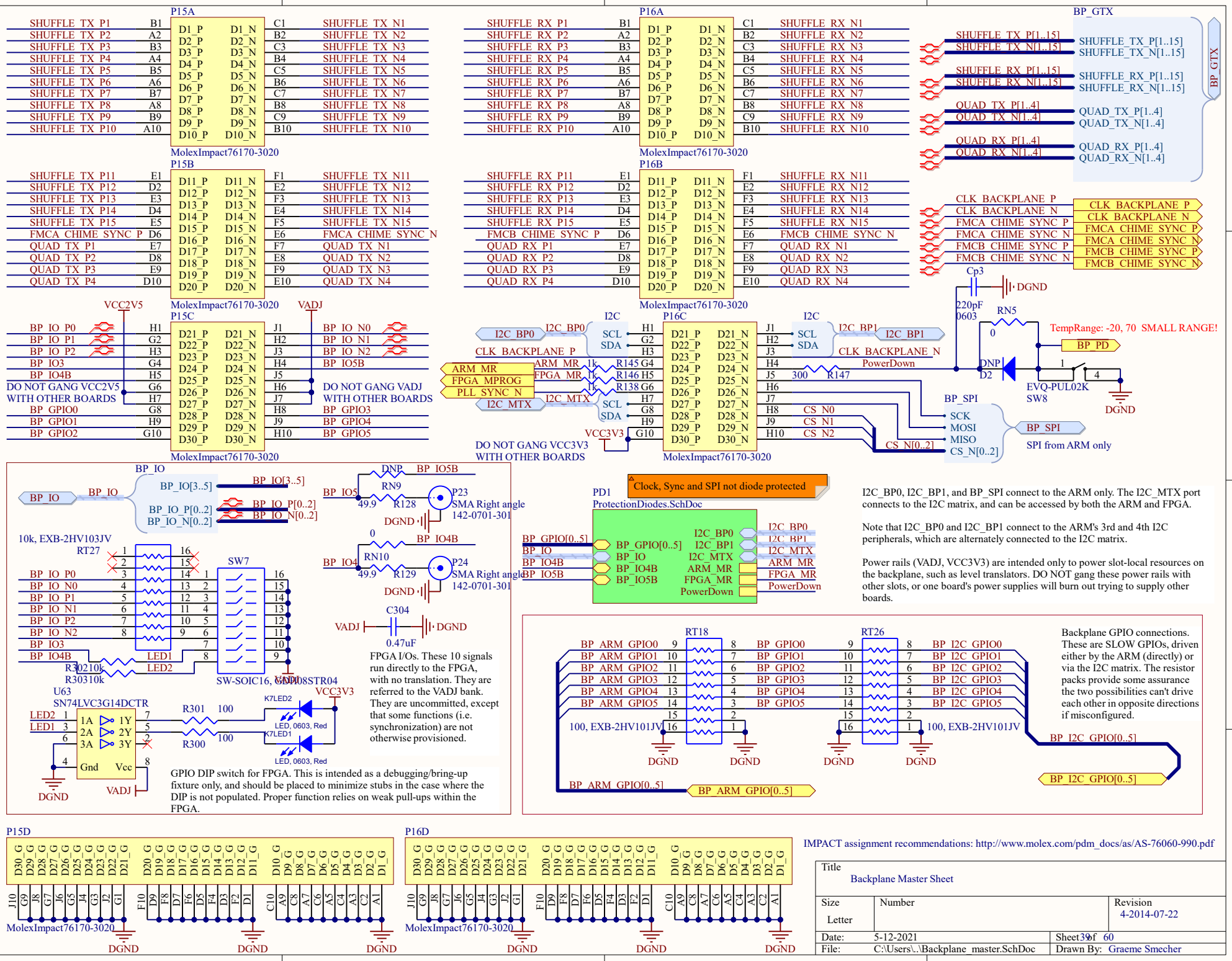
QSFP_FPGA_B
QSFPB_TX_N[1..4] QSFPB_TX_P[1..4] → TX_N[1..4] TX_P[1..4]
QSFPB_RX_N[1..4] QSFPB_RX_P[1..4] → RX_N[1..4] RX_P[1..4]

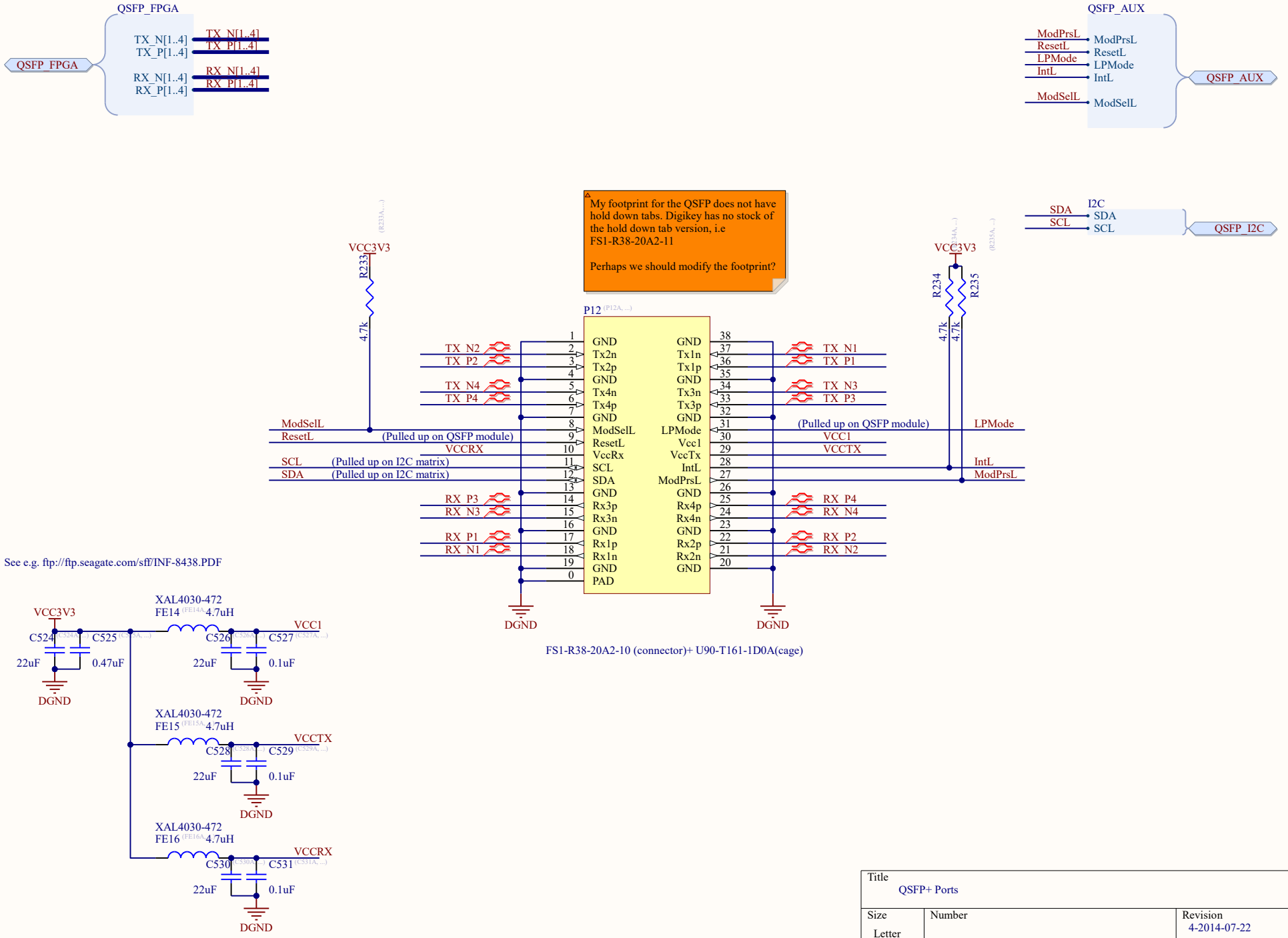
SFP_FPGA: TX_P TX_N RX_P RX_N

Selects between PCIe and SFP connections. Defaults to SFP.

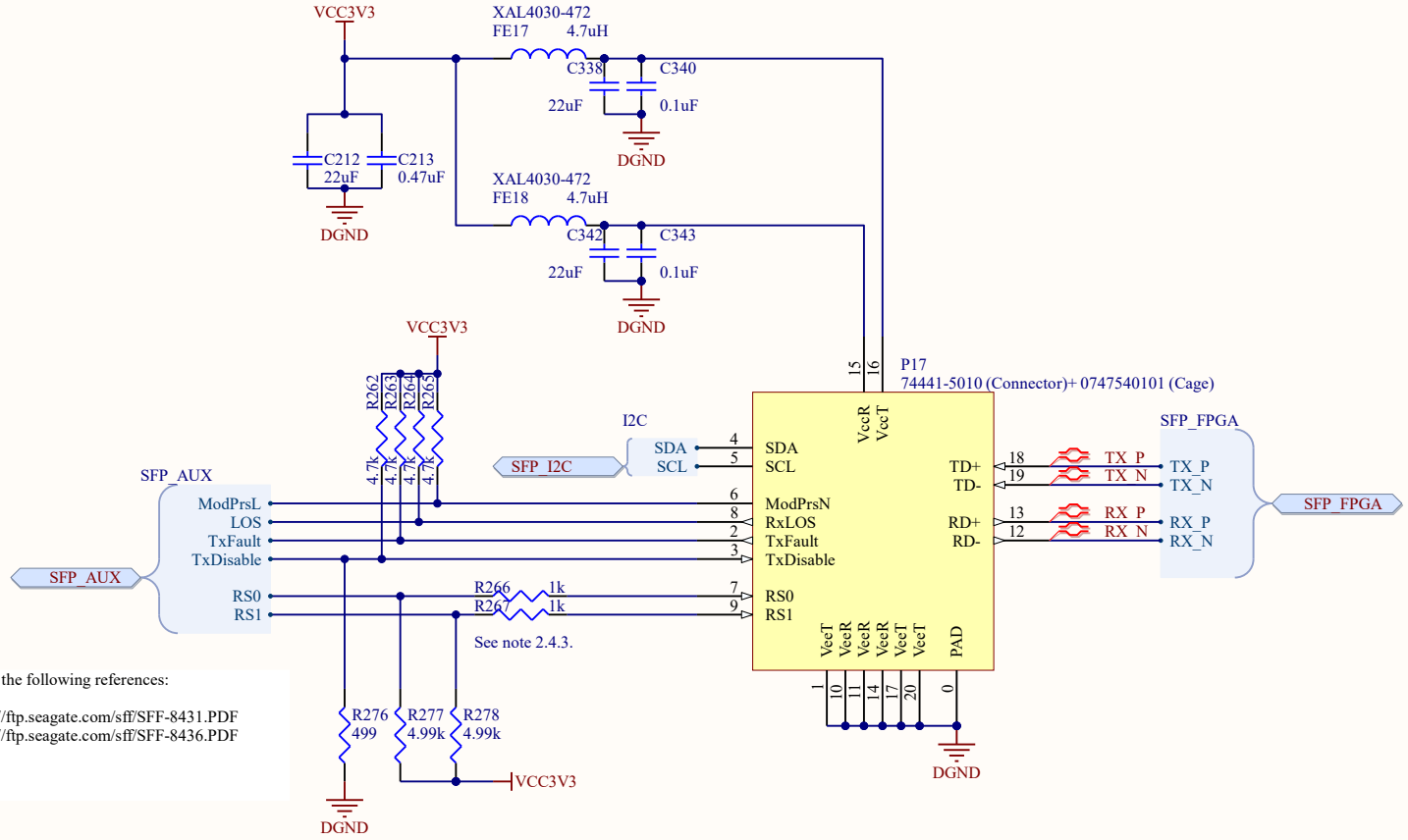
VCC3V3 — C314 47uF 6.3V T520B476M006ATE070 — C315 C316 C318 0.47uF — DGND

Title: FPGA GTX transceivers 115, 116, 117		
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File: C:\Users\..\FPGA_gtx115_gtx116_gtx117.SchDoc	Drawn By: Graeme Smecher	

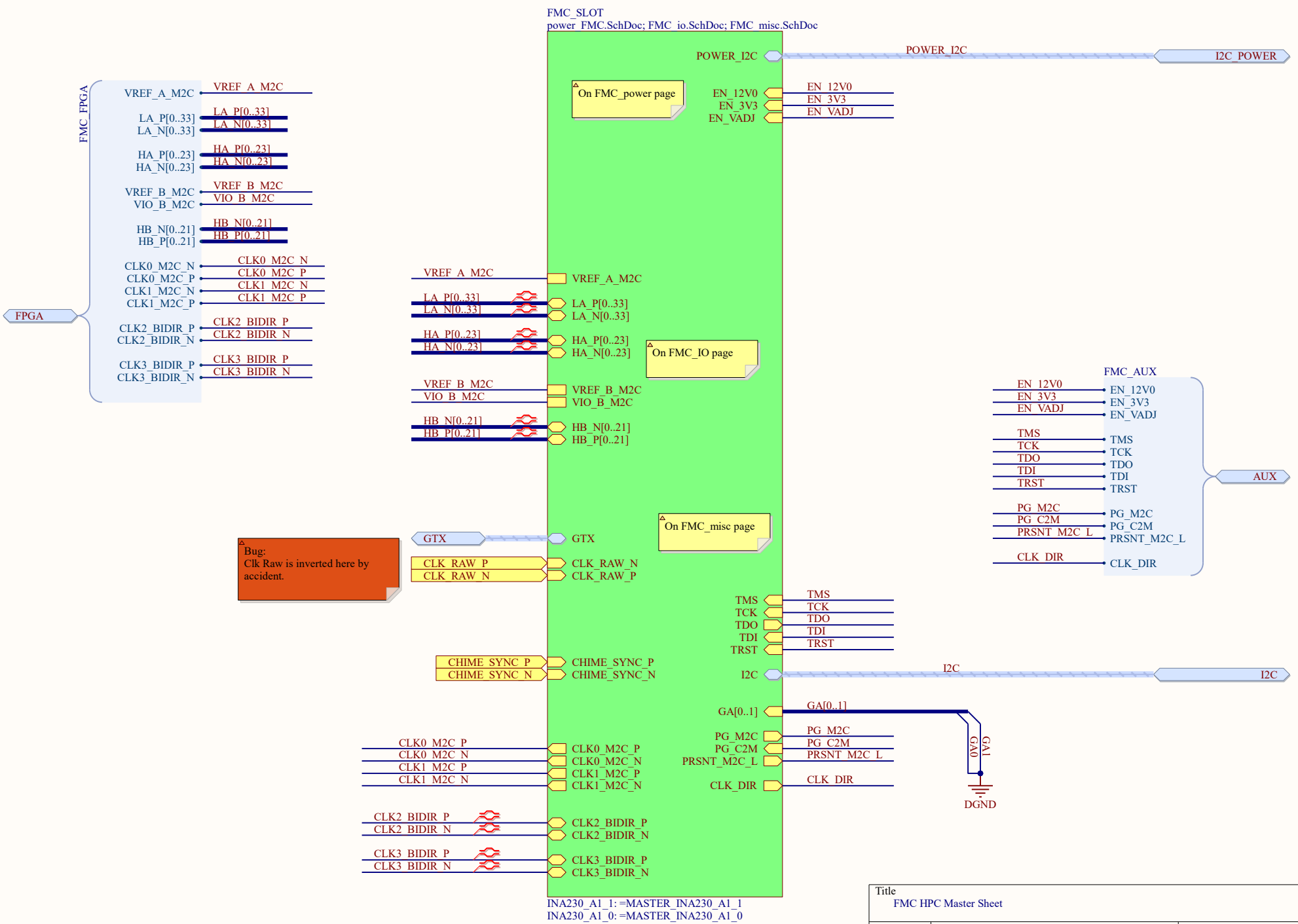

P15A
SHUFFLE_TX_P1 B1
SHUFFLE_TX_P2 A2
SHUFFLE_TX_P3 B3
SHUFFLE_TX_P4 A4
SHUFFLE_TX_P5 B5
SHUFFLE_TX_P6 A6
SHUFFLE_TX_P7 B7
SHUFFLE_TX_P8 A8
SHUFFLE_TX_P9 B9
SHUFFLE_TX_P10 A10
D1_P D1_N
D2_P D2_N
D3_P D3_N
D4_P D4_N
D5_P D5_N
D6_P D6_N
D7_P D7_N
D8_P D8_N
D9_P D9_N
D10_P D10_N
C1 SHUFFLE_TX_N1
B2 SHUFFLE_TX_N2
C3 SHUFFLE_TX_N3
B4 SHUFFLE_TX_N4
C5 SHUFFLE_TX_N5
B6 SHUFFLE_TX_N6
C7 SHUFFLE_TX_N7
B8 SHUFFLE_TX_N8
C9 SHUFFLE_TX_N9
B10 SHUFFLE_TX_N10
MolexImpact76170-3020
P16A
SHUFFLE_RX_P1 B1
SHUFFLE_RX_P2 A2
SHUFFLE_RX_P3 B3
SHUFFLE_RX_P4 A4
SHUFFLE_RX_P5 B5
SHUFFLE_RX_P6 A6
SHUFFLE_RX_P7 B7
SHUFFLE_RX_P8 A8
SHUFFLE_RX_P9 B9
SHUFFLE_RX_P10 A10
C1 SHUFFLE_RX_N1
B2 SHUFFLE_RX_N2
C3 SHUFFLE_RX_N3
B4 SHUFFLE_RX_N4
C5 SHUFFLE_RX_N5
B6 SHUFFLE_RX_N6
C7 SHUFFLE_RX_N7
B8 SHUFFLE_RX_N8
C9 SHUFFLE_RX_N9
B10 SHUFFLE_RX_N10
MolexImpact76170-3020
BP_GTX
SHUFFLE_TX_P[1..15]
SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15]
SHUFFLE_RX_N[1..15]
QUAD_TX_P[1..4]
QUAD_TX_N[1..4]
QUAD_RX_P[1..4]
QUAD_RX_N[1..4]
P15B
SHUFFLE_TX_P11 E1
SHUFFLE_TX_P12 D2
SHUFFLE_TX_P13 E3
SHUFFLE_TX_P14 D4
SHUFFLE_TX_P15 E5
FMCA_CHIME_SYNC_P D6
QUAD_TX_P1 E7
QUAD_TX_P2 D8
QUAD_TX_P3 E9
QUAD_TX_P4 D10
D11_P D11_N
D12_P D12_N
D13_P D13_N
D14_P D14_N
D15_P D15_N
D16_P D16_N
D17_P D17_N
D18_P D18_N
D19_P D19_N
D20_P D20_N
F1 SHUFFLE_TX_N11
E2 SHUFFLE_TX_N12
F3 SHUFFLE_TX_N13
E4 SHUFFLE_TX_N14
F5 SHUFFLE_TX_N15
E6 FMCA_CHIME_SYNC_N
F7 QUAD_TX_N1
E8 QUAD_TX_N2
F9 QUAD_TX_N3
E10 QUAD_TX_N4
MolexImpact76170-3020
P16B
SHUFFLE_RX_P11 E1
SHUFFLE_RX_P12 D2
SHUFFLE_RX_P13 E3
SHUFFLE_RX_P14 D4
SHUFFLE_RX_P15 E5
FMCB_CHIME_SYNC_P D6
QUAD_RX_P1 E7
QUAD_RX_P2 D8
QUAD_RX_P3 E9
QUAD_RX_P4 D10
F1 SHUFFLE_RX_N11
E2 SHUFFLE_RX_N12
F3 SHUFFLE_RX_N13
E4 SHUFFLE_RX_N14
F5 SHUFFLE_RX_N15
E6 FMCB_CHIME_SYNC_N
F7 QUAD_RX_N1
E8 QUAD_RX_N2
F9 QUAD_RX_N3
E10 QUAD_RX_N4
MolexImpact76170-3020
CLK_BACKPLANE_P
CLK_BACKPLANE_N
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
Cp3
220pF
0603
DGND
RN5
0
DNP
D2
TempRange: -20, 70 SMALL RANGE!
BP_PD
1
4
EVQ-PUL02K
SW8
DGND
P15C
VCC2V5
VADJ
BP_IO_P0 H1
BP_IO_P1 G2
BP_IO_P2 H3
BP_IO3 G4
BP_IO4B H5
DO NOT GANG VCC2V5
WITH OTHER BOARDS
G6
H7
BP_GPIO0 G8
BP_GPIO1 H9
BP_GPIO2 G10
D21_P D21_N
D22_P D22_N
D23_P D23_N
D24_P D24_N
D25_P D25_N
D26_P D26_N
D27_P D27_N
D28_P D28_N
D29_P D29_N
D30_P D30_N
J1 BP_IO_N0
H2 BP_IO_N1
J3 BP_IO_N2
H4 BP_IO5B
J5
H6 DO NOT GANG VADJ
J7 WITH OTHER BOARDS
H8 BP_GPIO3
J9 BP_GPIO4
H10 BP_GPIO5
MolexImpact76170-3020
P16C
I2C
I2C_BP0
SCL H1
SDA G2
CLK_BACKPLANE_P
ARM_MR
FPGA_MPROG
PLL_SYNC_N
I2C_MTX
SCL H7
SDA G8
H9
VCC3V3 G10
DO NOT GANG VCC3V3
WITH OTHER BOARDS
D21_P D21_N
D22_P D22_N
D23_P D23_N
D24_P D24_N
D25_P D25_N
D26_P D26_N
D27_P D27_N
D28_P D28_N
D29_P D29_N
D30_P D30_N
I2C
I2C_BP1
SCL J1
SDA H2
J3
CLK_BACKPLANE_N
H4 PowerDown
J5 300 R147
H6
J7
H8 CS_N0
J9 CS_N1
H10 CS_N2
CS_N[0..2]
MolexImpact76170-3020
BP_SPI
SCK
MOSI
MISO
CS_N[0..2]
BP_SPI
SPI from ARM only
BP_IO
BP_IO
BP_IO[3..5]
BP_IO_P[0..2]
BP_IO_N[0..2]
BP_IO[3..5]
BP_IO_P[0..2]
BP_IO_N[0..2]
10k, EXB-2HV103JV
RT27
SW7
BP_IO_P0
BP_IO_N0
BP_IO_P1
BP_IO_N1
BP_IO_P2
BP_IO_N2
BP_IO3
BP_IO4B
R30210k
R30310k
LED1
LED2
SW-SOIC16, GRID08STR04
VADJ
U63
SN74LVC3G14DCTR
LED2
LED1
1A 1Y
2A 2Y
3A 3Y
Gnd Vcc
R301 100
R300 100
K7LED2
K7LED1
LED, 0603, Red
LED, 0603, Red
VCC3V3
DGND
VADJ
BP_IO5
DNP BP_IO5B
RN9
49.9 R128
P23
SMA Right angle
142-0701-301
DGND
BP_IO4
0
BP_IO4B
RN10
49.9 R129
P24
SMA Right angle
142-0701-301
DGND
C304
VADJ
DGND
0.47uF
FPGA I/Os. These 10 signals run directly to the FPGA, with no translation. They are referred to the VADJ bank. They are uncommitted, except that some functions (i.e. synchronization) are not otherwise provisioned.
GPIO DIP switch for FPGA. This is intended as a debugging/bring-up fixture only, and should be placed to minimize stubs in the case where the DIP is not populated. Proper function relies on weak pull-ups within the FPGA.
Clock, Sync and SPI not diode protected
PD1
ProtectionDiodes.SchDoc
BP_GPIO[0..5]
BP_IO
BP_IO4B
BP_IO5B
I2C_BP0
I2C_BP1
I2C_MTX
ARM_MR
FPGA_MR
PowerDown
I2C_BP0
I2C_BP1
I2C_MTX
ARM_MR
FPGA_MR
PowerDown
I2C_BP0 and I2C_BP1 connect to the ARM only. The I2C_MTX port connects to the I2C matrix, and can be accessed by both the ARM and FPGA.
Note that I2C_BP0 and I2C_BP1 connect to the ARM's 3rd and 4th I2C peripherals, which are alternately connected to the I2C matrix.
Power rails (VADJ, VCC3V3) are intended only to power slot-local resources on the backplane, such as level translators. DO NOT gang these power rails with other slots, or one board's power supplies will burn out trying to supply other boards.
RT18
RT26
BP_ARM_GPIO0
BP_ARM_GPIO1
BP_ARM_GPIO2
BP_ARM_GPIO3
BP_ARM_GPIO4
BP_ARM_GPIO5
100, EXB-2HV101JV
BP_GPIO0
BP_GPIO1
BP_GPIO2
BP_GPIO3
BP_GPIO4
BP_GPIO5
100, EXB-2HV101JV
BP_I2C_GPIO0
BP_I2C_GPIO1
BP_I2C_GPIO2
BP_I2C_GPIO3
BP_I2C_GPIO4
BP_I2C_GPIO5
BP_I2C_GPIO[0..5]
BP_ARM_GPIO[0..5]
BP_ARM_GPIO[0..5]
DGND
Backplane GPIO connections. These are SLOW GPIOs, driven either by the ARM (directly) or via the I2C matrix. The resistor packs provide some assurance the two possibilities can't drive each other in opposite directions if misconfigured.
P15D
P16D
D30_G D29_G D28_G D27_G D26_G D25_G D24_G D23_G D22_G D21_G
D20_G D19_G D18_G D17_G D16_G D15_G D14_G D13_G D12_G D11_G
D10_G D9_G D8_G D7_G D6_G D5_G D4_G D3_G D2_G D1_G
J10 G9 J8 G7 J6 G5 J4 G3 J2 G1
F10 D9 F8 D7 F6 D5 F4 D3 F2 D1
C10 A9 C8 A7 C6 A5 C4 A3 C2 A1
MolexImpact76170-3020
DGND
IMPACT assignment recommendations: http://www.molex.com/pdm_docs/as/AS-76060-990.pdf
Title
Backplane Master Sheet
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
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File: C:\Users\..\Backplane_master.SchDoc
Drawn By: Graeme Smecher



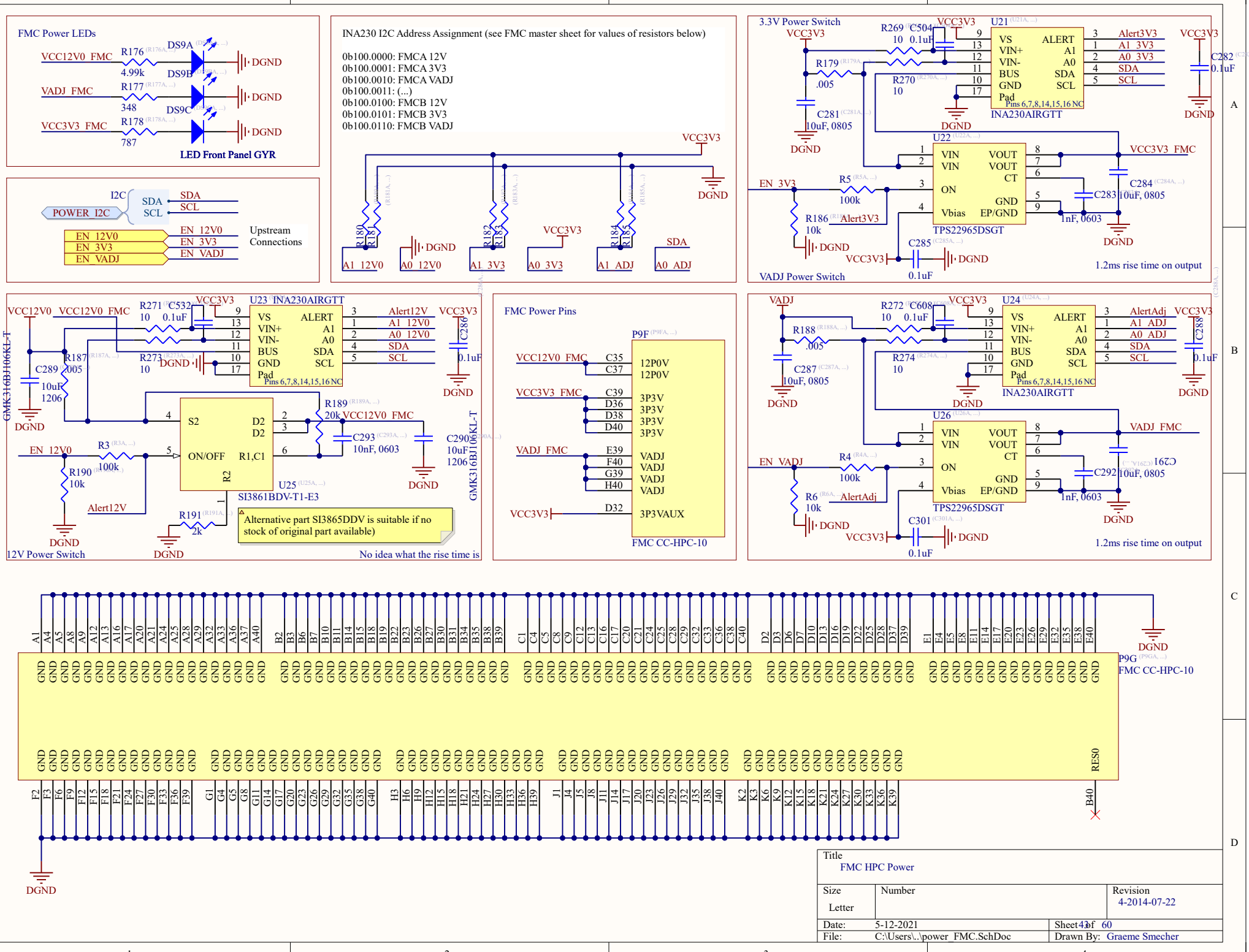
Title: QSFP+ Ports		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 40 of 60	
File: C:\Users\..\QSFP.SchDoc	Drawn By: Graeme Smecher	



Title: SFP+ Port		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 4 bf 60	
File: C:\Users\..\SFP.SchDoc	Drawn By: Graeme Smecher	



Title: FMC HPC Master Sheet		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 42 of 60
File: C:\Users\..\FMC_master.SchDoc		Drawn By: Graeme Smecher


FMC Power LEDs
VCC12V0_FMC
R176
4.99k
DS9A
DS9B
DGND
VADJ_FMC
R177
348
DS9C
VCC3V3_FMC
R178
787
LED Front Panel GYR
I2C
POWER_I2C
SDA
SCL
EN_12V0
EN_3V3
EN_VADJ
Upstream Connections
INA230 I2C Address Assignment (see FMC master sheet for values of resistors below)
0b100.0000: FMCA 12V
0b100.0001: FMCA 3V3
0b100.0010: FMCA VADJ
0b100.0011: (...)
0b100.0100: FMCB 12V
0b100.0101: FMCB 3V3
0b100.0110: FMCB VADJ
VCC3V3
DGND
A1_12V0
A0_12V0
A1_3V3
A0_3V3
A1_ADJ
A0_ADJ
3.3V Power Switch
R269
10
C504
0.1uF
U21
INA230AIRGTT
R179
.005
C281
10uF, 0805
R270
10
Alert3V3
A1_3V3
A0_3V3
C282
0.1uF
U22
TPS22965DSGT
VCC3V3_FMC
EN_3V3
R5
100k
R186
10k
C284
C283
10uF, 0805
1nF, 0603
C285
0.1uF
1.2ms rise time on output
VADJ Power Switch
VCC12V0_FMC
R271
10
C532
0.1uF
U23
INA230AIRGTT
Alert12V
A1_12V0
A0_12V0
C286
0.1uF
C289
10uF
1206
R187
.005
R273
10
R189
20k
C293
10nF, 0603
C290
10uF
1206
GMK316BJ106KL-T
R3
100k
EN_12V0
R190
10k
Alert12V
R191
2k
U25
SI3861BDV-T1-E3
Alternative part SI3865DDV is suitable if no stock of original part available)
12V Power Switch
No idea what the rise time is
FMC Power Pins
P9F
VCC12V0_FMC
C35
C37
12P0V
VCC3V3_FMC
C39
D36
D38
D40
3P3V
VADJ_FMC
E39
F40
G39
H40
VADJ
D32
3P3VAUX
FMC CC-HPC-10
VADJ
R272
10
C608
0.1uF
U24
INA230AIRGTT
R188
.005
C287
10uF, 0805
R274
10
AlertAdj
A1_ADJ
A0_ADJ
C288
0.1uF
U26
TPS22965DSGT
VADJ_FMC
EN_VADJ
R4
100k
R6
10k
C291
C292
10uF, 0805
1nF, 0603
C301
0.1uF
1.2ms rise time on output
P9G
FMC CC-HPC-10
B40
RES0
Title
FMC HPC Power
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 43 of 60
File:
C:\Users\..\power_FMC.SchDoc
Drawn By:
Graeme Smecher

FMC ERRATA: HA18 is also clock-capable, and should be labeled as such

VREF_A_M2C
VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
LA_P[0..33]
LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
HA_P[0..23]
HA_N[0..23]
VIO for port A is VADJ
VREF_B_M2C
VIO_B_M2C
VREF_B_M2C
VIO_B_M2C
HB_P[0..21]
HB_N[0..21]
HB_P[0..21]
HB_N[0..21]
FMC I/O connectivity

P9A (P9AA, ...)

HA_P0	F4	HA00_P_CC	HA12_P	F13	HA_P12
HA_N0	F5	HA00_N_CC	HA12_N	F14	HA_N12
HA_P1	E2	HA01_P_CC	HA13_P	E12	HA_P13
HA_N1	E3	HA01_N_CC	HA13_N	E13	HA_N13
HA_P2	K7	HA02_P	HA14_P	J15	HA_P14
HA_N2	K8	HA02_N	HA14_N	J16	HA_N14
HA_P3	J6	HA03_P	HA15_P	F16	HA_P15
HA_N3	J7	HA03_N	HA15_N	F17	HA_N15
HA_P4	F7	HA04_P	HA16_P	E15	HA_P16
HA_N4	F8	HA04_N	HA16_N	E16	HA_N16
HA_P5	E6	HA05_P	HA17_P_CC	K16	HA_P17
HA_N5	E7	HA05_N	HA17_N_CC	K17	HA_N17
HA_P6	K10	HA06_P	HA18_P_CC	J18	HA_P18
HA_N6	K11	HA06_N	HA18_N_CC	J19	HA_N18
HA_P7	J9	HA07_P	HA19_P	F19	HA_P19
HA_N7	J10	HA07_N	HA19_N	F20	HA_N19
HA_P8	F10	HA08_P	HA20_P	E18	HA_P20
HA_N8	F11	HA08_N	HA20_N	E19	HA_N20
HA_P9	E9	HA09_P	HA21_P	K19	HA_P21
HA_N9	E10	HA09_N	HA21_N	K20	HA_N21
HA_P10	K13	HA10_P	HA22_P	J21	HA_P22
HA_N10	K14	HA10_N	HA22_N	J22	HA_N22
HA_P11	J12	HA11_P	HA23_P	K22	HA_P23
HA_N11	J13	HA11_N	HA23_N	K23	HA_N23

FMC CC-HPC-10

P9C (P9CA, ...)

VREF_A_M2C	H1	VREF_A_M2C			
LA_P0	G6	LA00_P_CC	LA17_P_CC	D20	LA_P17
LA_N0	G7	LA00_N_CC	LA17_N_CC	D21	LA_N17
LA_P1	D8	LA01_P_CC	LA18_P_CC	C22	LA_P18
LA_N1	D9	LA01_N_CC	LA18_N_CC	C23	LA_N18
LA_P2	H7	LA02_P	LA19_P	H22	LA_P19
LA_N2	H8	LA02_N	LA19_N	H23	LA_N19
LA_P3	G9	LA03_P	LA20_P	G21	LA_P20
LA_N3	G10	LA03_N	LA20_N	G22	LA_N20
LA_P4	H10	LA04_P	LA21_P	H25	LA_P21
LA_N4	H11	LA04_N	LA21_N	H26	LA_N21
LA_P5	D11	LA05_P	LA22_P	G24	LA_P22
LA_N5	D12	LA05_N	LA22_N	G25	LA_N22
LA_P6	C10	LA06_P	LA23_P	D23	LA_P23
LA_N6	C11	LA06_N	LA23_N	D24	LA_N23
LA_P7	H13	LA07_P	LA24_P	H28	LA_P24
LA_N7	H14	LA07_N	LA24_N	H29	LA_N24
LA8_MUX_P	G12	LA08_P	LA25_P	G27	LA_P25
LA8_MUX_N	G13	LA08_N	LA25_N	G28	LA_N25
LA_P9	D14	LA09_P	LA26_P	D26	LA_P26
LA_N9	D15	LA09_N	LA26_N	D27	LA_N26
LA_P10	C14	LA10_P	LA27_P	C26	LA_P27
LA_N10	C15	LA10_N	LA27_N	C27	LA_N27
LA_P11	H16	LA11_P	LA28_P	H31	LA_P28
LA_N11	H17	LA11_N	LA28_N	H32	LA_N28
LA_P12	G15	LA12_P	LA29_P	G30	LA_P29
LA_N12	G16	LA12_N	LA29_N	G31	LA_N29
LA_P13	D17	LA13_P	LA30_P	H34	LA_P30
LA_N13	D18	LA13_N	LA30_N	H35	LA_N30
LA_P14	C18	LA14_P	LA31_P	G33	LA_P31
LA_N14	C19	LA14_N	LA31_N	G34	LA_N31
LA_P15	H19	LA15_P	LA32_P	H37	LA_P32
LA_N15	H20	LA15_N	LA32_N	H38	LA_N32
LA_P16	G18	LA16_P	LA33_P	G36	LA_P33
LA_N16	G19	LA16_N	LA33_N	G37	LA_N33

FMC CC-HPC-10

LA_P8
LA_N8
R231 (R231A, ...) 0
R232 (R232A, ...) 0
R241 (R241A, ...) DNP
R236 (R236A, ...) DNP
CHIME_SYNC_P
CHIME_SYNC_N

CHIME-SPECIFIC:
LA08_[PN] are used for sync pulses from the backplane to the FMCs. This network permits the signal to be routed directly, bypassing the FPGA.

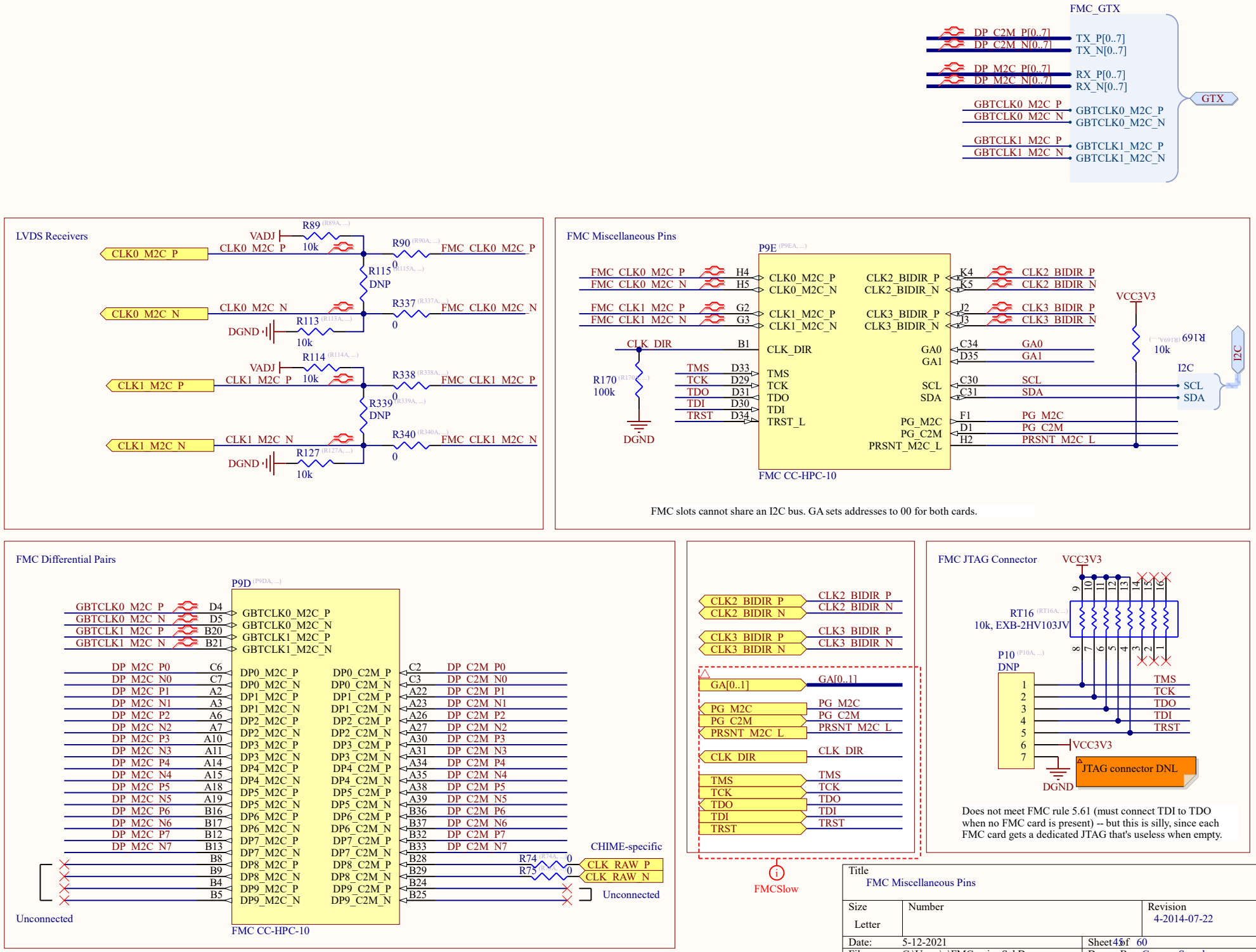
Placement: try and minimize stubs for both cases.

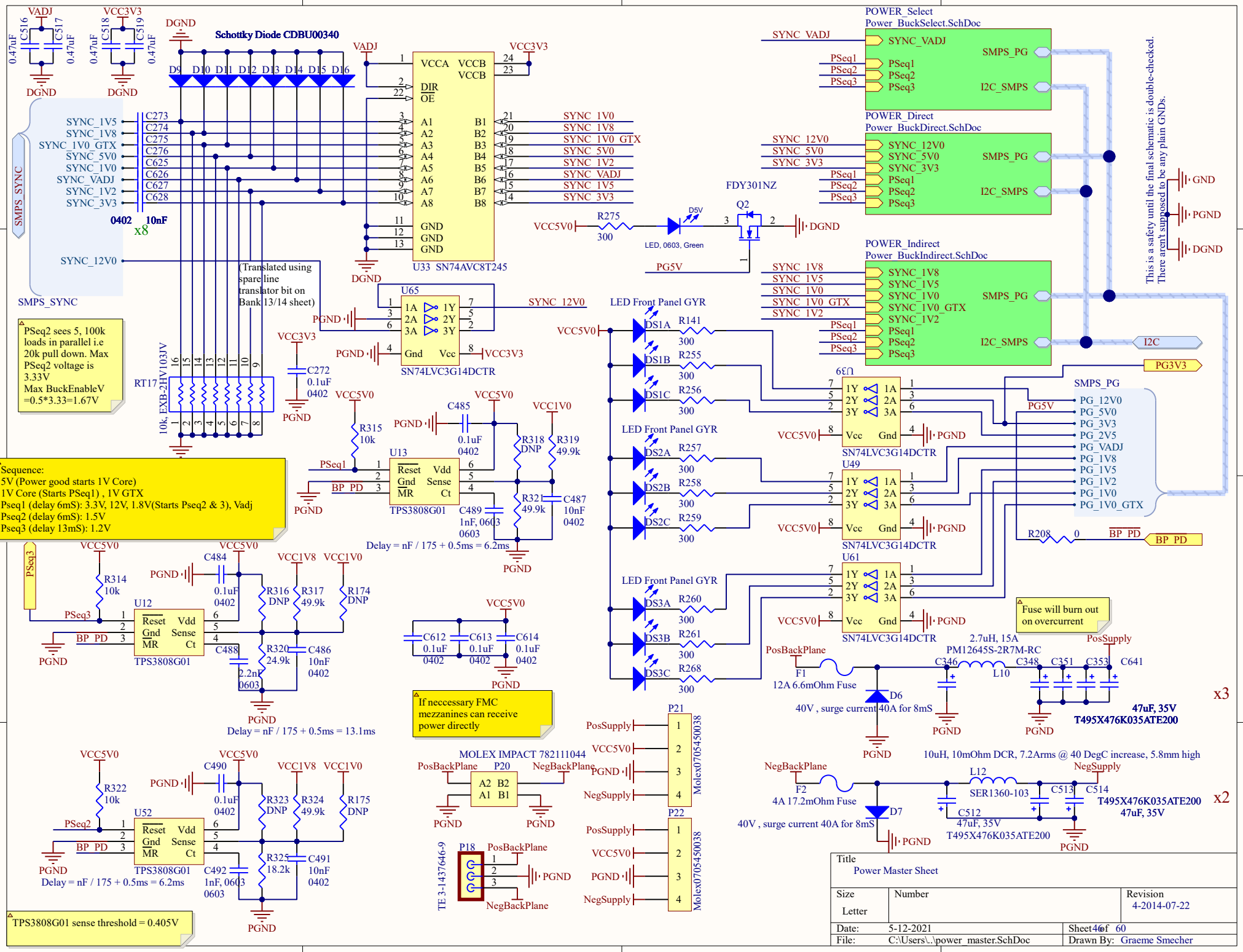
P9B (P9BA, ...)

VREF_B_M2C	K1	VREF_B_M2C	VIO_B_M2C	J39	VIO_B_M2C
			VIO_B_M2C	K40	
HB_P0	K25	HB00_P_CC	HB11_P	J30	HB_P11
HB_N0	K26	HB00_N_CC	HB11_N	J31	HB_N11
HB_P1	J24	HB01_P	HB12_P	F31	HB_P12
HB_N1	J25	HB01_N	HB12_N	F32	HB_N12
HB_P2	F22	HB02_P	HB13_P	E30	HB_P13
HB_N2	F23	HB02_N	HB13_N	E31	HB_N13
HB_P3	E21	HB03_P	HB14_P	K34	HB_P14
HB_N3	E22	HB03_N	HB14_N	K35	HB_N14
HB_P4	F25	HB04_P	HB15_P	J33	HB_P15
HB_N4	F26	HB04_N	HB15_N	J34	HB_N15
HB_P5	E24	HB05_P	HB16_P	F34	HB_P16
HB_N5	E25	HB05_N	HB16_N	F35	HB_N16
HB_P6	K28	HB06_P_CC	HB17_P_CC	K37	HB_P17
HB_N6	K29	HB06_N_CC	HB17_N_CC	K38	HB_N17
HB_P7	J27	HB07_P	HB18_P	J36	HB_P18
HB_N7	J28	HB07_N	HB18_N	J37	HB_N18
HB_P8	F28	HB08_P	HB19_P	E33	HB_P19
HB_N8	F29	HB08_N	HB19_N	E34	HB_N19
HB_P9	E27	HB09_P	HB20_P	F37	HB_P20
HB_N9	E28	HB09_N	HB20_N	F38	HB_N20
HB_P10	K31	HB10_P	HB21_P	E36	HB_P21
HB_N10	K32	HB10_N	HB21_N	E37	HB_N21

FMC CC-HPC-10

Title: FMC HPC I/O Pins		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 44 of 60
File: C:\Users\..\FMC_io.SchDoc		Drawn By: Graeme Smecher


FMC_GTX
DP_C2M_P[0..7]
DP_C2M_N[0..7]
TX_P[0..7]
TX_N[0..7]
DP_M2C_P[0..7]
DP_M2C_N[0..7]
RX_P[0..7]
RX_N[0..7]
GBTCLK0_M2C_P
GBTCLK0_M2C_N
GBTCLK0_M2C_P
GBTCLK0_M2C_N
GBTCLK1_M2C_P
GBTCLK1_M2C_N
GBTCLK1_M2C_P
GBTCLK1_M2C_N
GTX
LVDS Receivers
CLK0_M2C_P
CLK0_M2C_N
CLK1_M2C_P
CLK1_M2C_N
VADJ
10k
R89
R90
R115
DNP
R337
R113
R114
R338
R339
R340
R127
0
FMC_CLK0_M2C_P
FMC_CLK0_M2C_N
FMC_CLK1_M2C_P
FMC_CLK1_M2C_N
DGND
FMC Miscellaneous Pins
P9E
FMC_CLK0_M2C_P
FMC_CLK0_M2C_N
FMC_CLK1_M2C_P
FMC_CLK1_M2C_N
CLK_DIR
TMS
TCK
TDO
TDI
TRST
R170
100k
CLK0_M2C_P
CLK0_M2C_N
CLK1_M2C_P
CLK1_M2C_N
CLK_DIR
TMS
TCK
TDO
TDI
TRST_L
CLK2_BIDIR_P
CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
GA0
GA1
SCL
SDA
PG_M2C
PG_C2M
PRSNT_M2C_L
VCC3V3
R169
I2C
FMC CC-HPC-10
FMC slots cannot share an I2C bus. GA sets addresses to 00 for both cards.
FMC Differential Pairs
P9D
GBTCLK0_M2C_P
GBTCLK0_M2C_N
GBTCLK1_M2C_P
GBTCLK1_M2C_N
DP_M2C_P0
DP_M2C_N0
DP_M2C_P1
DP_M2C_N1
DP_M2C_P2
DP_M2C_N2
DP_M2C_P3
DP_M2C_N3
DP_M2C_P4
DP_M2C_N4
DP_M2C_P5
DP_M2C_N5
DP_M2C_P6
DP_M2C_N6
DP_M2C_P7
DP_M2C_N7
DP_C2M_P0
DP_C2M_N0
DP_C2M_P1
DP_C2M_N1
DP_C2M_P2
DP_C2M_N2
DP_C2M_P3
DP_C2M_N3
DP_C2M_P4
DP_C2M_N4
DP_C2M_P5
DP_C2M_N5
DP_C2M_P6
DP_C2M_N6
DP_C2M_P7
DP_C2M_N7
CHIME-specific
CLK_RAW_P
CLK_RAW_N
R74
R75
Unconnected
Unconnected
FMC CC-HPC-10
CLK2_BIDIR_P
CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
GA[0..1]
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR
TMS
TCK
TDO
TDI
TRST
FMCSlow
FMC JTAG Connector
VCC3V3
RT16
10k, EXB-2HV103JV
P10
DNP
JTAG connector DNL
DGND
Does not meet FMC rule 5.61 (must connect TDI to TDO when no FMC card is present) -- but this is silly, since each FMC card gets a dedicated JTAG that's useless when empty.
Title
FMC Miscellaneous Pins
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 45 of 60
File:
C:\Users\..\FMC_misc.SchDoc
Drawn By:
Graeme Smecher


Schottky Diode CDBU00340
U33 SN74AVC8T245
SMPS_SYNC
0402 10nF x8
(Translated using spare line translator bit on Bank 13/14 sheet)
U65
SN74LVC3G14DCTR
RT17
10k, EXB-2HV103V
PSeq2 sees 5, 100k loads in parallel i.e 20k pull down. Max PSeq2 voltage is 3.33V
Max BuckEnableV =0.5*3.33=1.67V
Sequence:
5V (Power good starts 1V Core)
1V Core (Starts PSeq1) , 1V GTX
Pseq1 (delay 6mS): 3.3V, 12V, 1.8V(Starts Pseq2 & 3), Vadj
Pseq2 (delay 6mS): 1.5V
Pseq3 (delay 13mS): 1.2V
U13
TPS3808G01
Delay = nF / 175 + 0.5ms = 6.2ms
U12
Delay = nF / 175 + 0.5ms = 13.1ms
U52
Delay = nF / 175 + 0.5ms = 6.2ms
TPS3808G01 sense threshold = 0.405V
If neccessary FMC mezzanines can receive power directly
MOLEX IMPACT 782111044
TE 3-1437646-9
Molex0705450038
FDY301NZ
LED, 0603, Green
POWER_Select
Power_BuckSelect.SchDoc
POWER_Direct
Power_BuckDirect.SchDoc
POWER_Indirect
Power_BuckIndirect.SchDoc
LED Front Panel GYR
SN74LVC3G14DCTR
Fuse will burn out on overcurrent
2.7uH, 15A PM12645S-2R7M-RC
F1 12A 6.6mOhm Fuse
D6 40V , surge current 40A for 8mS
47uF, 35V T495X476K035ATE200
10uH, 10mOhm DCR, 7.2Arms @ 40 DegC increase, 5.8mm high
F2 4A 17.2mOhm Fuse
D7 40V , surge current 40A for 8mS
SER1360-103
This is a safety until the final schematic is double-checked. There aren't supposed to be any plain GNDs.
Title
Power Master Sheet
Size Letter
Number
Revision 4-2014-07-22
Date: 5-12-2021
File: C:\Users\..\power_master.SchDoc
Sheet 46 of 60
Drawn By: Graeme Smecher

Vo=Vref x (1+RfbA/RfbB)

On this page we have all the buck regulators that come from the main supply voltage

Note that the 5V buck regulator is used to generate: two of 1V , 1.2V, 1.5V & 1.8V
These buck converts are seen on the "Power_BuckIndirect" page

Total capacitance 900uF

PSeq1 R163 0
PSeq2 R171 DNP
PSeq3 R131 DNP
PosSupply 1.8A @ 16V in
4x C446 C447 C448 C449 10uF 1206 GMK316BJ106KL-T
PGND
VCC5V0
INA230_I2C_3V3

B1 BuckRegulator.SchDoc
DeviceType: IR3895
SYNC_3V3
BuckSync
Vout
Enable
MasterVo
PVin
Vin
SwitchNode
VCC/LDO
INA230_I2C
PG
IR3895

8A
93%
L1 SER1360-182
PG_3V3
VCC3V3
C444 C220 C221
3x 100uF 10V, 1206 C3216X5R1A107M160AC
T520Y337M010ATE035 330uF C644
PGND

1.8W Loss in IC

1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high

Vin=16V, 93% @ 4A, 93% @ 8A, LIR=19%,
4A step ->4mV deviation, 1mV ripple
0.2W core+winding loss in inductor at 8A

RenA: CRCW060349K9FKEA - 50k
RenB: CRCW060349K9FKEA - 50k
RfbA: CRCW060320K0FKEA - 20k
RfbB: CRCW06033K57FKEA - 3.57k
RcompB: CRCW060326K7FKEA - 26.7k
CcompB: GRM188R71E182KA01D - 1.8nF

RcA1: ERJ-3EKF2260V - 226
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: CRCW060320K0FKEA - 20k
RsenseB: CRCW06033K57FKEA - 3.57k
CcompC: GRM1885C1H120FA01J - 12pF

VCC3V3
C452 100uF 10V, 1206 C3216X5R1A107M160AC
R11 10k
U54
Vin Vout
EN Adj
Gnd EP/GND
LP38512TJ-ADJ/NOPB
R12 4.02k
C453 1nF, 0603
R13 1k
VCC2V5
R78 DNL
PG_2V5
C223 100uF 10V, 1206 C3216X5R1A107M160AC
PGND

Vout=0.5*(1+Rtop/Rbot)

Decided that it was advantagous space wise to use the LDO instead of a buck regulator

VCC5V0
RN6 20k
PG_12V0
PG_5V0
PG_3V3
PG_2V5
SMPS_PG
PG_12V0
PG_5V0
PG_3V3
PG_2V5
PG_VADJ
PG_1V8
PG_1V5
PG_1V2
PG_1V0
PG_1V0_GTX
SMPS_PG

Total capacitance 70uF

PSeq1 R132 0
PSeq2 R149 DNP
PSeq3 R150 DNP
PosSupply 2.3A @ 16V in
2x C637 C457 10uF 1206 GMK316BJ106KL-T
PGND
VCC5V0
INA230_I2C_12V0

B3 BuckRegulator.SchDoc
DeviceType: IR3894
SYNC_12V0
BuckSync
Vout
Enable
MasterVo
PVin
Vin
SwitchNode
VCC/LDO
INA230_I2C
PG
IR3894

3A
98%
L3 SER1360-602
PG_12V0
VCC12V0 1.8mm high
5x C456 C458 C225 C226 C227 10uF 1206 GMK316BJ106KL-T
PGND

6.0uH, 5.5mOhm DCR, 9.4Arms @ 40 DegC increase, 5.8mm high

RenA: CRCW060349K9FKEA - 50k
RenB: CRCW060349K9FKEA - 50k
RfbA: CRCW060316K5FKEA - 16.5k
RfbB: CRCW0603715RFKEA - 715
RcompB: CRCW06036K34FKEA - 6.34k
CcompB: C0603C103K5RACTU - 10nF

RcA1: CRCW0603127RFKEA - 127
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: CRCW060316K5FKEA - 16.5k
RsenseB: CRCW0603715RFKEA - 715
CcompC: C0603C470K5GACTU - 47pF

Vin=16V, 97% @ 4A, 98% @ 8A, LIR=17%,
1.5A step ->27mV deviation, 0.9mV ripple
0.10W core+winding loss in inductor at 3A

0.7W Loss in IC

Actually 5.5V - linear regulated down for USB

Total capacitance 1700uF

2.8W Loss in IC

PSeq1 R163 0
PSeq2 R171 DNP
PSeq3 R131 DNP
PosSupply 3.6A @ 16V in
6x C461 C462 C463 C464 C465 C466 10uF 1206 GMK316BJ106KL-T
PGND
INA230_I2C_5V0

B4 BuckRegulator.SchDoc
DeviceType: IR3895
SYNC_5V0
BuckSync
Vout
Enable
MasterVo
PVin
Vin
SwitchNode
VCC/LDO
INA230_I2C
PG
IR3895

11A
94.5%
L4 SER1360-182
PG_5V0
VCC5V0
5X C228 C598 C599 C600 C601 100uF 1210 C3216X5R1A107M160AC
2X C642 C643 T520Y337M010ATE035 330uF
PGND

1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high

RenA: CRCW060349K9FKEA - 50k
RenB: CRCW060310K0FKEA - 10k
RfbA: CRCW060327K4FKEA - 27.4k
RfbB: CRCW06032K74FKEA - 2.74k
RcompB: CRCW060347R5FKEA - 47.5k
CcompB: C0603C122K5RACTU - 1.2nF

RcA1: CRCW0603383RFKEA - 383
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: CRCW060327K4FKEA - 27.4k
RsenseB: CRCW06032K74FKEA - 2.74k
CcompC: C0603C689K5GACTU - 6.8pF

Vin=16V, 94% @ 5.5A, 94% @ 11A, LIR=18%,
5.5A step ->13mV deviation, <2mVp-p ripple
0.40W core+winding loss in inductor at 11A

All switchers designed with aid of "SupIRBuck Design Tool"
http://ir.transim.com/mypower/Pages/DesignReq.aspx

Inductors chosen with http://www.coilcraft.com/apps/selector/selector_1.cfm tool

5V output buck has 7.2V UVLO on supply voltage

3.3, 2.5 and 12V rails have 4V UVLO on 5V rail

VCC3V3
12v: 0b1000111 INA230_I2C_12V0.A1
5v: 0b1001000 INA230_I2C_5V0.A1 SDA
3.3v: 0b1001001 INA230_I2C_3V3.A1 SDA
INA230_I2C_12V0.A0 SCL
INA230_I2C_5V0.A0
INA230_I2C_3V3.A0
VCC3V3
DGND

SYNC_3V3 SYNC_3V3
SYNC_12V0 SYNC_12V0
SYNC_5V0 SYNC_5V0

PSeq1 PSeq1
PSeq2 PSeq2
PSeq3 PSeq3

I2C
INA230_I2C_12V0.SDA
INA230_I2C_5V0.SDA
INA230_I2C_3V3.SDA
SDA
SDA
INA230_I2C_12V0.SCL
INA230_I2C_5V0.SCL
INA230_I2C_3V3.SCL
SCL
SCL
I2C_SMPS

Title	Power: Direct Converters		
Size	Number		Revision
Letter			4-2014-07-22
Date:	5-12-2021	Sheet 47 of 60	
File:	C:\Users\..\Power_BuckDirect.SchDoc	Drawn By:	Adam Gilbert

Only 1% more efficient if run at 750kHz

LIR = Ip-p / I avg

PSeq1 PSeq2 PSeq3

SMPS_PG: PG_12V0, PG_5V0, PG_3V3, PG_2V5, PG_VADJ, PG_1V8, PG_1V5, PG_1V2, PG_1V0, PG_1V0_GTX

B5 BuckRegulator.SchDoc — DeviceType: IR3895

2.7W Loss in IC | VCC5V0 | 3.7A | 2x C445 100uF 1206 | PGND

SYNC_1V0_GTX | GTXSeq | INA230_I2C_1V0_GTX

BuckSync, Vout, Enable, MasterVo, PVin, Vin, SwitchNode, VCC/LDO, INA230_I2C, PG — IR3895

16A | 87% | L5 SER1360-331 | PG_1V0_GTX | VCC1V0_GTX

0.33uH, 0.85mOhm DCR, 36Arms @ 40 DegC increase, 5.8mm high

C230			RcA1: ERJ-3EKF2260V - 226
C231	RenA: CRCW060349K9FKEA - 50k		RseqA: DNP
7x C469	RenB: CRCW060349K9FKEA - 50k		RseqB: DNP
C443	RfbA: CRCW06038K45FKEA - 8.45k		Cseq: DNP
100uF C450	RfbB: CRCW06037K68FKEA - 7.68k		RsenseA: CRCW06038K45FKEA - 8.45k
1206 C455	RcompB: CRCW060314K0FKEA - 14k		RsenseB: CRCW06037K68FKEA - 7.68k
C459	CcompB: C0603C122K5RACTU - 1.2nF		CcompC: C0603C220K4GACTU - 22pF

Vin=5.5V, 88% @ 8A, 85% @ 16A, LIR=16%, 8A step ->14mV deviation, <2mV ripple, 0.2W core+winding loss in inductor at 16A

Total capacitance 1100uF

B6 BuckRegulator.SchDoc — DeviceType: IR3894

1.2W Loss in IC | VCC5V0 | 2.2A | 2x C471 100uF 1206 | C473 | PGND

SYNC_1V2 | PSeq3 | INA230_I2C_1V2

BuckSync, Vout, Enable, MasterVo, PVin, Vin, SwitchNode, VCC/LDO, INA230_I2C, PG — IR3895

8A | 89% | L6 SER1360-651 | PG_1V2 | VCC1V2

0.65uH, 0.85mOhm DCR, 17Arms @ 40 DegC increase, 5.8mm high

	RenA: CRCW060349K9FKEA - 50k		RcA1: CRCW0603127RFKEA - 127
C460	RenB: CRCW060349K9FKEA - 50k		RseqA: DNP
6x C467	RfbA: CRCW060315K0FKEA - 15k		RseqB: DNP
100uF C470	RfbB: CRCW060311K0FKEA - 11k		Cseq: DNP
1206 C472	RcompB: CRCW060315K8FKEA - 15.8k		RsenseA: CRCW060315K0FKEA - 15k
C474	CcompB: GRM188R71E392KA01D - 3.9nF		RsenseB: CRCW060311K0FKEA - 11k
			CcompC: C0603C220K4GACTU - 22pF

Vin=5.5V, 88% @ 8A, 91% @ 4A, LIR=19%, 4A step ->8mV deviation, 0.5mV ripple, 0.1W core+winding loss in inductor at 8A

Total capacitance 1100uF

B7 BuckRegulator.SchDoc — DeviceType: IR3894

0.4W Loss in IC | VCC5V0 | 1A | 2x C638 100uF 1206 C479 | PGND

SYNC_1V5 | PSeq2 | INA230_I2C_1V5

BuckSync, Vout, Enable, MasterVo, PVin, Vin, SwitchNode, VCC/LDO, INA230_I2C, PG — IR3894

3A | 93% | L7 SER1360-182 | PG_1V5 | VCC1V5

1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high

	RenA: CRCW060349K9FKEA - 50k		RcA1: CRCW0603127RFKEA - 127
C476	RenB: CRCW060349K9FKEA - 50k		RseqA: DNP
3x C477	RfbA: CRCW060319K1FKEA - 19.1k		RseqB: DNP
C478	RfbB: CRCW06039K76FKEA - 9.76k		Cseq: DNP
100uF	RcompB: CRCW060325K5FKEA - 25.5k		RsenseA: CRCW060319K1FKEA - 19.1k
1206 C631 DNL:	CcompB: GRM188R71E332KA01D - 3.3nF		RsenseB: CRCW06039K76FKEA - 9.76k
C652			CcompC: GRM1885C1H120FA01J - 12pF

Vin=5.5V, 92%@ 3A, 92% @ 1.5A, LIR=20%, 1.5A step ->6mV deviation, 0.4mV ripple, No inductor losses

Total capacitance 580uF

B8 BuckRegulator.SchDoc — DeviceType: IR3895

2.7W Loss in IC | VCC5V0 | 3.7A | 2x C639 100uF 1206 C481 | PGND

SYNC_1V0 | PG_5V0 | INA230_I2C_1V0

BuckSync, Vout, Enable, MasterVo, PVin, Vin, SwitchNode, VCC/LDO, INA230_I2C, PG — IR3894

16A | 87% | L8 SER1360-331 | PG_1V0 | VCC1V0

0.33uH, 0.85mOhm DCR, 36Arms @ 40 DegC increase, 5.8mm high

C480			
C229			RcA1: CRCW0603127RFKEA - 127
C654	RenA: CRCW060349K9FKEA - 50k		RseqA: DNP
8x C655	RenB: CRCW060349K9FKEA - 50k		RseqB: DNP
C656	RfbA: CRCW060312K1FKEA - 12.1k		Cseq: DNP
100uF C657	RfbB: CRCW060312K1FKEA - 12.1k		RsenseA: CRCW060312K1FKEA - 12.1k
1206 C658	RcompB: CRCW060310K2FKEA - 10.2k		RsenseB: CRCW060312K1FKEA - 12.1k
C468	CcompB: C0603C562K5RACTU - 5.6nF		CcompC: GRM1885C1H330FA01D - 33pF

Vin=5.5V, 84% @ 16A, 87% @ 8A, LIR=16%, 8A step ->11mV deviation, 200mW core+winding loss in inductor at 16A

Total capacitance > 3200uF

1.8V rail must be > 1.5V rail at all times

B9 BuckRegulator.SchDoc — DeviceType: IR3894

0.15W Loss in IC | VCC5V0 | 0.4A | 2x C483 100uF 1206 C640 | PGND

SYNC_1V8 | PSeq1 | VCC3V3 | INA230_I2C_1V8

BuckSync, Vout, Enable, MasterVo, PVin, Vin, SwitchNode, VCC/LDO, INA230_I2C, PG — IR3894

1A | 94% | L9 SER1360-602 | PG_1V8 | VCC1V8

6.0uH, 5.5mOhm DCR, 9.4Arms @ 40 DegC increase, 5.8mm high

			RcA1: CRCW0603127RFKEA - 127
	RenA: CRCW060349K9FKEA - 50k		RseqA: DNP
3x	RenB: CRCW060349K9FKEA - 50k		RseqB: DNP
C482	RfbA: CRCW060320K0FKEA - 20k		Cseq: DNP
100uF C632	RfbB: CRCW06037K68FKEA - 7.68k		RsenseA: CRCW060320K0FKEA - 20k
1206 C653	RcompB: CRCW060329K4FKEA - 29.4k		RsenseB: CRCW06037K68FKEA - 7.68k
	CcompB: C0603C272K5RACTU - 2.7nF		CcompC: C0603C100K3GACTU - 10pF

Vin=5.5V, 88%@ 0.5A, 92%@ 1A, LIR=17%, 0.5A step ->4mV deviation, 0.1mV ripple, No inductor losses

Total capacitance > 900uF

I2C

INA230_I2C_1V0_GTX.SDA, INA230_I2C_1V0.SDA, INA230_I2C_1V2.SDA, INA230_I2C_1V5.SDA, INA230_I2C_1V8.SDA → SDA

INA230_I2C_1V0_GTX.SCL, INA230_I2C_1V0.SCL, INA230_I2C_1V2.SCL, INA230_I2C_1V5.SCL, INA230_I2C_1V8.SCL → SCL

I2C_SMPS

I2C Address Connections

1.8v: 0b1001011	INA230_I2C_1V8.A1	SDA
1.5v: 0b1001100	INA230_I2C_1V5.A1	SCL
1.2v: 0b1001101	INA230_I2C_1V2.A1	SCL
1.0v: 0b1001110	INA230_I2C_1V0.A1	SCL
1.0v (GTX): 0b1001111	INA230_I2C_1V0_GTX.A1	SCL

INA230_I2C_1V8.A0		SCL
INA230_I2C_1V5.A0		
INA230_I2C_1V2.A0		
INA230_I2C_1V0.A0		SDA
INA230_I2C_1V0_GTX.A0		SCL

VCC3V3

Enable pin >1.2V to turn on Device.

Vo=Vref x (1+RfbA/RfbB)

Type III compensation - RcompB, CcompB and CcompC. Note that RcompA=127Ohms, CcompA=2.2nF

UVLO = 4V on 5V rail

SYNC_1V2 → SYNC_1V2

SYNC_1V0_GTX, SYNC_1V0 → SYNC_1V0_GTX, SYNC_1V0

SYNC_1V5, SYNC_1V8 → SYNC_1V5, SYNC_1V8

DGND

PG_5V0 R151 0

PSeq1 R152 DNP

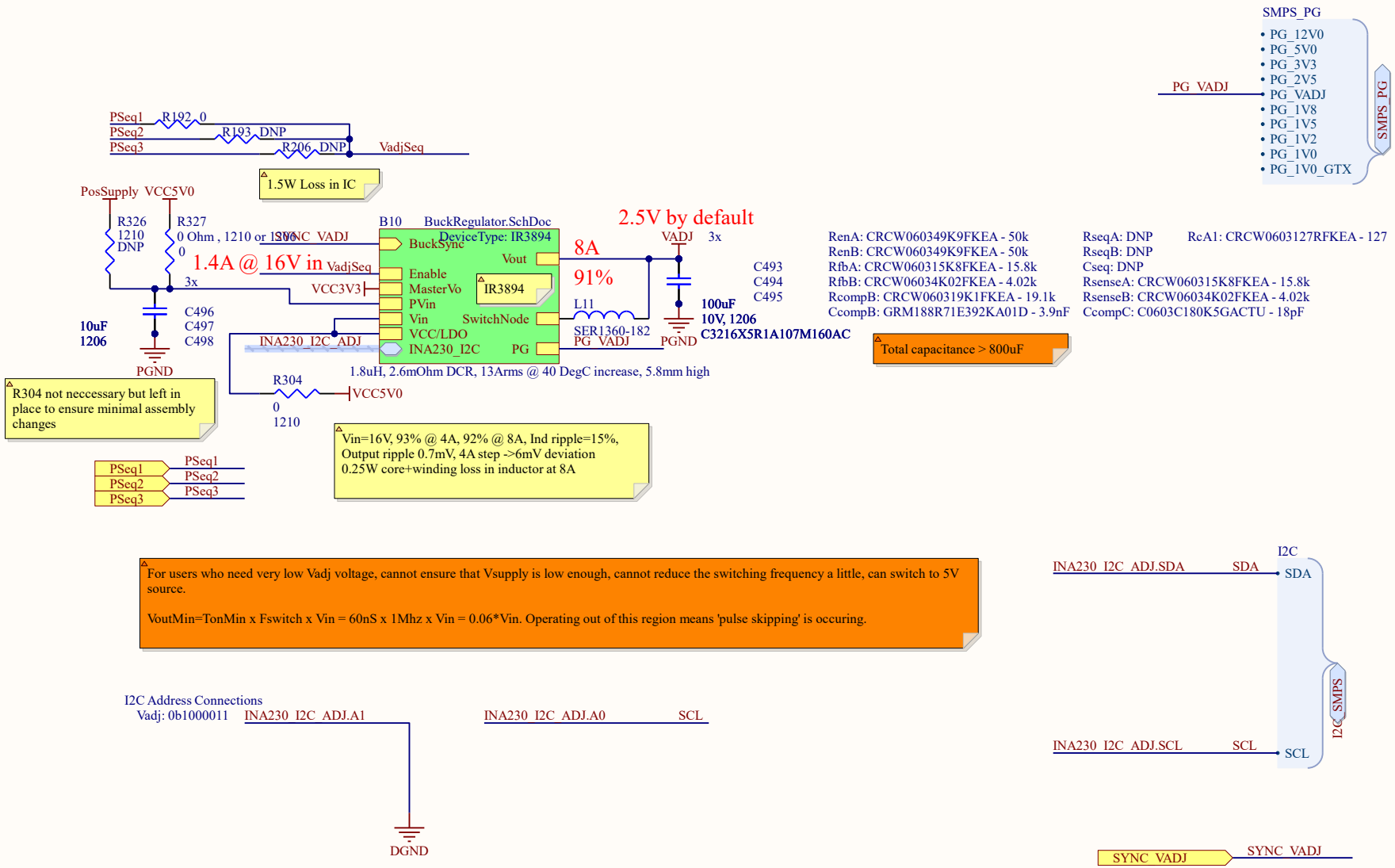
PSeq2 R164 DNP

PSeq3 R165 DNP

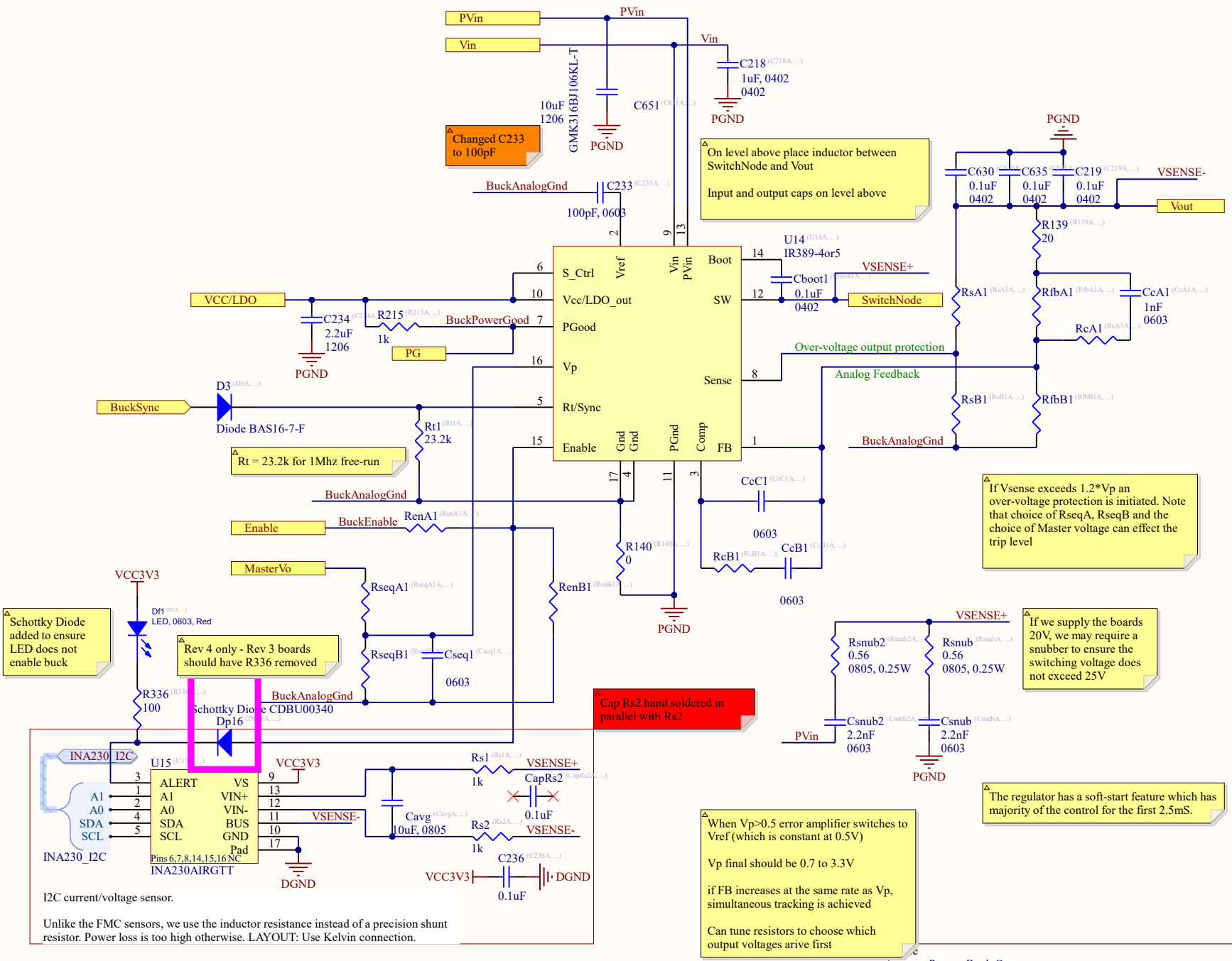
GTXSeq

Title: Power: Indirect Converters

Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 48 of 60	
File: C:\Users\..\Power_BuckIndirect.SchDoc	Drawn By: Adam Gilbert	

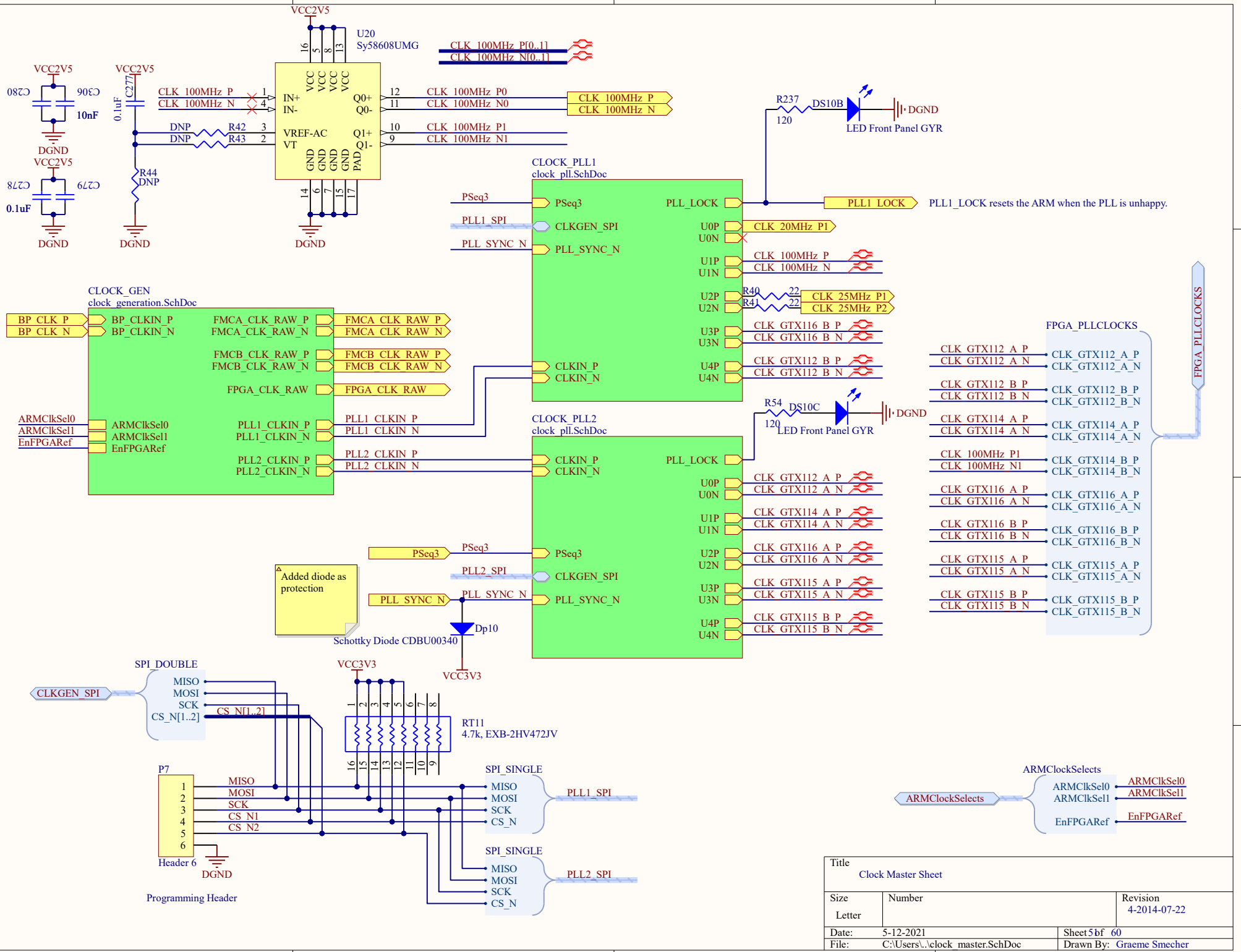


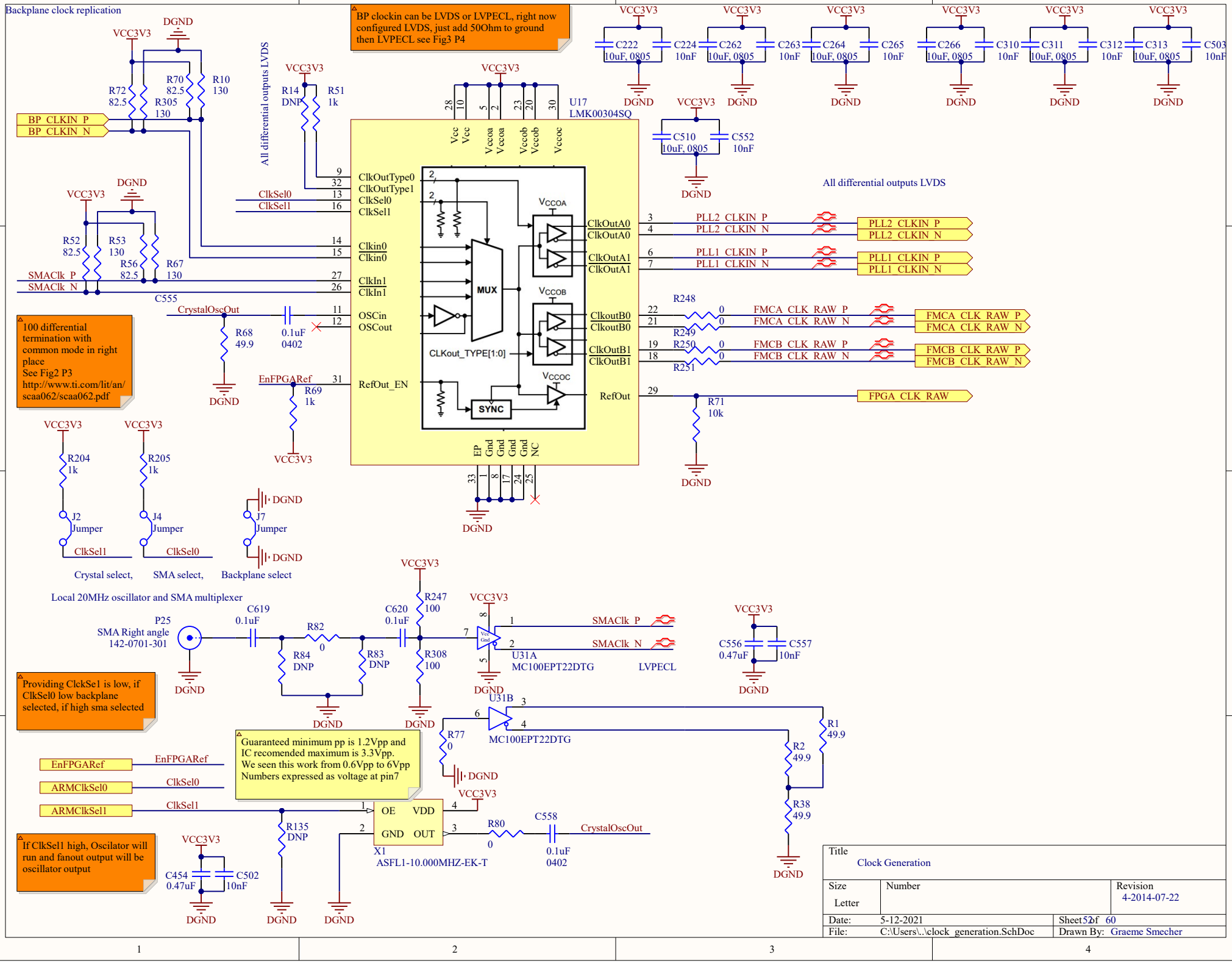
Title: Selectable Buck Regulators		
Size: Letter	Number	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 49 of 60	
File: C:\Users\..\Power_BuckSelect.SchDoc	Drawn By: Adam Gilbert	

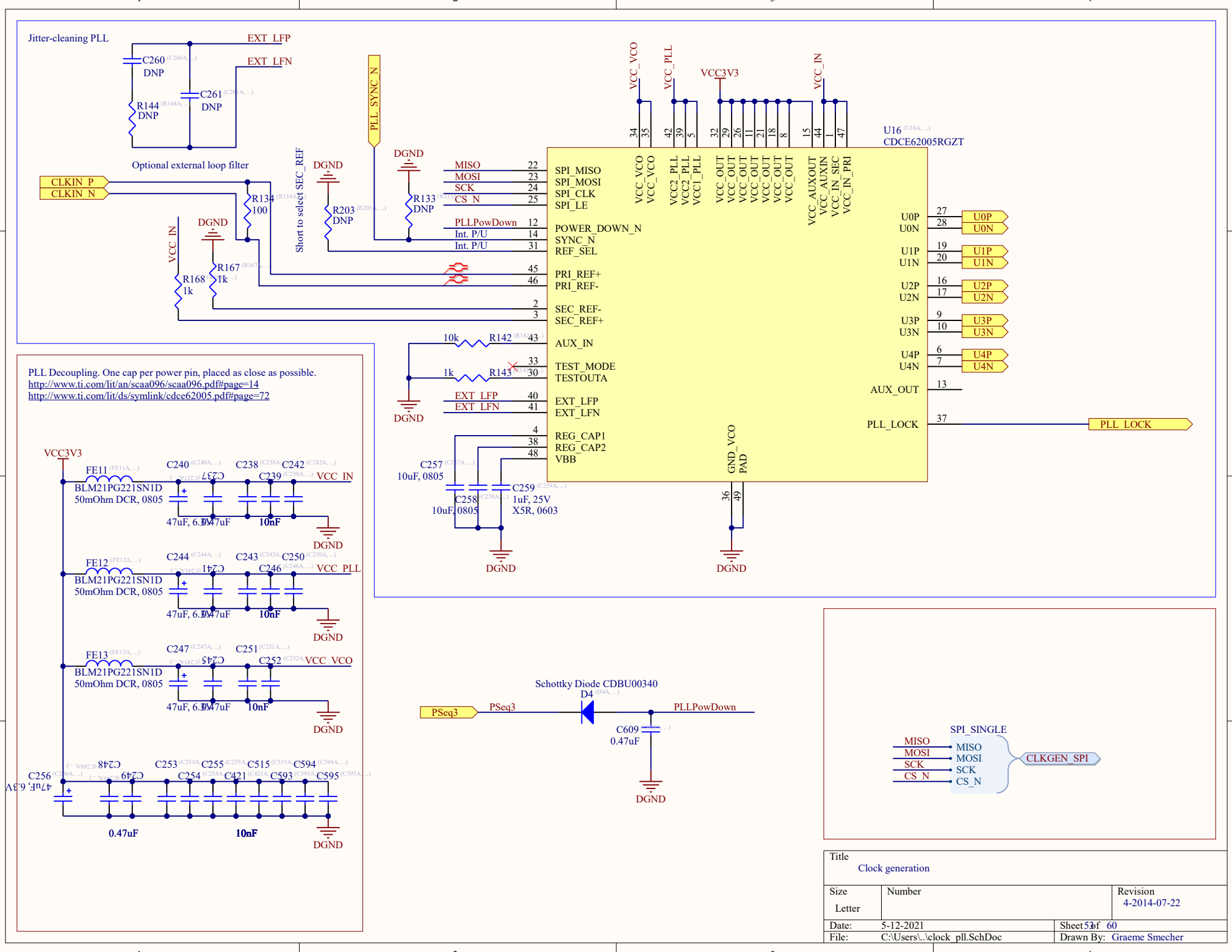


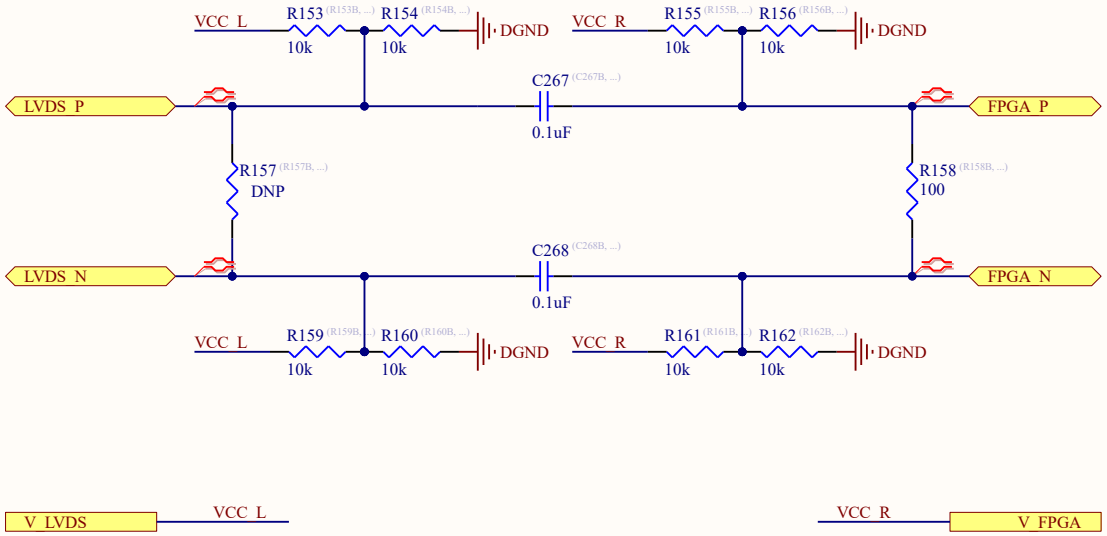
Power: Buck Converter

Size	Number	Revision
Letter		4-2014-07-22
Date: 5-12-2021		Sheet 50 of 60
File: C:\Users\..\BuckRegulator.SchDoc		Drawn By: Adam Gilbert


VCC2V5
U20
Sy58608UMG
CLK_100MHz_P[0..1]
CLK_100MHz_N[0..1]
C280
C306
10nF
C277
0.1uF
C278
C279
0.1uF
DGND
CLK_100MHz_P
CLK_100MHz_N
DNP
R42
R43
R44
DNP
IN+
IN-
VREF-AC
VT
Q0+
Q0-
Q1+
Q1-
CLK_100MHz_P0
CLK_100MHz_N0
CLK_100MHz_P1
CLK_100MHz_N1
R237
120
DS10B
LED Front Panel GYR
CLOCK_PLL1
clock_pll.SchDoc
PSeq3
PLL1_SPI
PLL_SYNC_N
PLL_LOCK
CLKGEN_SPI
PLL_SYNC_N
U0P
U0N
U1P
U1N
U2P
U2N
U3P
U3N
U4P
U4N
CLKIN_P
CLKIN_N
CLK_20MHz_P1
CLK_100MHz_P
CLK_100MHz_N
R40
R41
22
CLK_25MHz_P1
CLK_25MHz_P2
CLK_GTX116_B_P
CLK_GTX116_B_N
CLK_GTX112_B_P
CLK_GTX112_B_N
PLL1_LOCK
PLL1_LOCK resets the ARM when the PLL is unhappy.
CLOCK_GEN
clock_generation.SchDoc
BP_CLK_P
BP_CLK_N
BP_CLKIN_P
BP_CLKIN_N
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
FPGA_CLK_RAW
ARMClkSel0
ARMClkSel1
EnFPGARef
PLL1_CLKIN_P
PLL1_CLKIN_N
PLL2_CLKIN_P
PLL2_CLKIN_N
CLOCK_PLL2
clock_pll.SchDoc
R54
120
DS10C
LED Front Panel GYR
DGND
CLK_GTX112_A_P
CLK_GTX112_A_N
CLK_GTX114_A_P
CLK_GTX114_A_N
CLK_100MHz_P1
CLK_100MHz_N1
CLK_GTX116_A_P
CLK_GTX116_A_N
CLK_GTX115_A_P
CLK_GTX115_A_N
CLK_GTX115_B_P
CLK_GTX115_B_N
FPGA_PLLCLOCKS
Added diode as protection
Schottky Diode CDBU00340
Dp10
VCC3V3
SPI_DOUBLE
CLKGEN_SPI
MISO
MOSI
SCK
CS_N[1..2]
RT11
4.7k, EXB-2HV472JV
P7
Header 6
CS_N1
CS_N2
SPI_SINGLE
PLL1_SPI
PLL2_SPI
Programming Header
ARMClockSelects
Title
Clock Master Sheet
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 5 of 60
File:
C:\Users\..\clock_master.SchDoc
Drawn By:
Graeme Smecher


Backplane clock replication
BP clockin can be LVDS or LVPECL, right now configured LVDS, just add 50Ohm to ground then LVPECL see Fig3 P4
All differential outputs LVDS
U17
LMK00304SQ
LMK00304SQ
MUX
SYNC
CLKout_TYPE[1:0]
All differential outputs LVDS
100 differential termination with common mode in right place
See Fig2 P3
http://www.ti.com/lit/an/scaa062/scaa062.pdf
Crystal select,
SMA select,
Backplane select
Local 20MHz oscillator and SMA multiplexer
SMA Right angle
142-0701-301
U31A
MC100EPT22DTG
LVPECL
U31B
MC100EPT22DTG
Providing ClckSe1 is low, if ClkSel0 low backplane selected, if high sma selected
Guaranteed minimum pp is 1.2Vpp and IC recomended maximum is 3.3Vpp.
We seen this work from 0.6Vpp to 6Vpp
Numbers expressed as voltage at pin7
If ClkSel1 high, Oscilator will run and fanout output will be oscillator output
X1
ASFL1-10.000MHZ-EK-T
Title
Clock Generation
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 52 of 60
File:
C:\Users\..\clock_generation.SchDoc
Drawn By:
Graeme Smecher

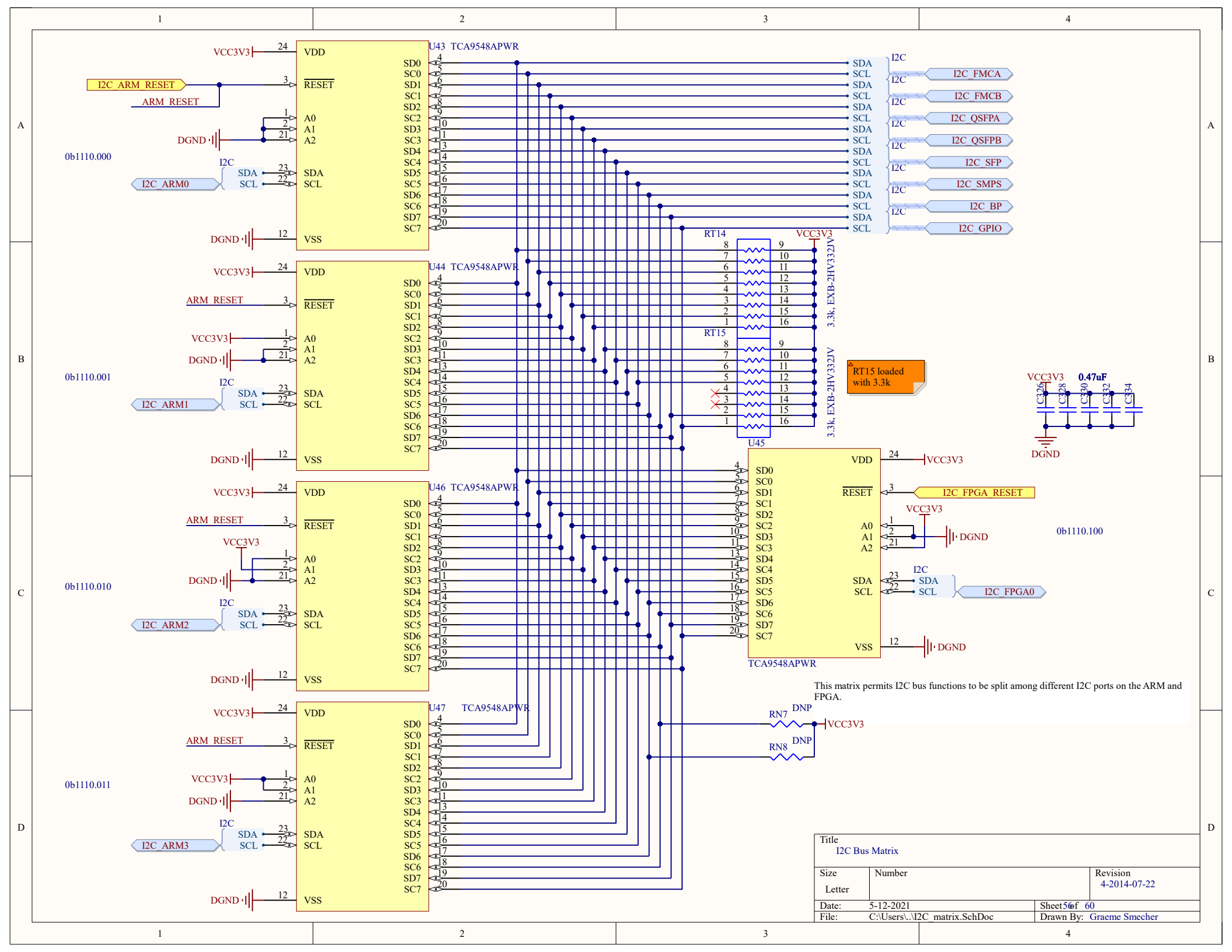

Jitter-cleaning PLL
EXT_LFP
EXT_LFN
C260 DNP
C261 DNP
R144 DNP
Optional external loop filter
CLKIN_P
CLKIN_N
R134 100
Short to select SEC_REF
R203 DNP
R133 DNP
PLL_SYNC_N
DGND
VCC_IN
R167 1k
R168 1k
MISO
MOSI
SCK
CS_N
PLLPowDown
Int. P/U
SPI_MISO
SPI_MOSI
SPI_CLK
SPI_LE
POWER_DOWN_N
SYNC_N
REF_SEL
PRI_REF+
PRI_REF-
SEC_REF-
SEC_REF+
AUX_IN
TEST_MODE
TESTOUTA
EXT_LFP
EXT_LFN
REG_CAP1
REG_CAP2
VBB
10k R142
1k R143
C257 10uF, 0805
C258 10uF, 0805
C259 1uF, 25V X5R, 0603
VCC_VCO
VCC_PLL
VCC3V3
VCC_IN
VCC2_PLL
VCC1_PLL
VCC_OUT
VCC_AUXOUT
VCC_AUXIN
VCC_IN_SEC
VCC_IN_PRI
GND_VCO PAD
U16 CDCE62005RGZT
U0P
U0N
U1P
U1N
U2P
U2N
U3P
U3N
U4P
U4N
AUX_OUT
PLL_LOCK
PLL Decoupling. One cap per power pin, placed as close as possible.
http://www.ti.com/lit/an/scaa096/scaa096.pdf#page=14
http://www.ti.com/lit/ds/symlink/cdce62005.pdf#page=72
FE11 BLM21PG221SN1D 50mOhm DCR, 0805
FE12 BLM21PG221SN1D 50mOhm DCR, 0805
FE13 BLM21PG221SN1D 50mOhm DCR, 0805
47uF, 6.3V
0.47uF
10nF
Schottky Diode CDBU00340
D4
PSeq3
C609 0.47uF
SPI_SINGLE
CLKGEN_SPI
Title
Clock generation
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
File: C:\Users\..\clock_pll.SchDoc
Sheet 53 of 60
Drawn By: Graeme Smecher

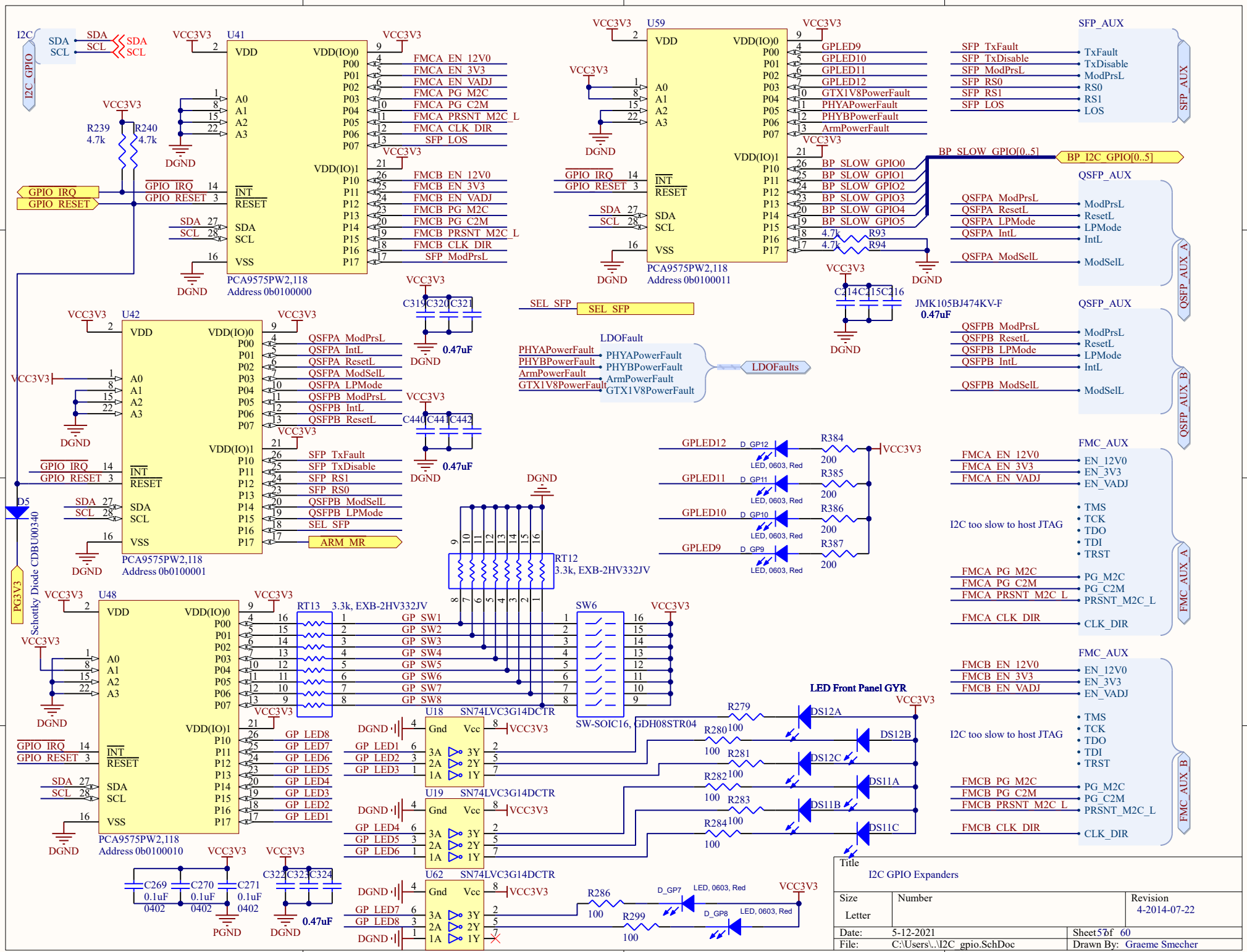


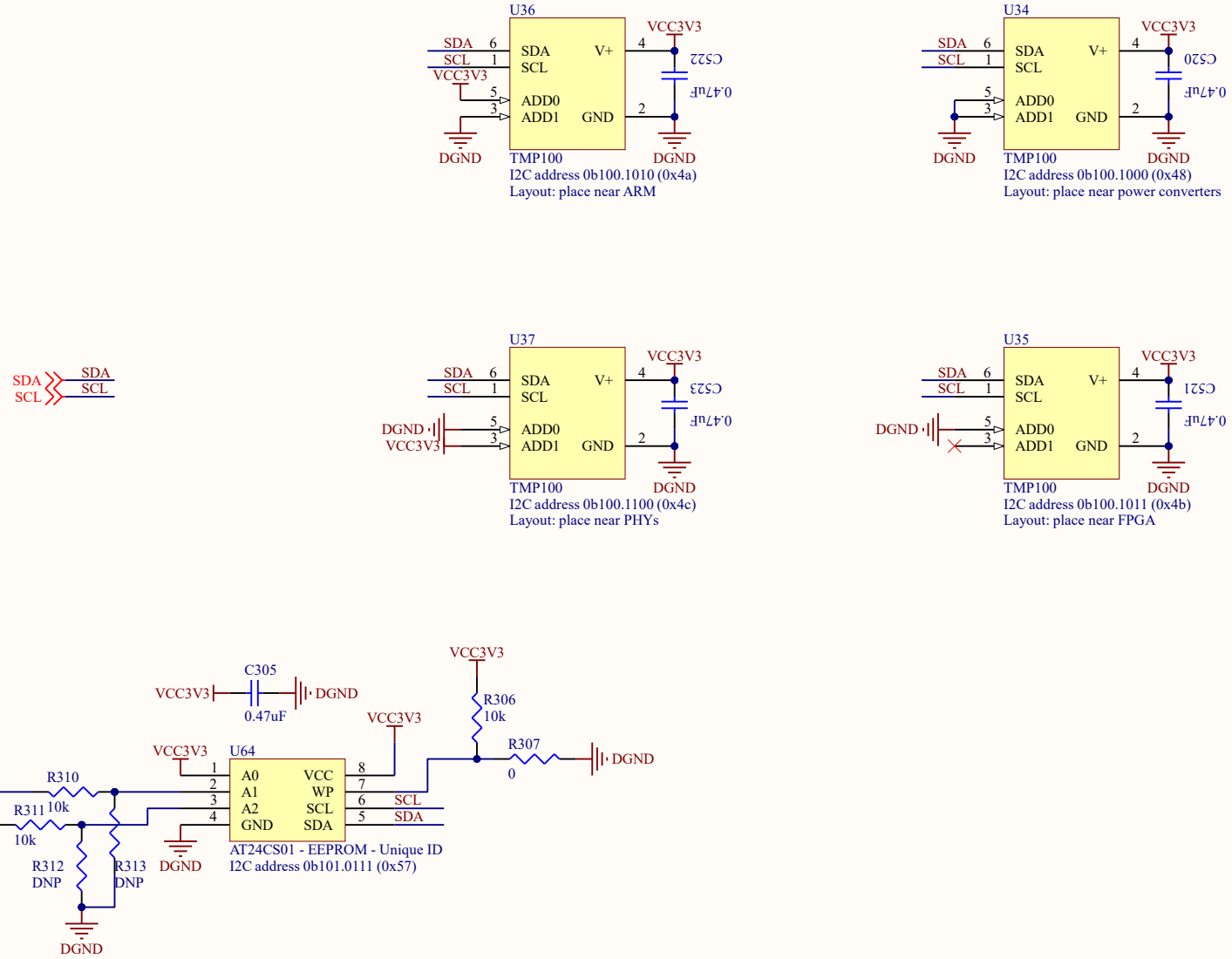
FMC clocks are 2.5v LVDS, and are hooked up to FPGA banks powered via either VADJ or VCCIOB. In both cases, the actual bank voltage isn't known a-priori, so it's necessary to provide a level-translation or coupling network. We follow the advice of UG471, p.92.

Note that some clocks are actually bidirectional, so I've modified the recommended biasing network to be symmetric.

Title: LVDS Clock Coupling		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 55 of 60	
File: C:\Users\..\clock_passive_coupling.SchDoc	Drawn By: Graeme Smecher	


U43 TCA9548APWR
U44 TCA9548APWR
U46 TCA9548APWR
U47 TCA9548APWR
U45
TCA9548APWR
VDD
VSS
RESET
A0
A1
A2
SDA
SCL
SD0
SC0
SD1
SC1
SD2
SC2
SD3
SC3
SD4
SC4
SD5
SC5
SD6
SC6
SD7
SC7
VCC3V3
DGND
I2C_ARM_RESET
ARM_RESET
I2C_ARM0
I2C_ARM1
I2C_ARM2
I2C_ARM3
0b1110.000
0b1110.001
0b1110.010
0b1110.011
0b1110.100
I2C_FMCA
I2C_FMCB
I2C_QSFPA
I2C_QSFPB
I2C_SFP
I2C_SMPS
I2C_BP
I2C_GPIO
I2C_FPGA_RESET
I2C_FPGA0
RT14
RT15
3.3k, EXB-2HV332JV
RT15 loaded with 3.3k
C326
C328
C330
C332
C334
0.47uF
RN7
RN8
DNP
This matrix permits I2C bus functions to be split among different I2C ports on the ARM and FPGA.
Title
I2C Bus Matrix
Size
Letter
Number
Revision
4-2014-07-22
Date:
5-12-2021
Sheet 56 of 60
File:
C:\Users\..\I2C_matrix.SchDoc
Drawn By:
Graeme Smecher


I2C
SDA
SCL
I2C_GPIO
U41
PCA9575PW2,118
Address 0b0100000
U59
PCA9575PW2,118
Address 0b0100011
U42
PCA9575PW2,118
Address 0b0100001
U48
PCA9575PW2,118
Address 0b0100010
VCC3V3
DGND
PGND
R239 4.7k
R240 4.7k
GPIO_IRQ
GPIO_RESET
INT
RESET
FMCA_EN_12V0
FMCA_EN_3V3
FMCA_EN_VADJ
FMCA_PG_M2C
FMCA_PG_C2M
FMCA_PRSNT_M2C_L
FMCA_CLK_DIR
SFP_LOS
FMCB_EN_12V0
FMCB_EN_3V3
FMCB_EN_VADJ
FMCB_PRSNT_M2C_L
FMCB_CLK_DIR
SFP_ModPrsL
GPLED9
GPLED10
GPLED11
GPLED12
GTX1V8PowerFault
PHYAPowerFault
PHYBPowerFault
ArmPowerFault
BP_SLOW_GPIO0
BP_SLOW_GPIO1
BP_SLOW_GPIO2
BP_SLOW_GPIO3
BP_SLOW_GPIO4
BP_SLOW_GPIO5
BP_SLOW_GPIO[0..5]
BP_I2C_GPIO[0..5]
R93 4.7k
R94 4.7k
C314 C315 C316
JMK105BJ474KV-F
0.47uF
C319 C320 C321
C440 C441 C442
QSFPA_ModPrsL
QSFPA_IntL
QSFPA_ResetL
QSFPA_ModSelL
QSFPA_LPMode
QSFPB_ModPrsL
QSFPB_IntL
QSFPB_ResetL
SFP_TxFault
SFP_TxDisable
SFP_RS1
SFP_RS0
QSFPB_ModSelL
QSFPB_LPMode
SEL_SFP
ARM_MR
LDOFault
LDOFaults
SFP_AUX
QSFP_AUX
QSFP_AUX_A
QSFP_AUX_B
FMC_AUX
FMC_AUX_A
FMC_AUX_B
TxFault
TxDisable
ModPrsL
RS0
RS1
LOS
ResetL
LPMode
IntL
ModSelL
EN_12V0
EN_3V3
EN_VADJ
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR
TMS
TCK
TDO
TDI
TRST
I2C too slow to host JTAG
D5
PG3V3
Schottky Diode CDBU00340
RT12
RT13
3.3k, EXB-2HV332JV
SW6
SW-SOIC16, GDH08STR04
GP_SW1
GP_SW2
GP_SW3
GP_SW4
GP_SW5
GP_SW6
GP_SW7
GP_SW8
GP_LED1
GP_LED2
GP_LED3
GP_LED4
GP_LED5
GP_LED6
GP_LED7
GP_LED8
U18
U19
U62
SN74LVC3G14DCTR
Gnd
Vcc
3A
2A
1A
3Y
2Y
1Y
LED Front Panel GYR
R279
R280 100
R281
R282 100
R283
R284 100
R286
R299 100
DS12A
DS12B
DS12C
DS11A
DS11B
DS11C
D_GP7
D_GP8
LED, 0603, Red
R384
R385
R386
R387
200
D_GP12
D_GP11
D_GP10
D_GP9
C269 C270 C271
C322 C323 C324
0.1uF
0402
0.47uF
Title
I2C GPIO Expanders
Size
Letter
Number
Revision
4-2014-07-22
Date: 5-12-2021
File: C:\Users\..\I2C_gpio.SchDoc
Sheet 5 of 60
Drawn By: Graeme Smecher



Title: I2C Temperature Sensors		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 58 of 60	
File: C:\Users\..\I2C_temp_eeprom.SchDoc	Drawn By: Graeme Smecher	

Fiducial marks

FM1 Fiducial
FM2 Fiducial
FM3 Fiducial
FM4 Fiducial
FM5 Fiducial
FM6 Fiducial
FM7 Fiducial
FM8 Fiducial
FM9 Fiducial
FM10 Fiducial

HS5 Heatsink
HS6 Heatsink

PLL heatsinks should be glued down
Can use epoxy or sticky stab

HS1 Heatsink
HS2 MPW37.5-12
HS3 Heatsink

HS4 Heatsink + Pushpins + Compound

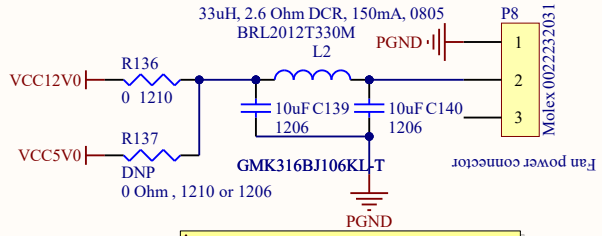
Changing BOM to include the custom Cool innovations part

Have ensured that the PCB will be compatible with these heatsinks.

The MPA31-12 is 11.6mm in height and has 3.84, 2.43 and 1.82 degC/W at 200, 400 and 600 LFM respectively

The MPW37.5-12 is 11.6mm in height and has 2.74, 1.77 and 1.4 decC/W at 200, 400 and 600 LFM respectively

The SLF-40 is 13.8mm in height and has 2 degC/W. Note that in a crate this height will cause a collision.



Pi filter here will only attenuate high frequency noise

ASSEMBLY NOTE:
We would like to have a 100nF capacitor >50V capacitor soldered in parallel with ever instances of Rs2 (e.g Rs2A, Rs2B up to Rs2I) 9 in total. The part can be either soldered on the side of Rs2, or directly on top. We would require that the solution be lower profile than the bottom side 22uF 1206 capacitors. Note that Rs2 A..I are present under each of the buck converters next to U15 A.. I the buck power sensor ICs.

Logos

Logo1
LOGO
ChimeLogo1
Logo2
LOGO
ChimeLogo1
Logo3
LOGO
McGillLogo1
Logo4
LOGO
McGillLogo1
Logo5
LOGO
McGillLogo1

FMC Front Panel Bezels

FMC Bezel
Bezel1
FMC Bezel
Bezel2
FMC Bezel
Bezel3
FMC Bezel
Bezel4

These FMC Front panel Bezels have been added to aid the mechanical design of the board

FP1
9U Front panel
FrontPanel 9U

Front panel Support

PS1
Front panel bracket
61-960
PS2
Front panel bracket
61-960
PS3
Front panel bracket
61-960

ASSEMBLY ISSUE: Board stiffener warped board

Board Stiffener

Board Stiffener
BusBarNet
BoardStiffener1
PGND

14 inches long, 1 inch spacing, half hardness, non epoxy coated, lowest profile 250mil

ITX Case mounting holes

ITXMount
ITX Mounting Holes
0
Gnd
ITXmountingholes
PGND

ARM / PHY / Memory Shield

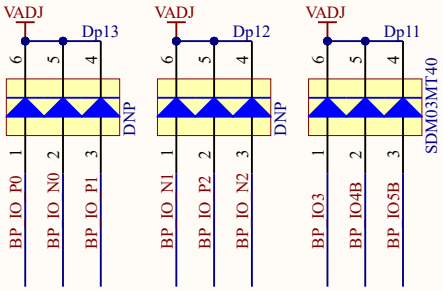
Arm Shield
SH1
20S-CBS-4x2x0.2
PGND

Shield is 200mil high

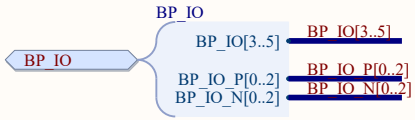
W1 Washer
W2 Washer
Please install washer on back of P15/P16

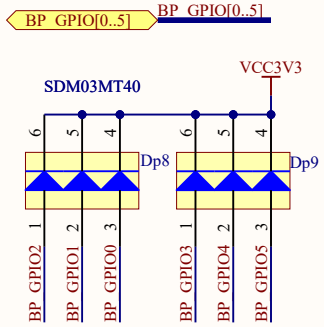
Title: Mechanical		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021		Sheet 59 of 60
File: C:\Users\..\mechanical.SchDoc		Drawn By: Adam Gilbert

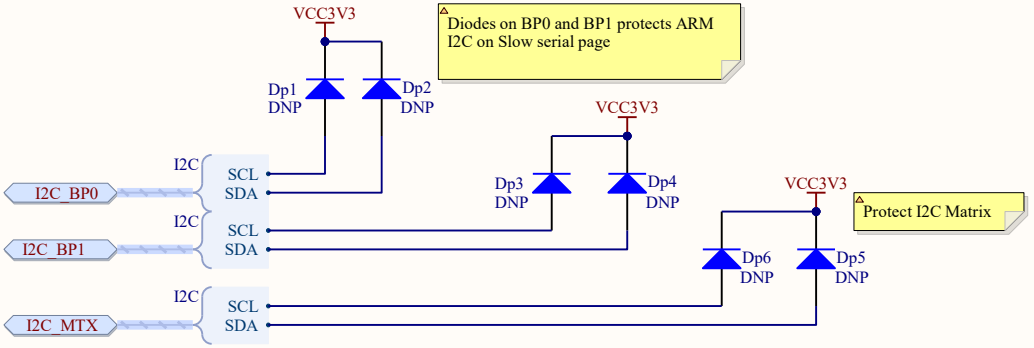
This page is new to Rev 4 - i.e not present on Rev 3
All diodes 30mA, <0.4Vf, 40Vr

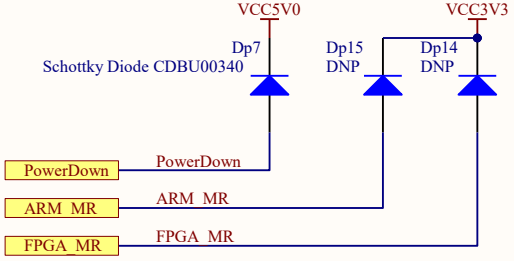


Pins go to FPGA bank 13 and 14









Title: Protection Diodes		
Size: Letter	Number:	Revision: 4-2014-07-22
Date: 5-12-2021	Sheet 60 of 60	
File: C:\Users\..\ProtectionDiodes.SchDoc	Drawn By: Graeme Smecher	